

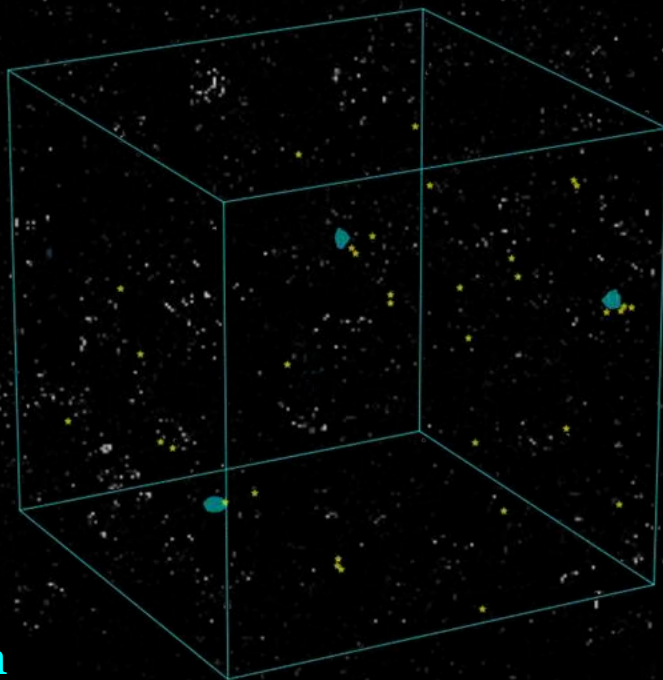


SCUOLA
NORMALE
SUPERIORE

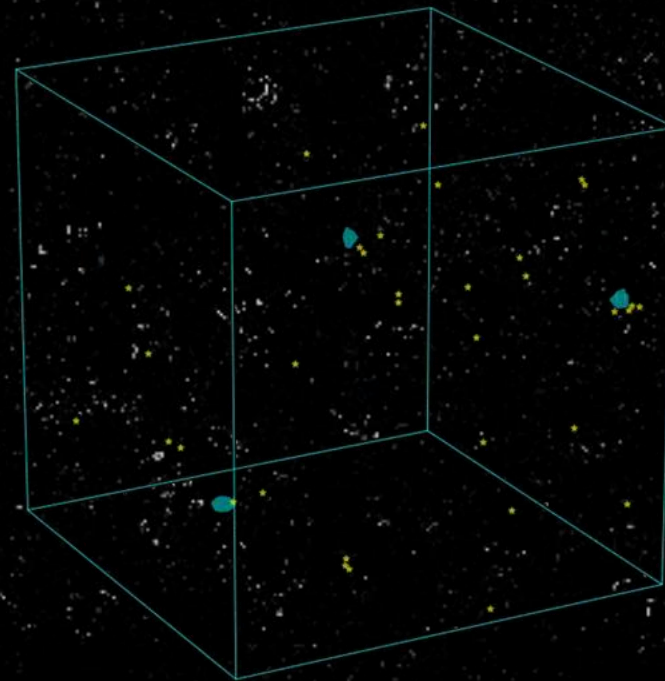
TILING: Bayesian reconstruction of cosmological initial conditions during the Epoch of Reionization

Nikolaos (Nikos) Triantafyllou

Andrei Mesinger, Steven Murray, Samuel Gagnon-Hartman



Truth



Reconstructed

Simulated with



Context

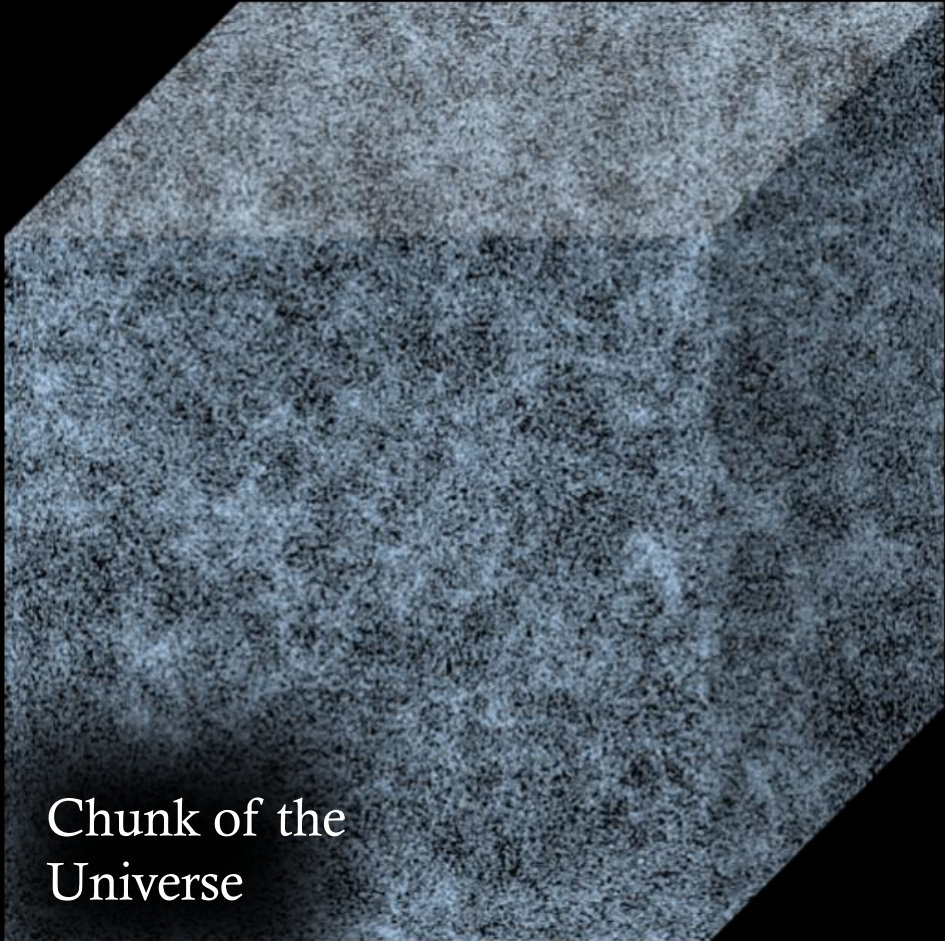


$\gamma\phi$

Big bang

$z \sim$ a few hundred

$z \sim$ a few hundred

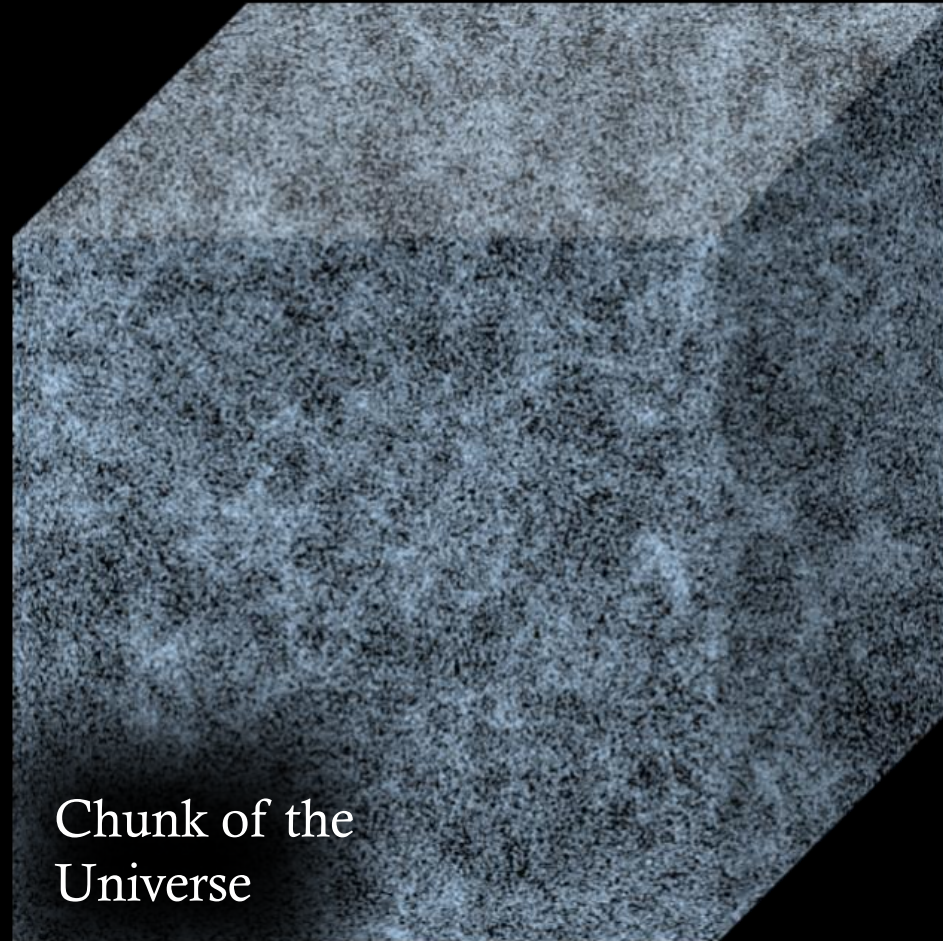


Density



$z \sim$ a few hundred

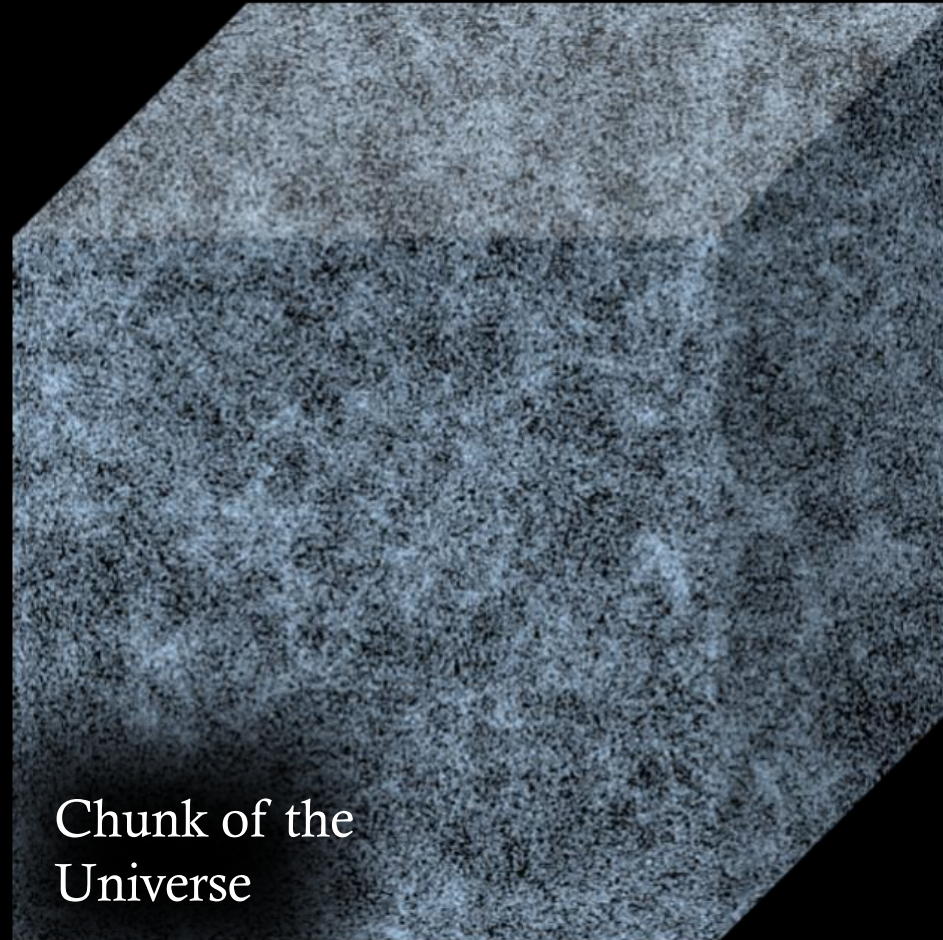
Initial Conditions (Gaussian Random Field)



Density



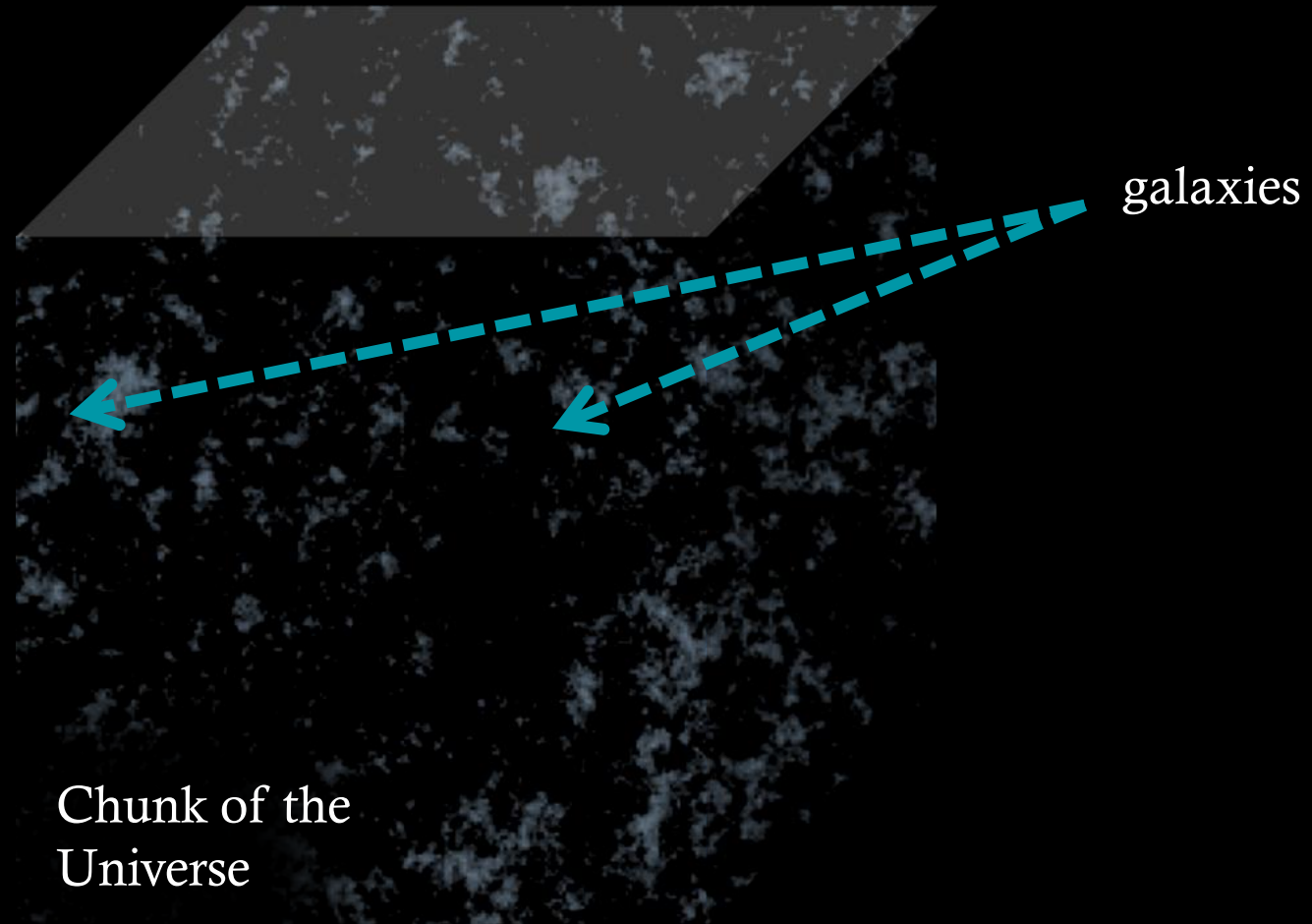
Evolving density



Density



Evolving density



Density



Galaxies ionizing their surrounding regions

All this
just in the 1st billion years
of the Universe

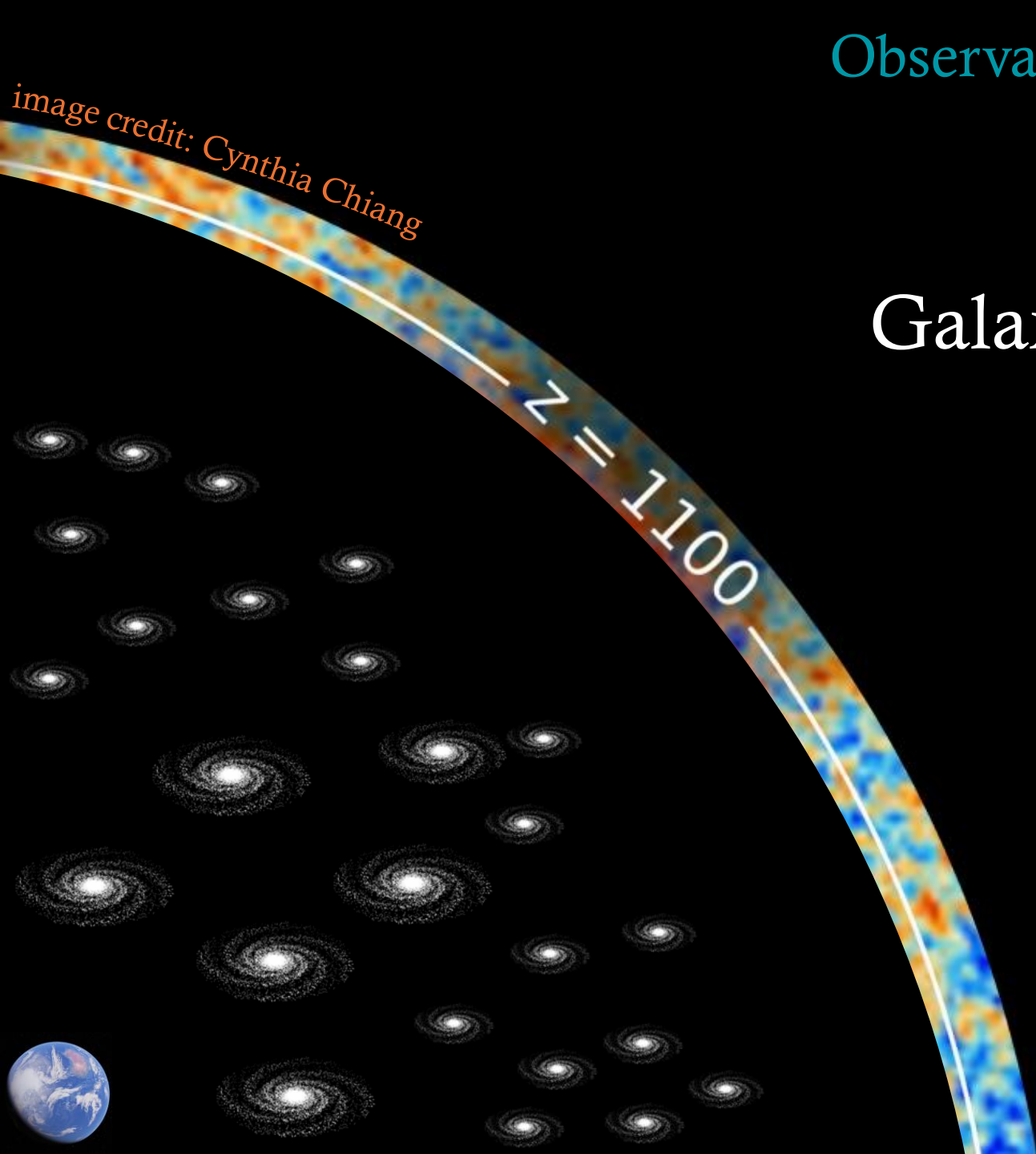
~1/2 of the Observable
Universe

Huge discovery space
mostly uncharted

Observations

image credit: Cynthia Chiang

Galaxies

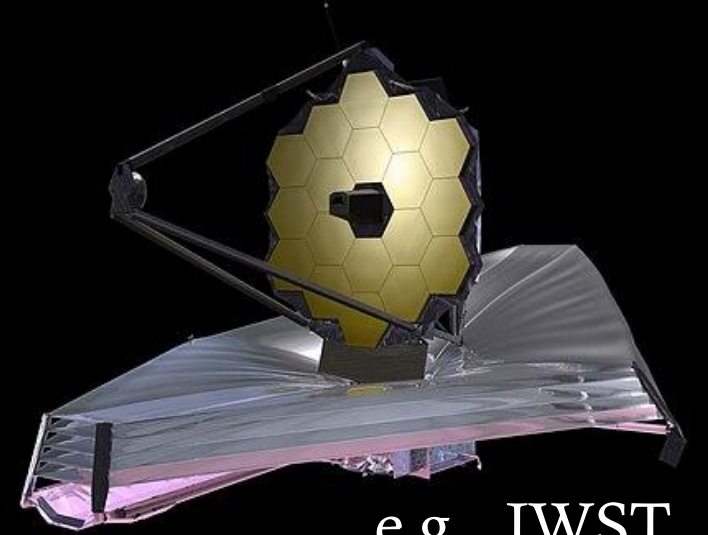


Observations

Galaxies

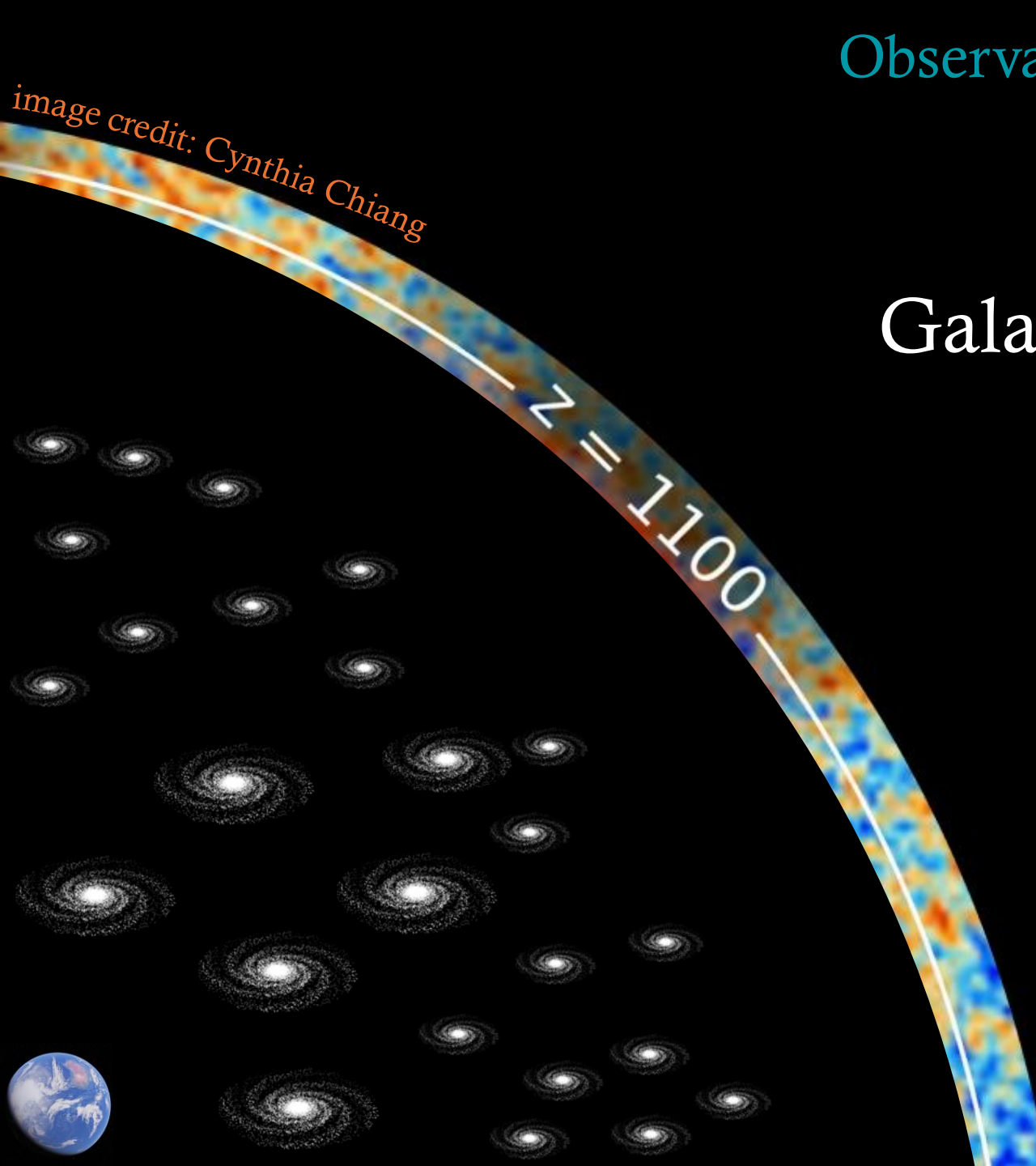
image credit: Cynthia Chiang

$z = 1100$



e.g., JWST

e.g., Bagley+23, Finkelstein+25,
Donnan+24, Eisenstein+25, Casey+22,
Oesch+23



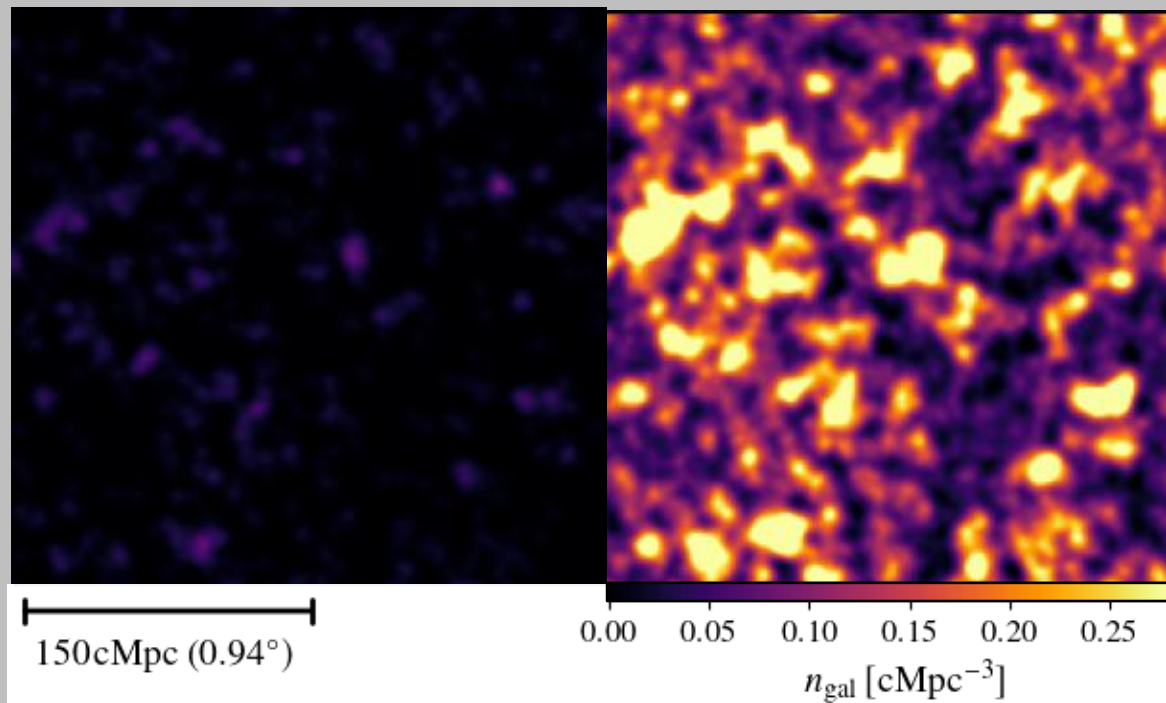
Observations

image credit: Cynthia Chiang

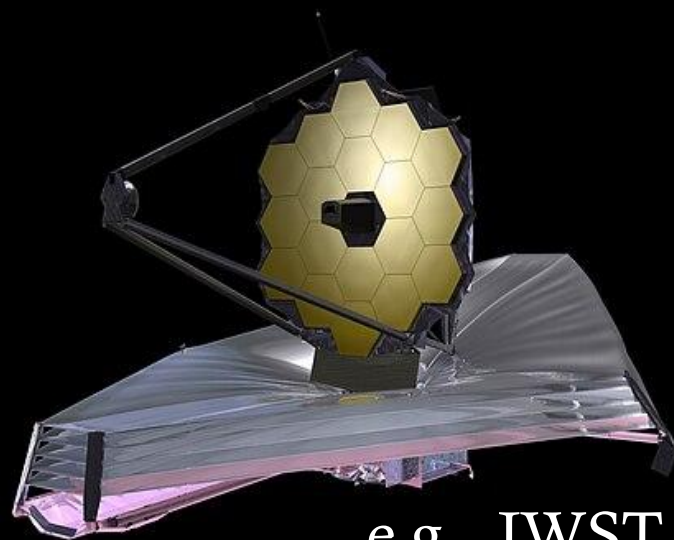
We only see a **few** bright galaxies!

What we see

What is actually there



NT+in prep



e.g., JWST

e.g., Bagley+23, Finkelstein+25,
Donnan+24, Eisenstein+25, Casey+22,
Oesch+23

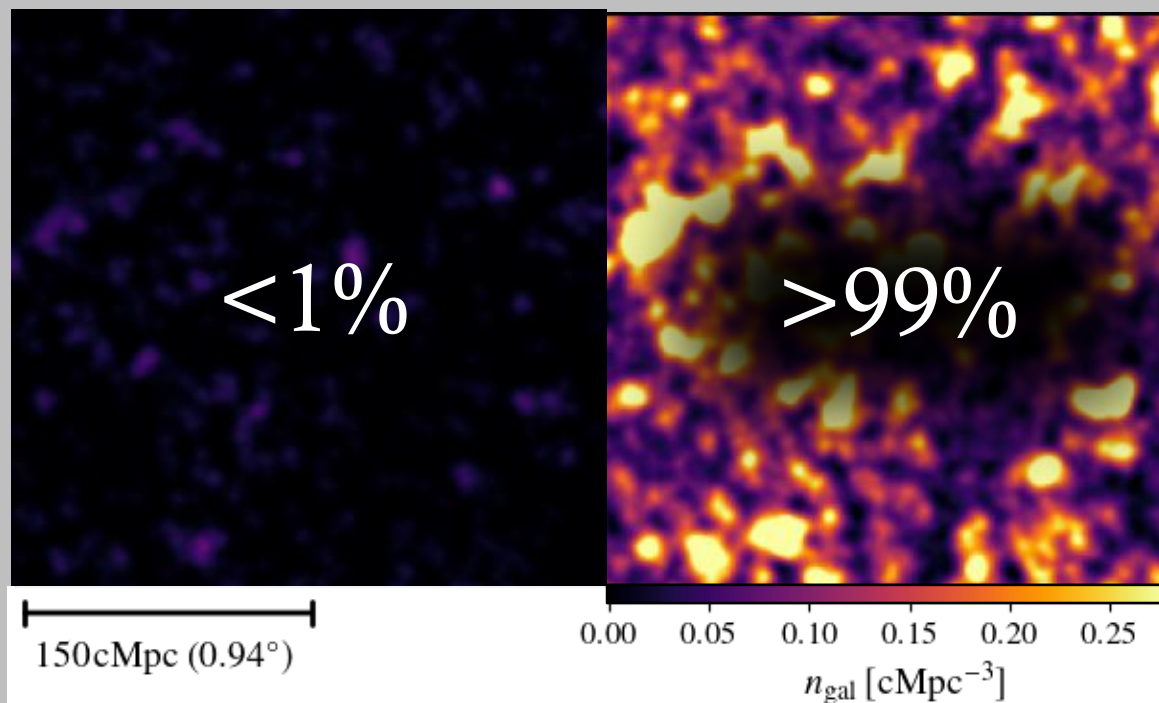
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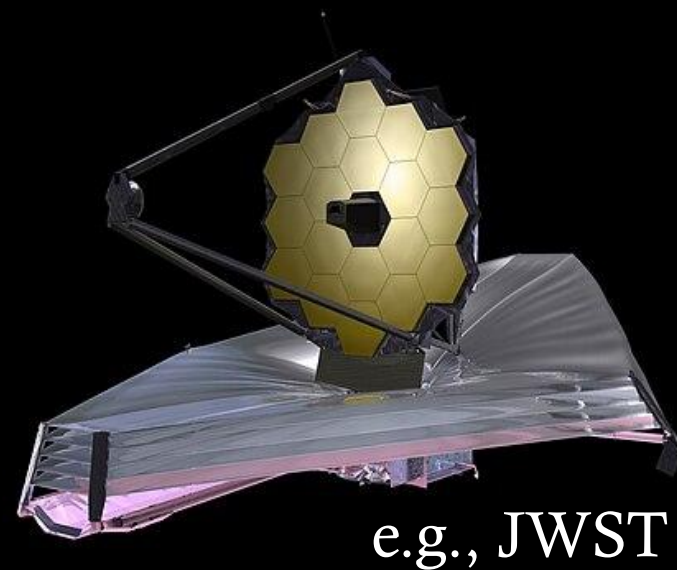
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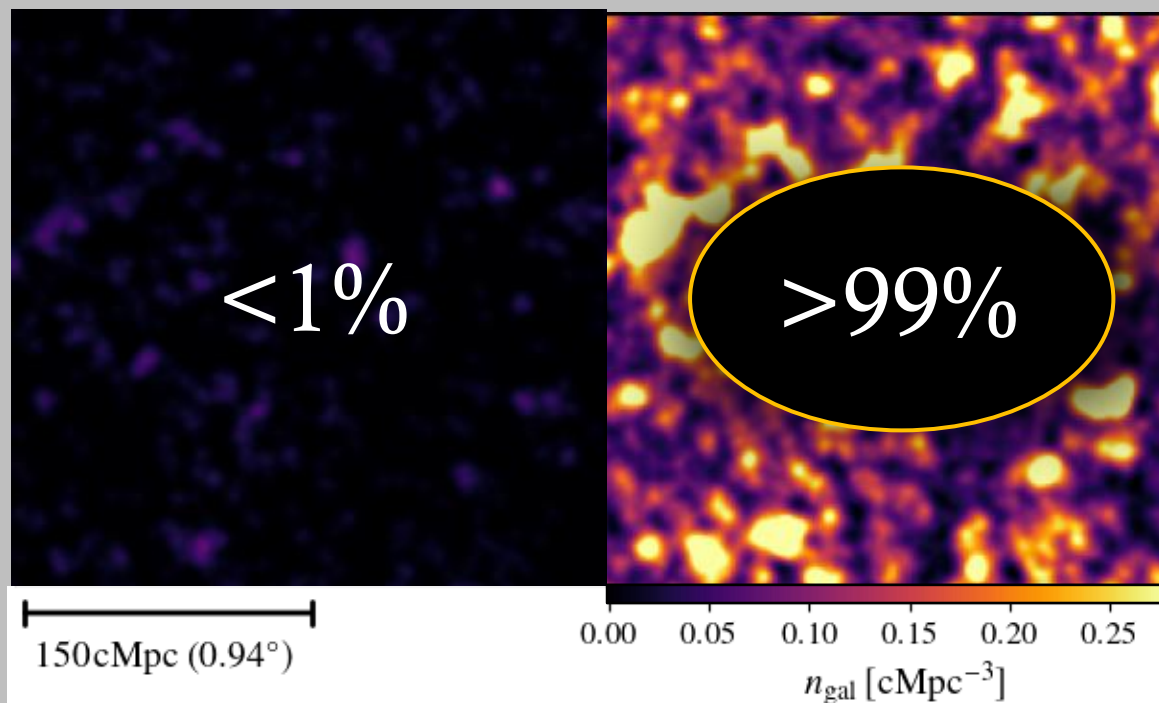
Observations

image credit: Cynthia Chiang

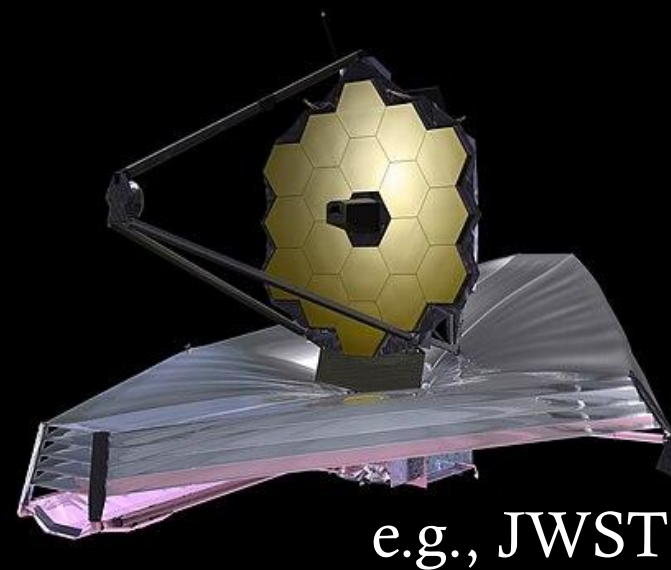
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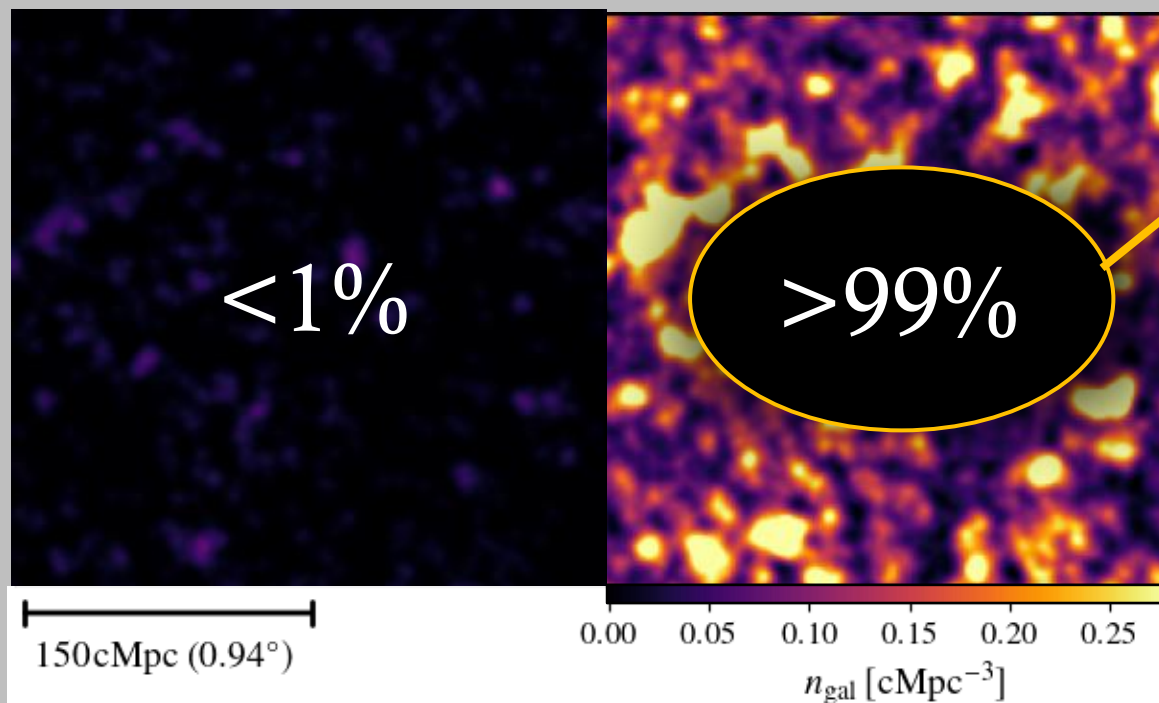
Observations

image credit: Cynthia Chiang

We only see a **few** bright galaxies!

What we see

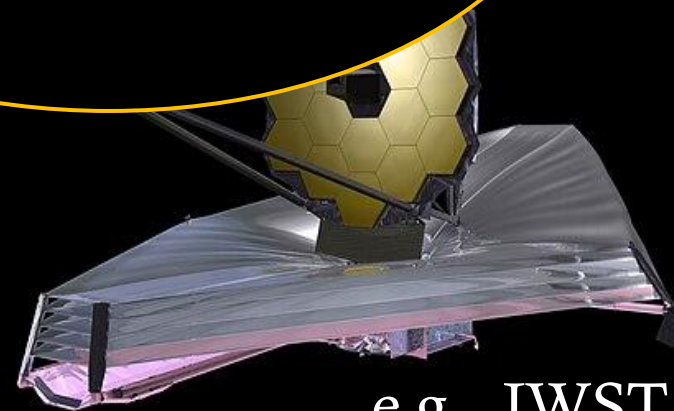
What is actually there



NT+in prep

Responsible for
Reionization

e.g., Qin+2026



e.g., JWST

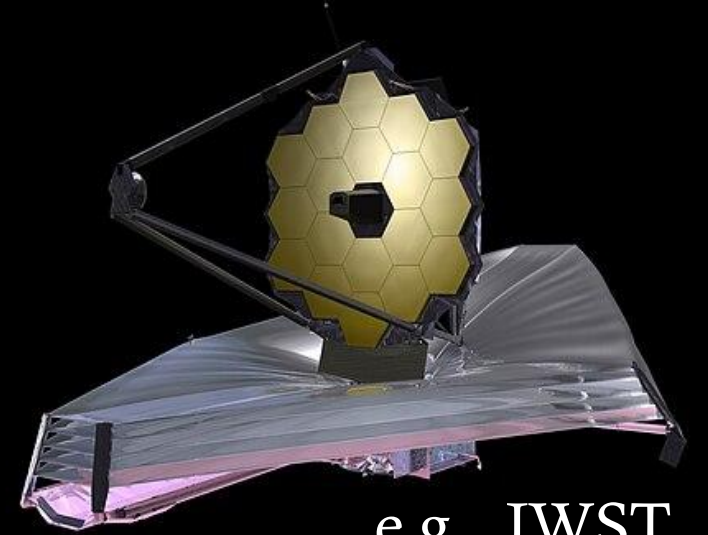
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Observations

Galaxies

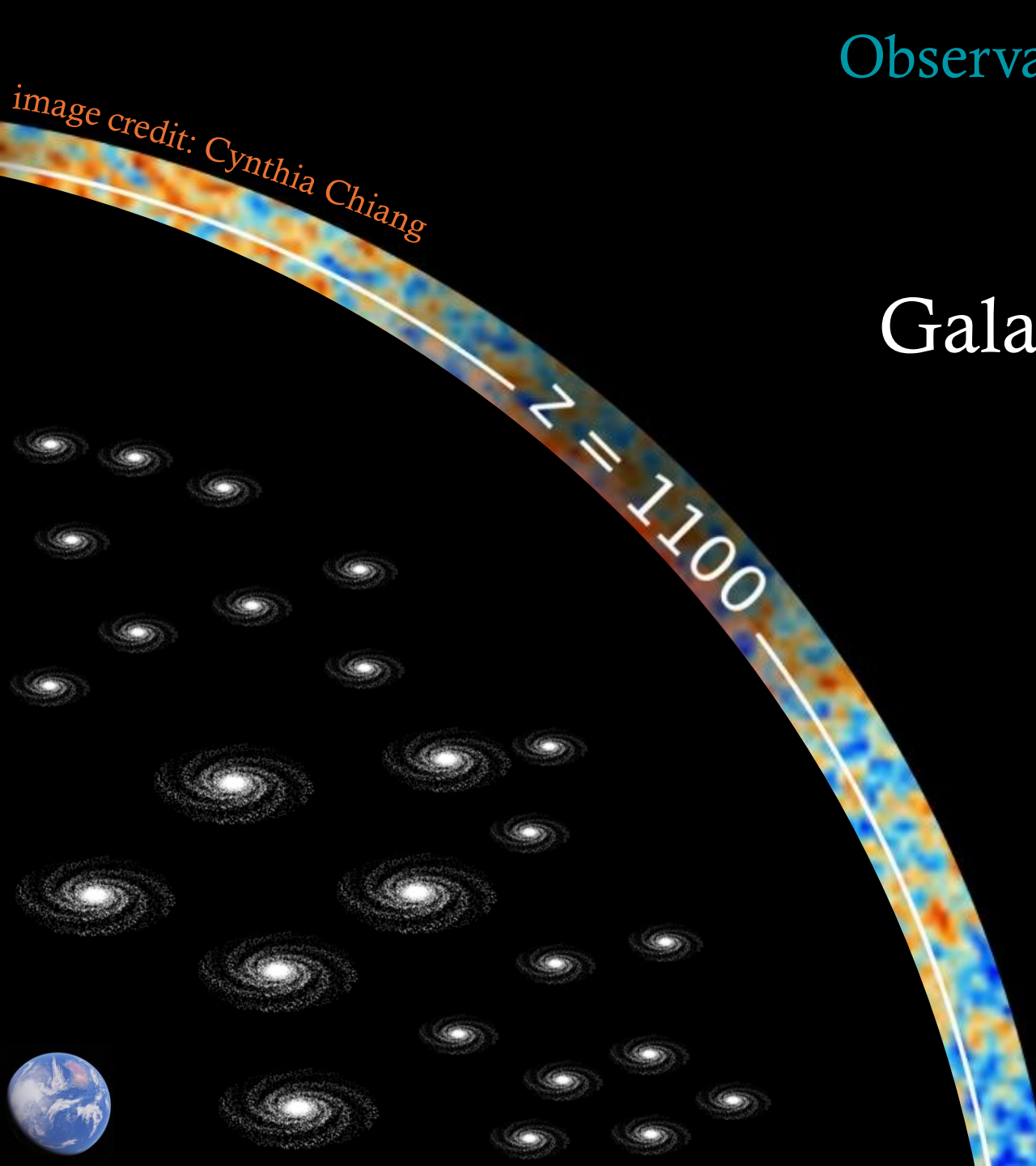
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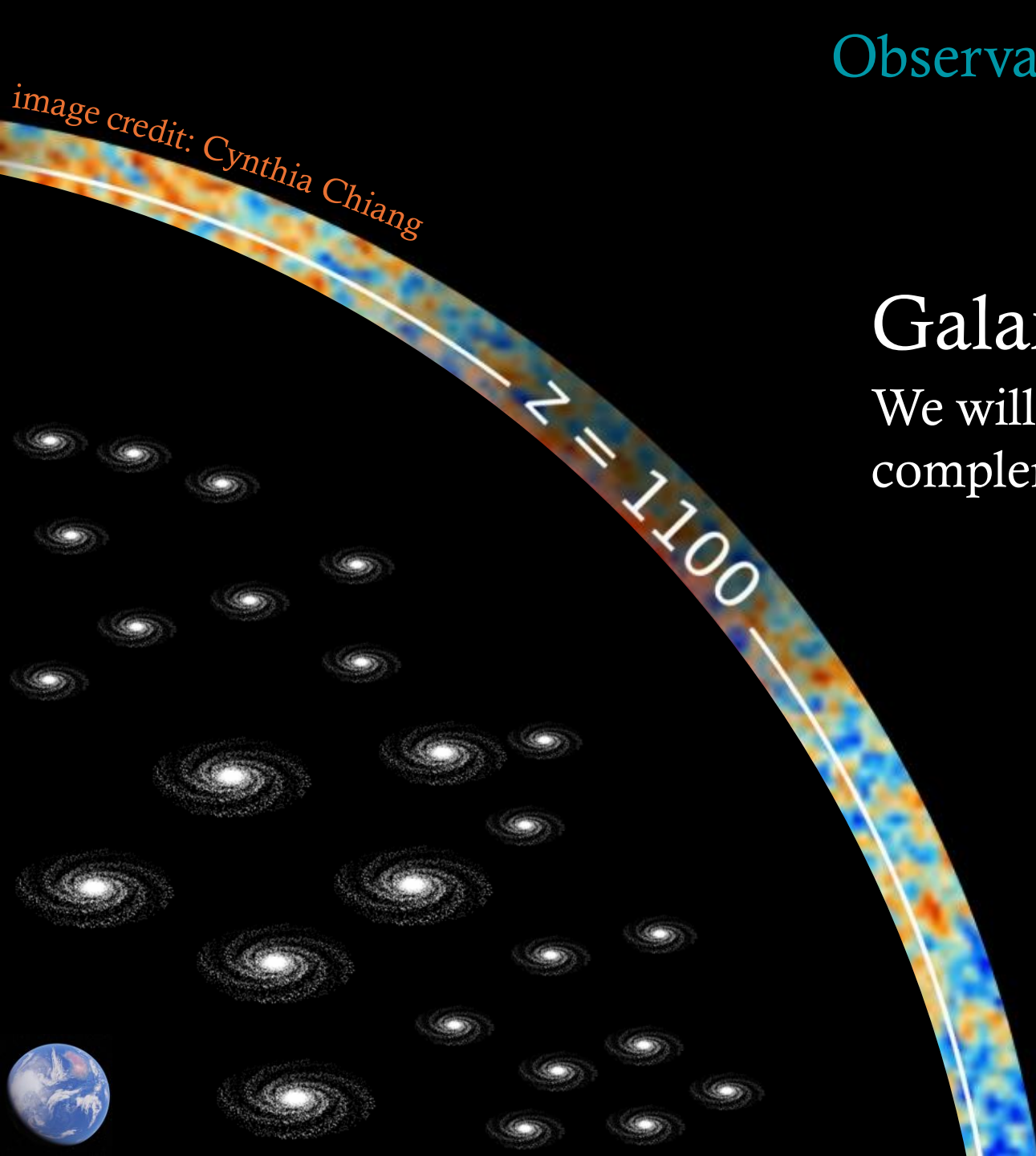


Observations

image credit: Cynthia Chiang

Galaxies+

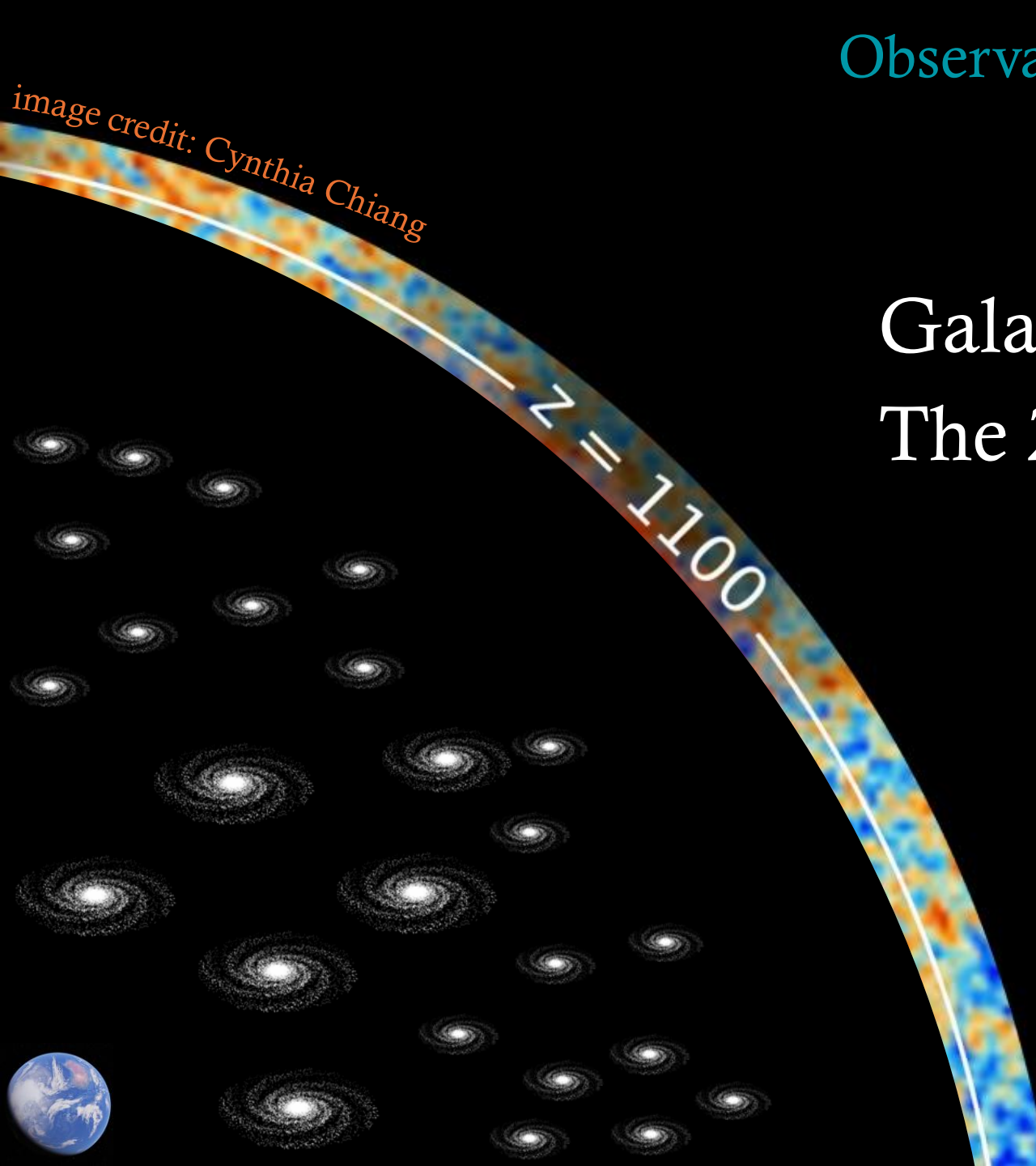
We will soon have a
complementary probe



Observations

image credit: Cynthia Chiang

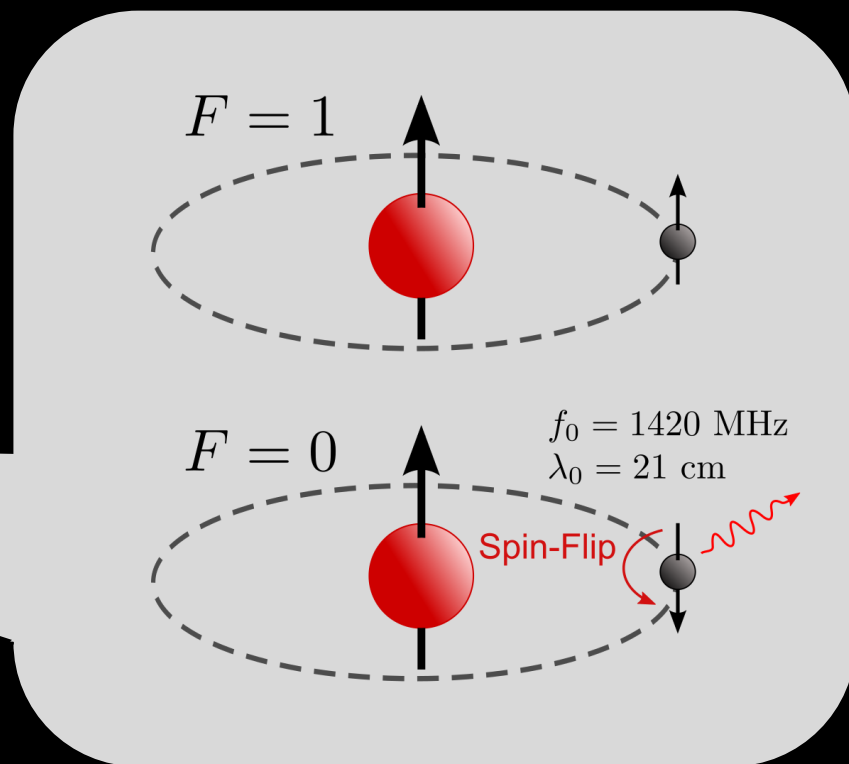
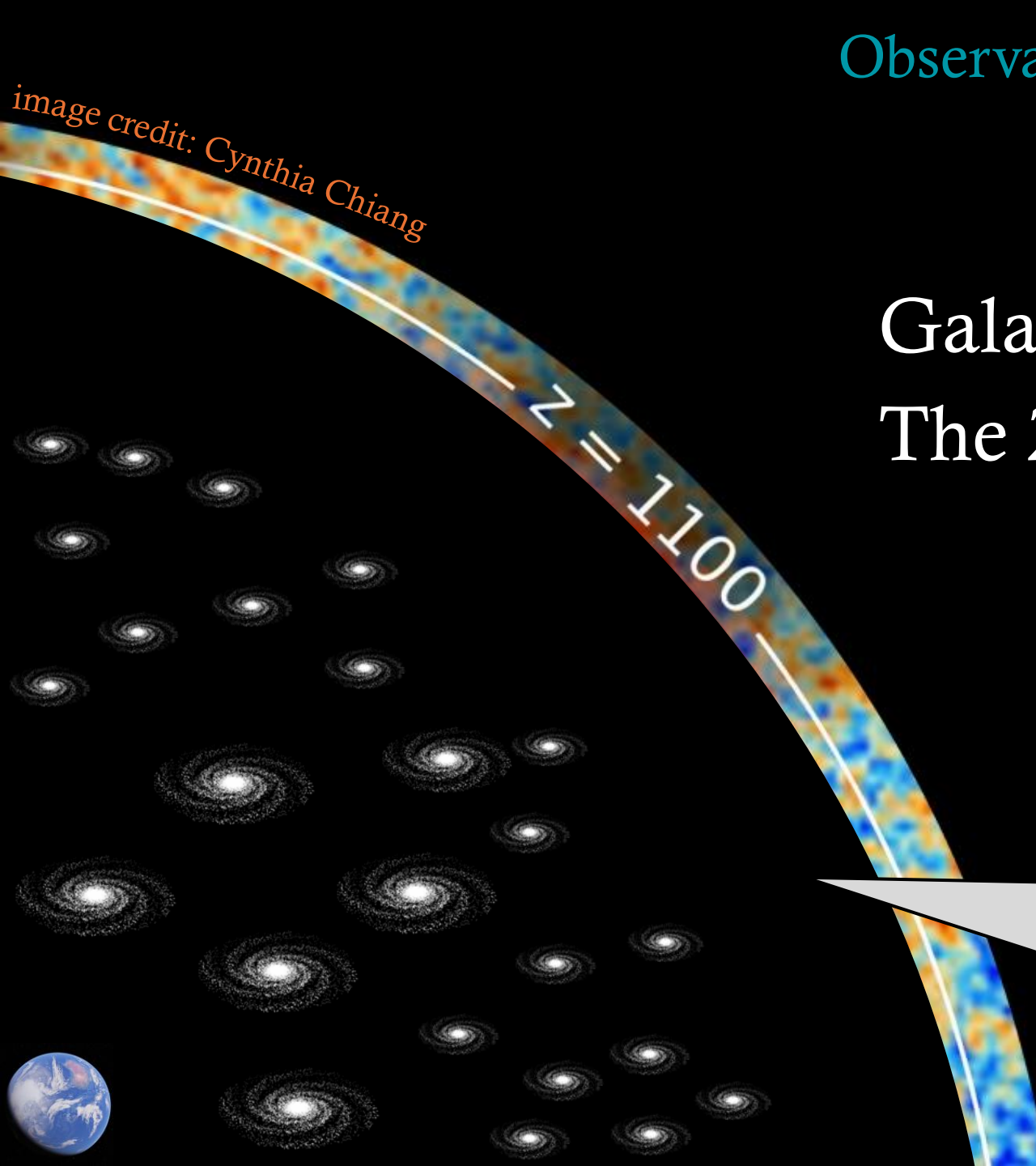
Galaxies+
The 21cm signal



Observations

image credit: Cynthia Chiang

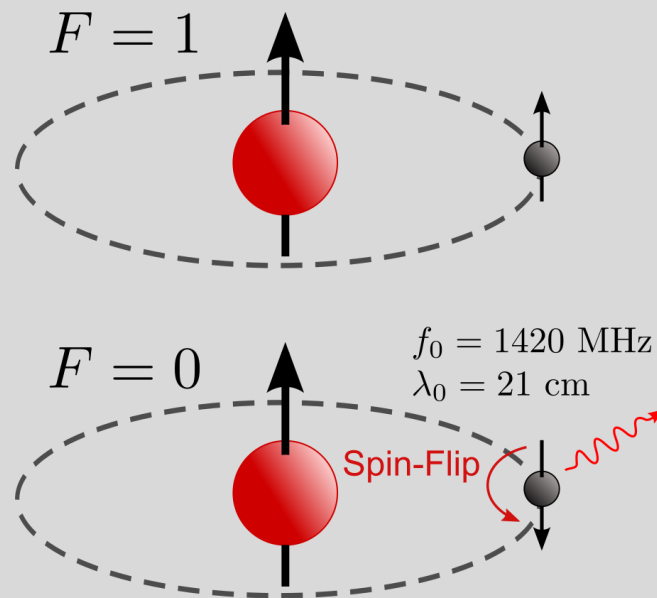
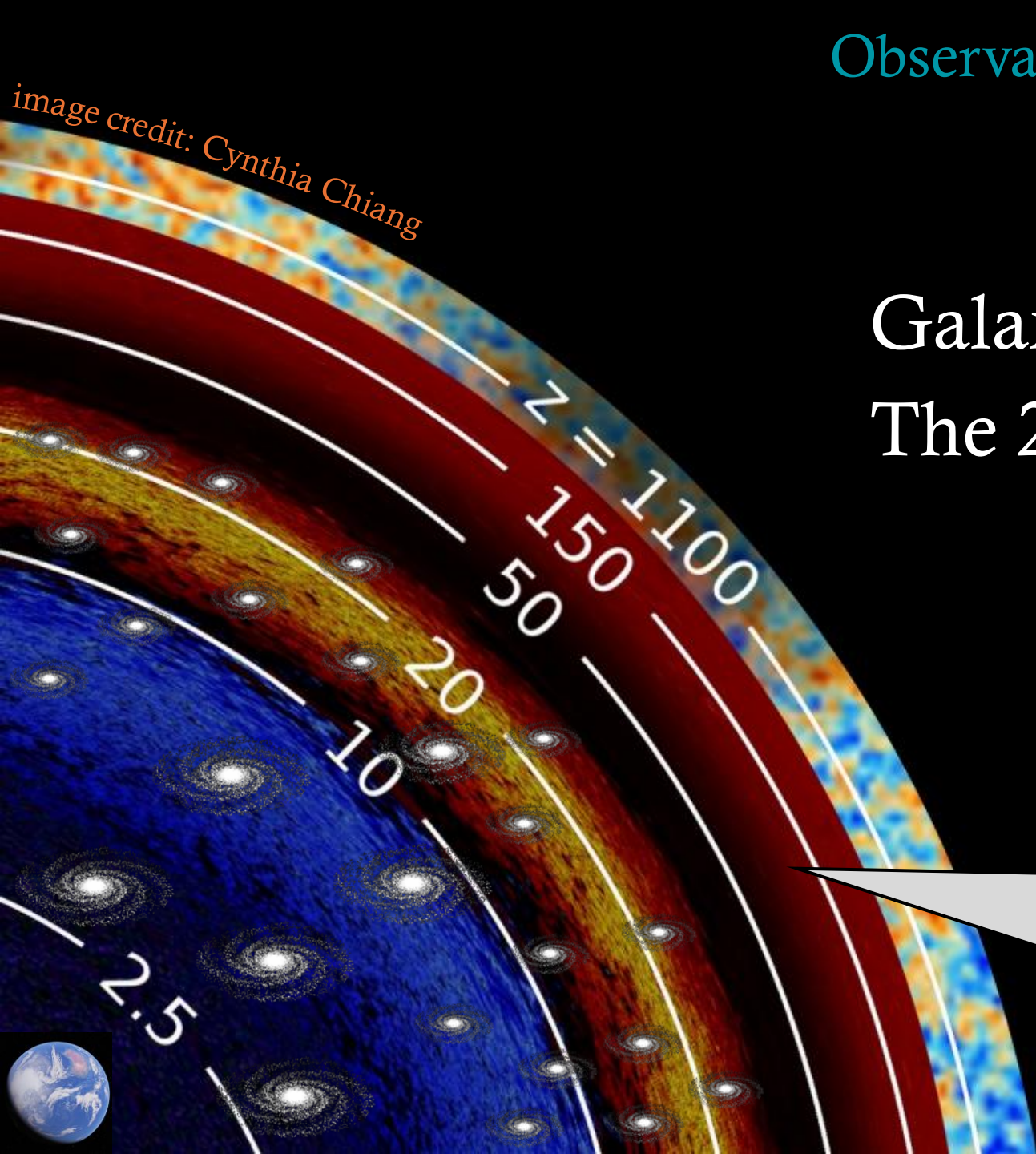
Galaxies+ The 21cm signal



Observations

image credit: Cynthia Chiang

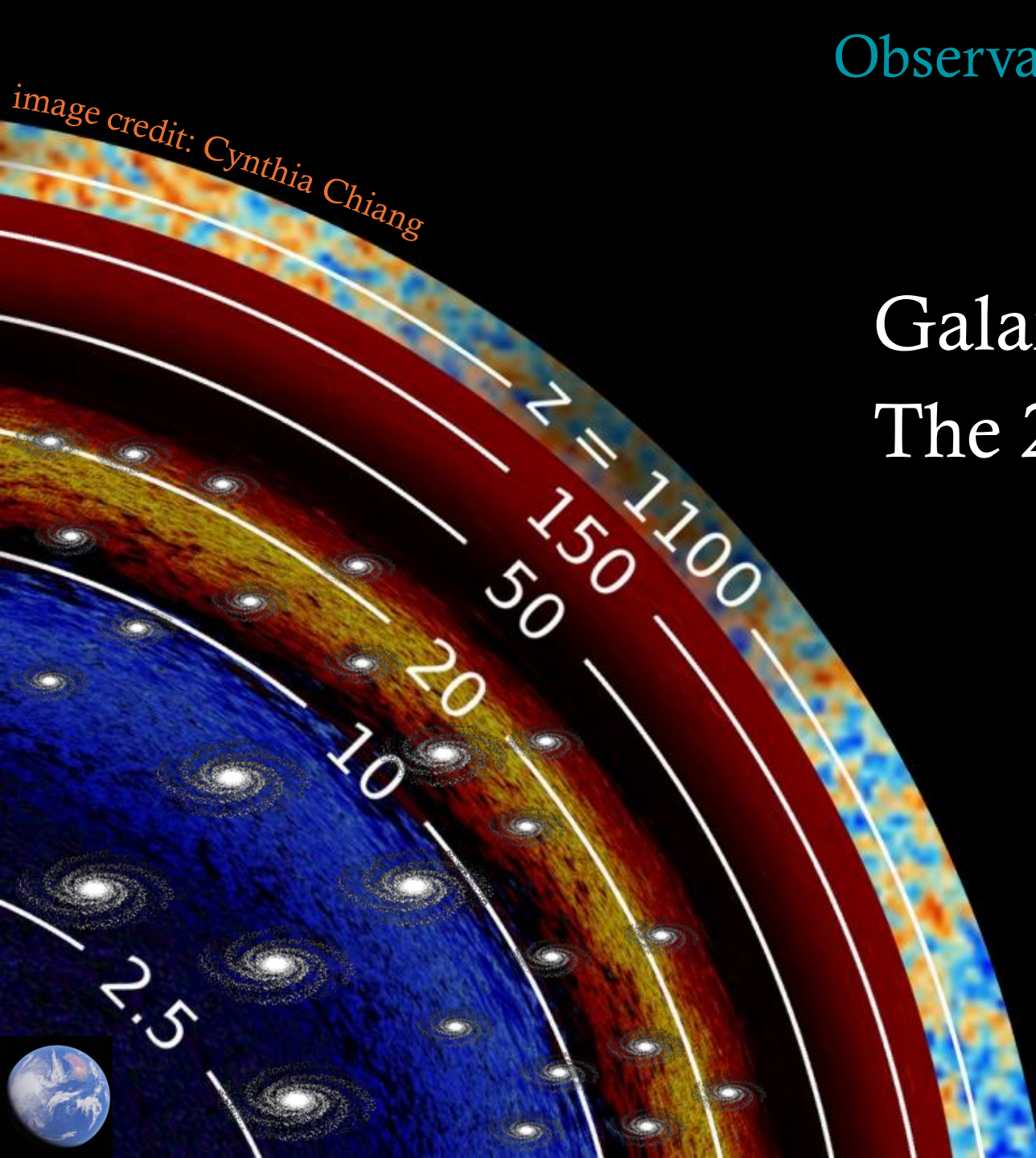
Galaxies+ The 21cm signal



Observations

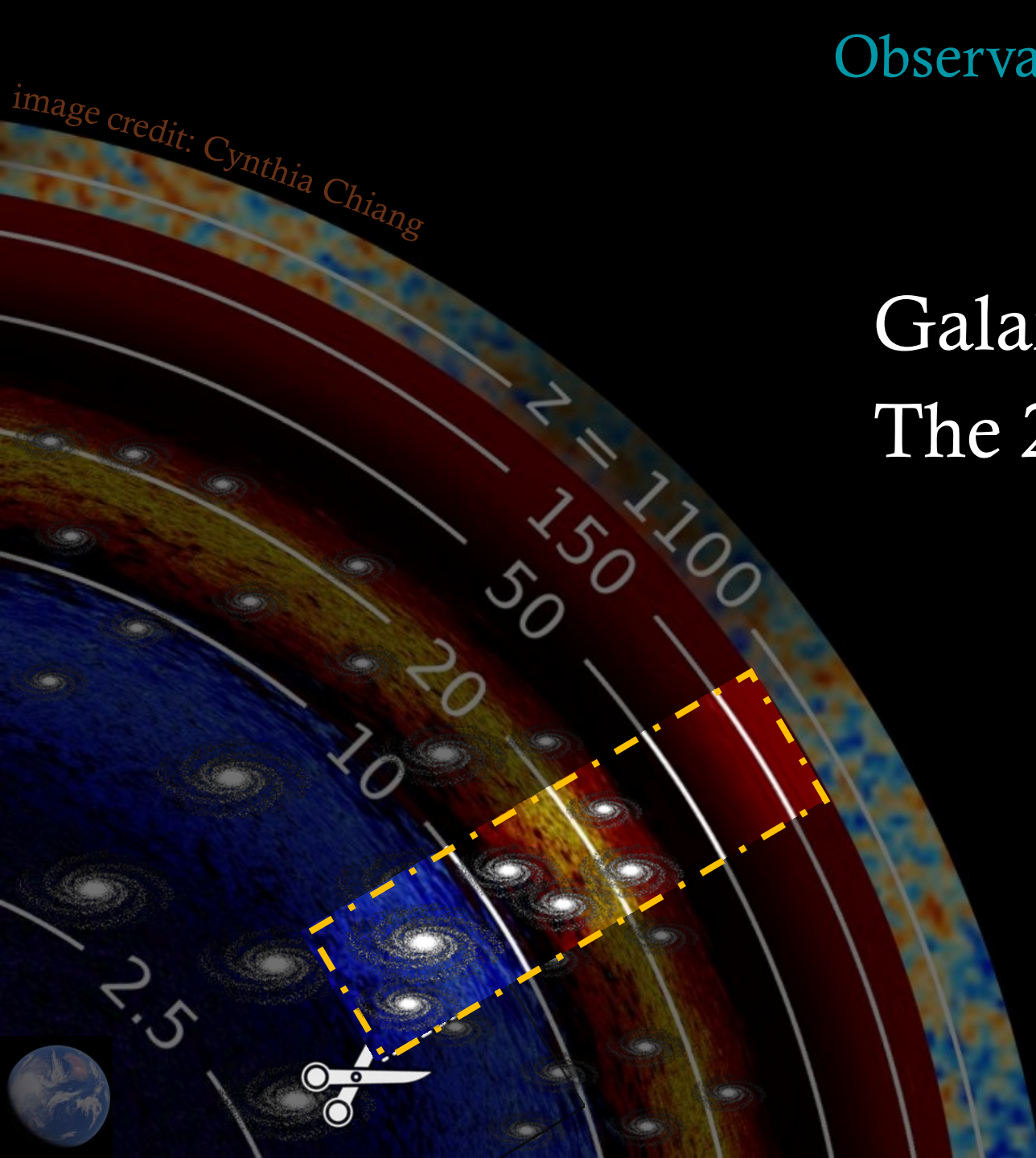
image credit: Cynthia Chiang

Galaxies+
The 21cm signal



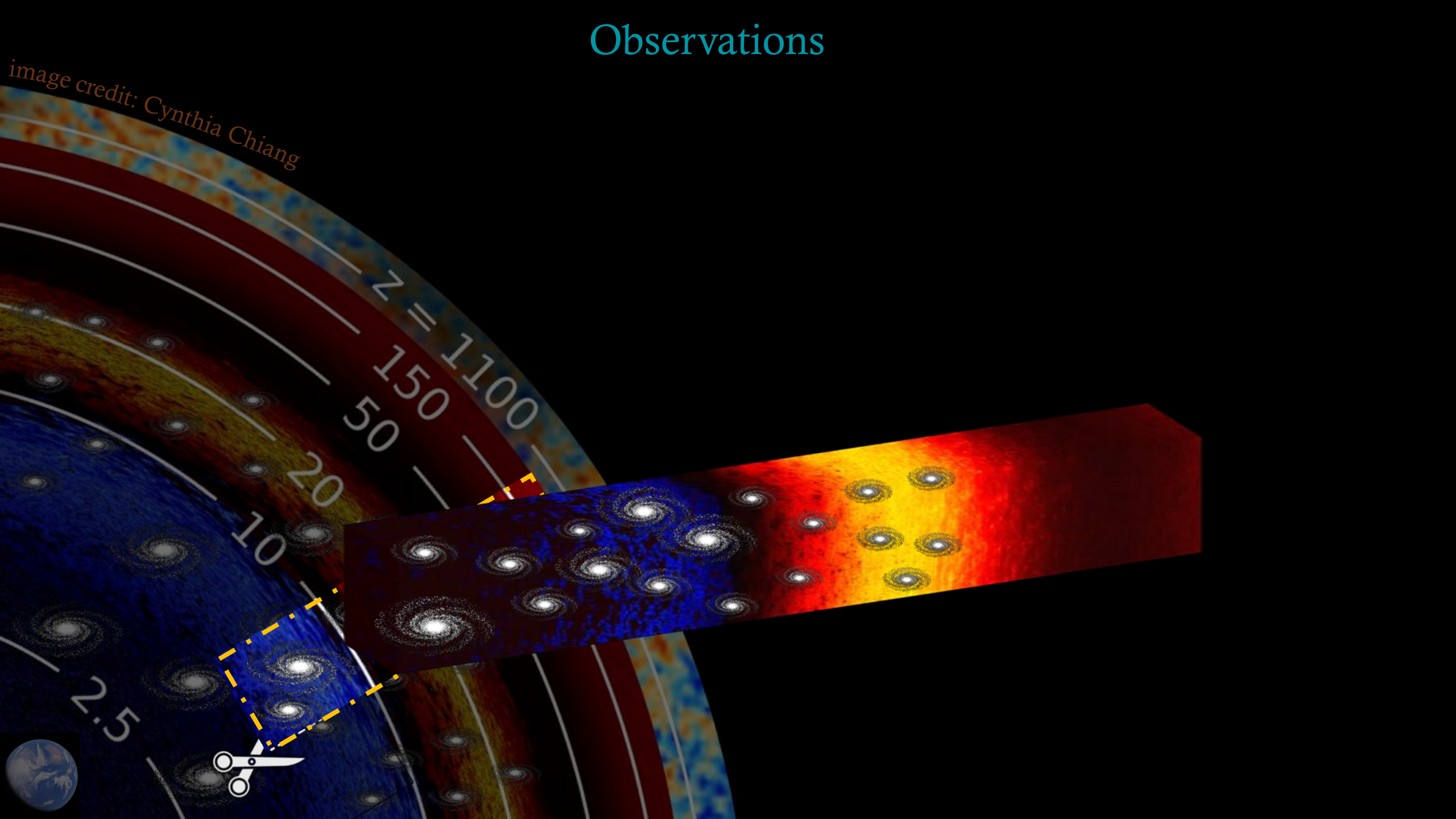
Observations

Galaxies+
The 21cm signal



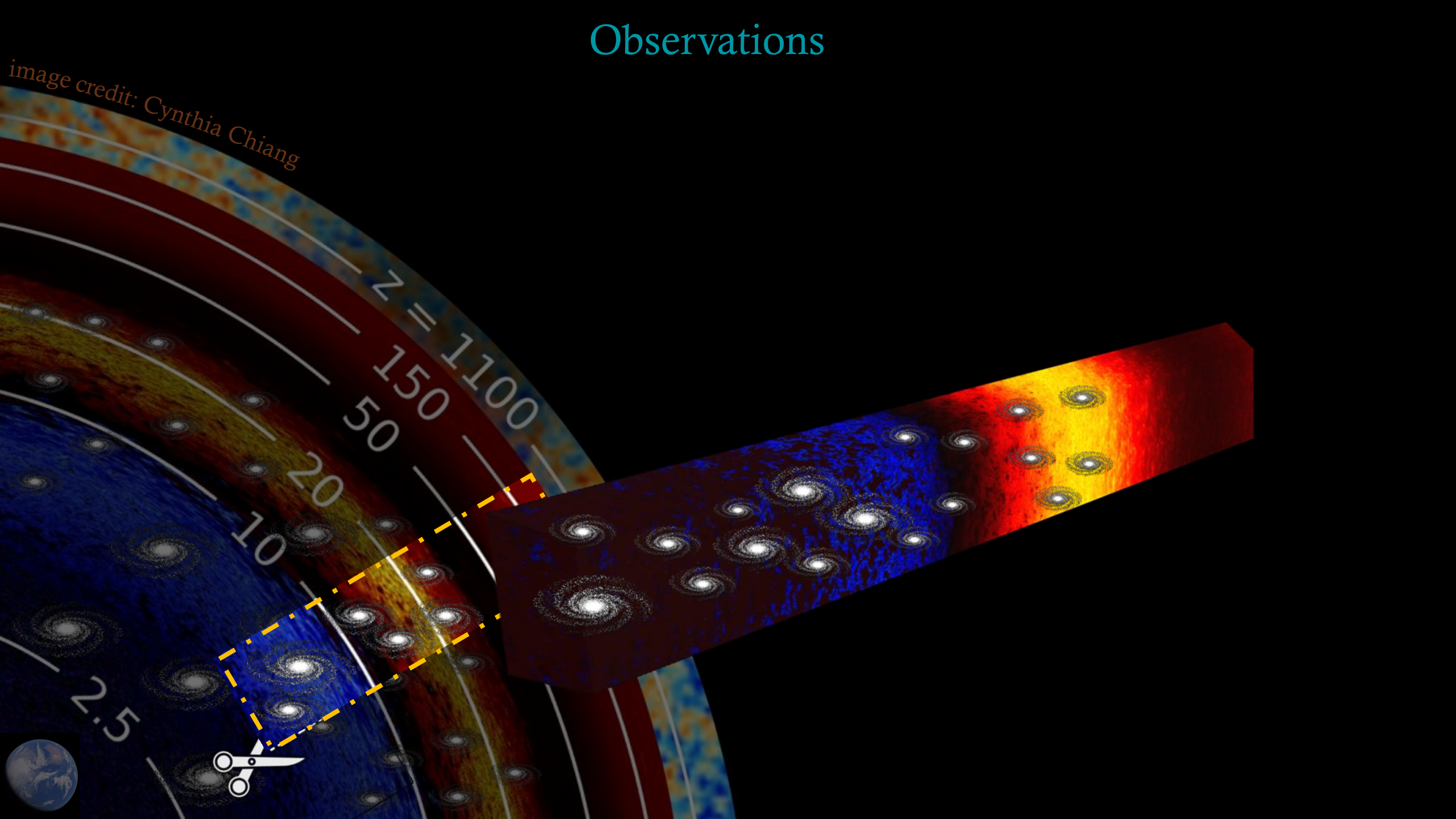
Observations

image credit: Cynthia Chiang



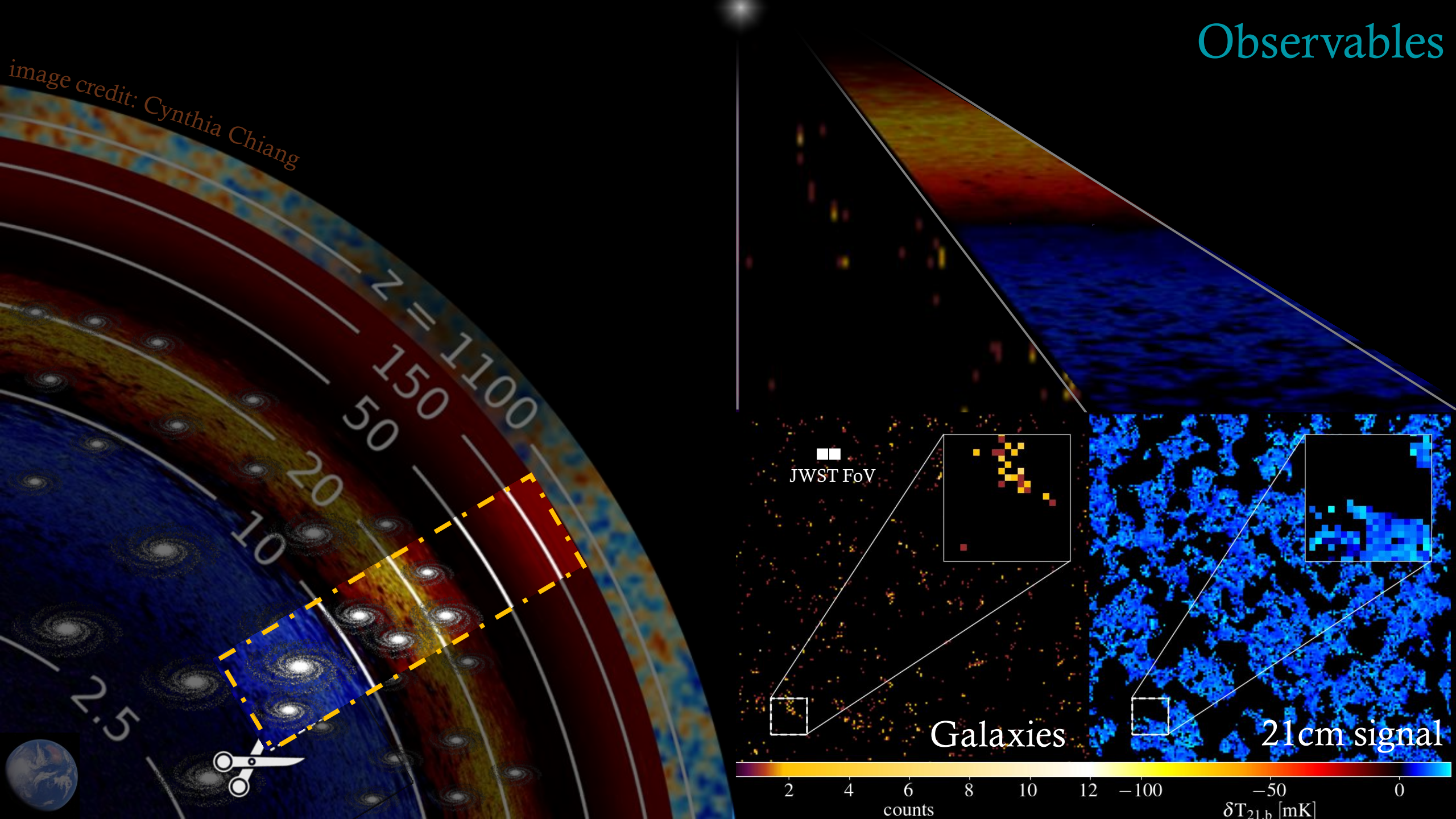
Observations

image credit: Cynthia Chiang



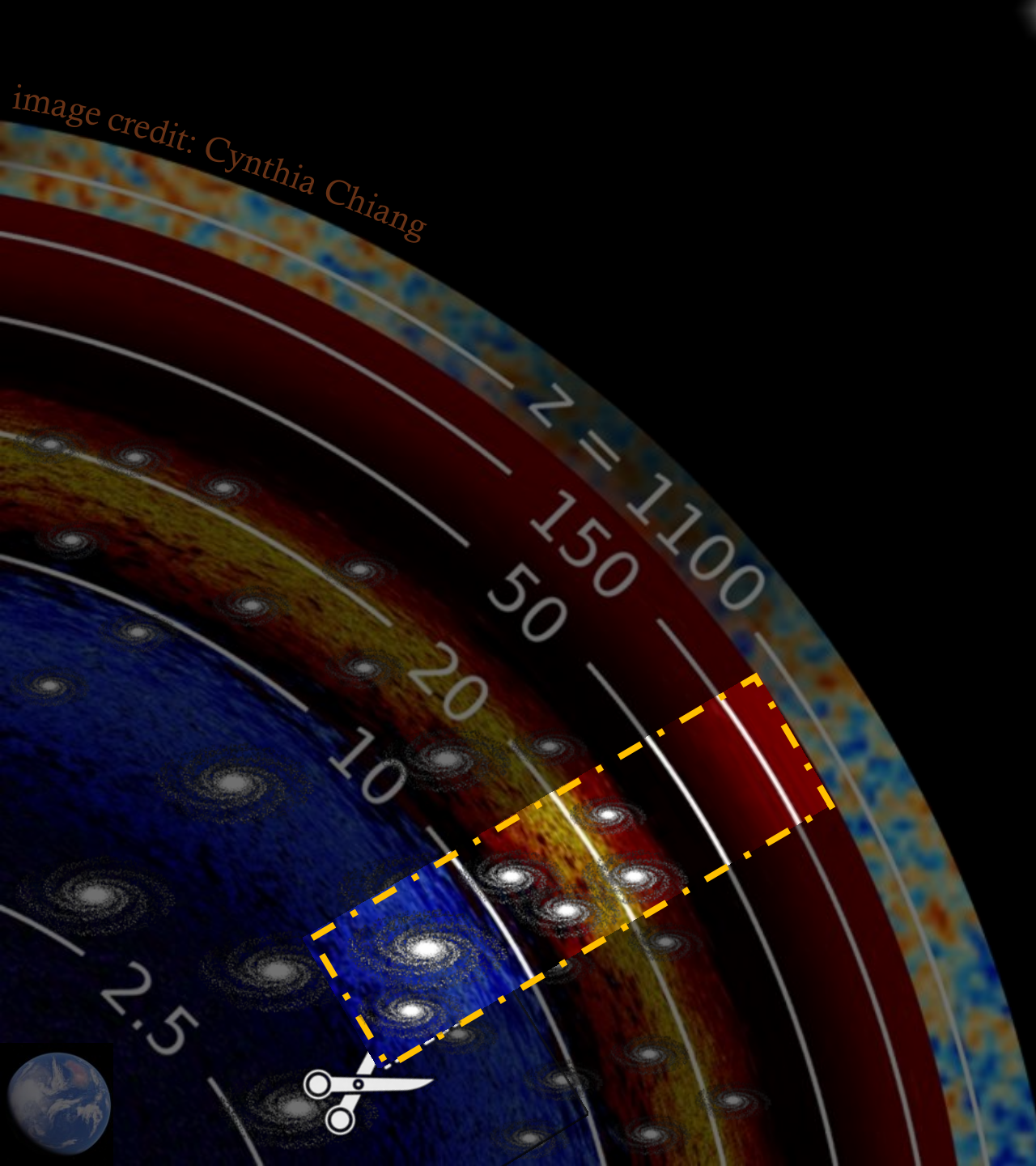
Observables

image credit: Cynthia Chiang

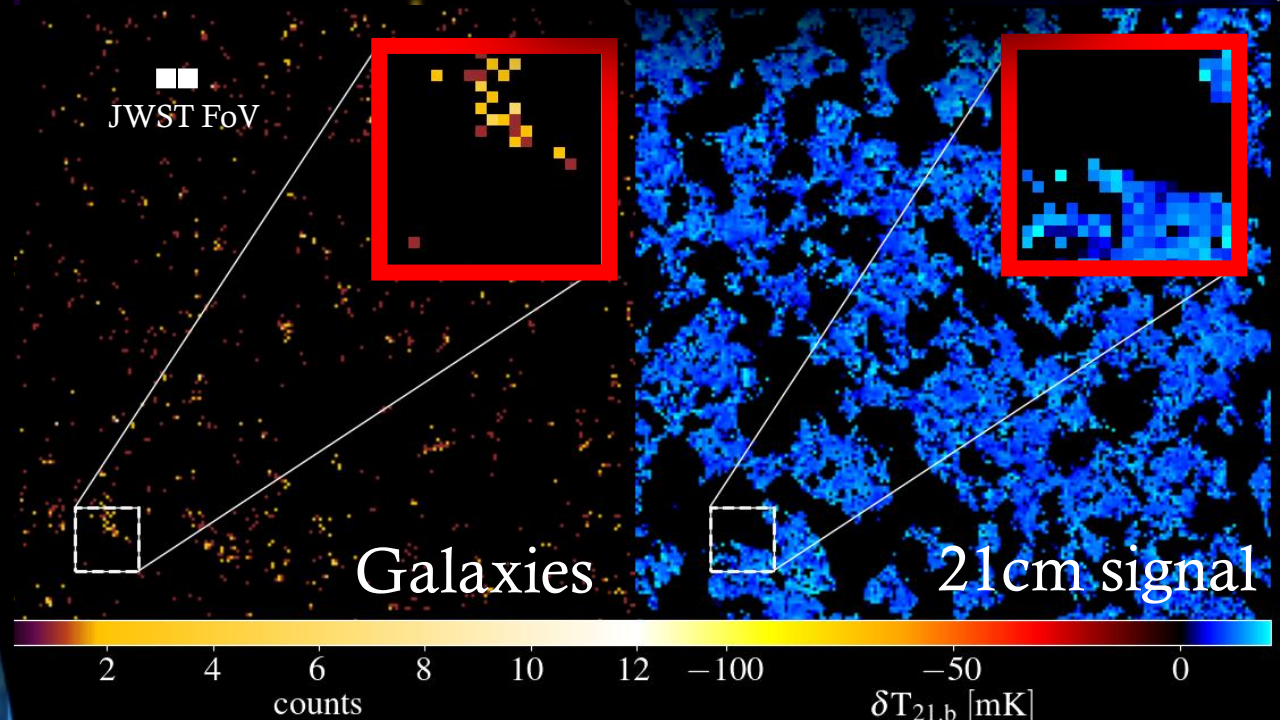


Observables

image credit: Cynthia Chiang



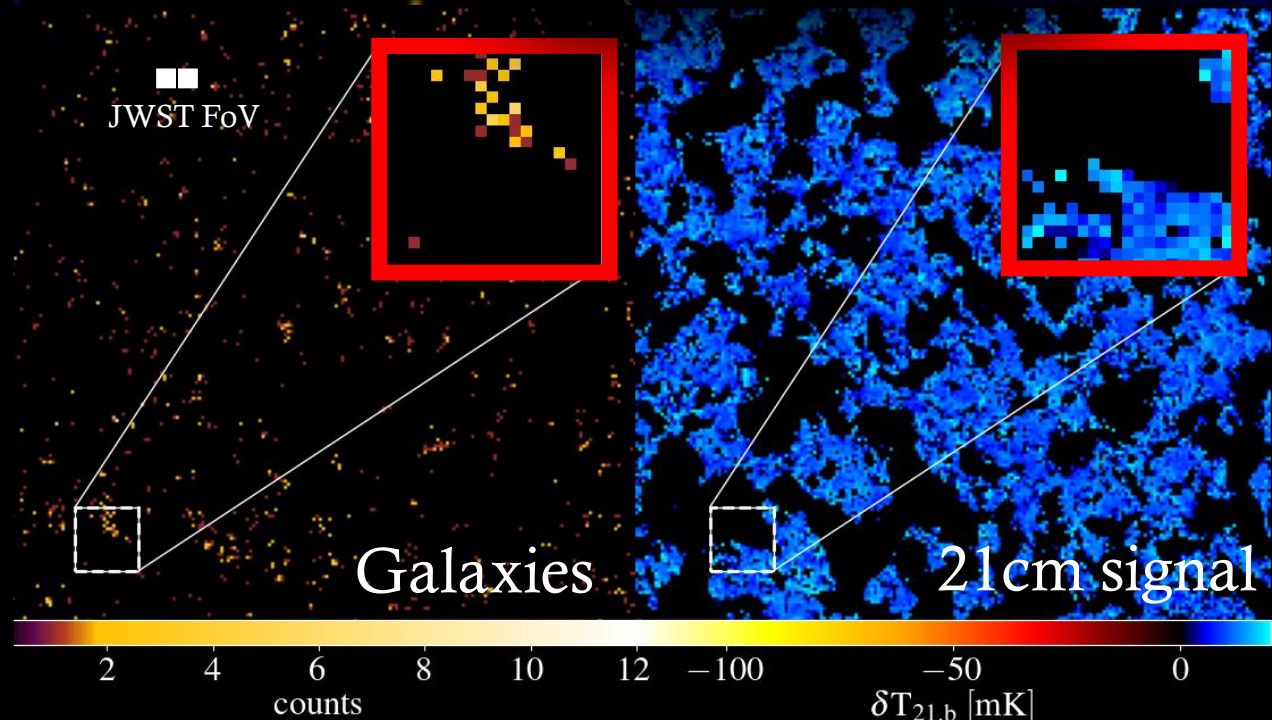
Highly
complementary!!!



We want to leverage this
complementarity

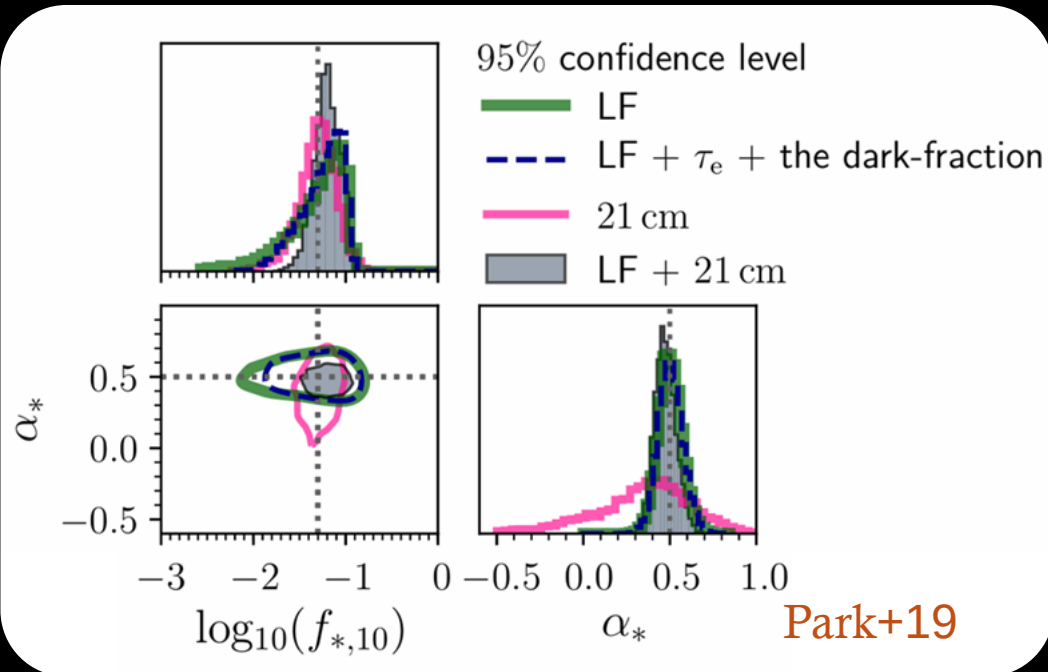
Observables

Highly
complementary!!!



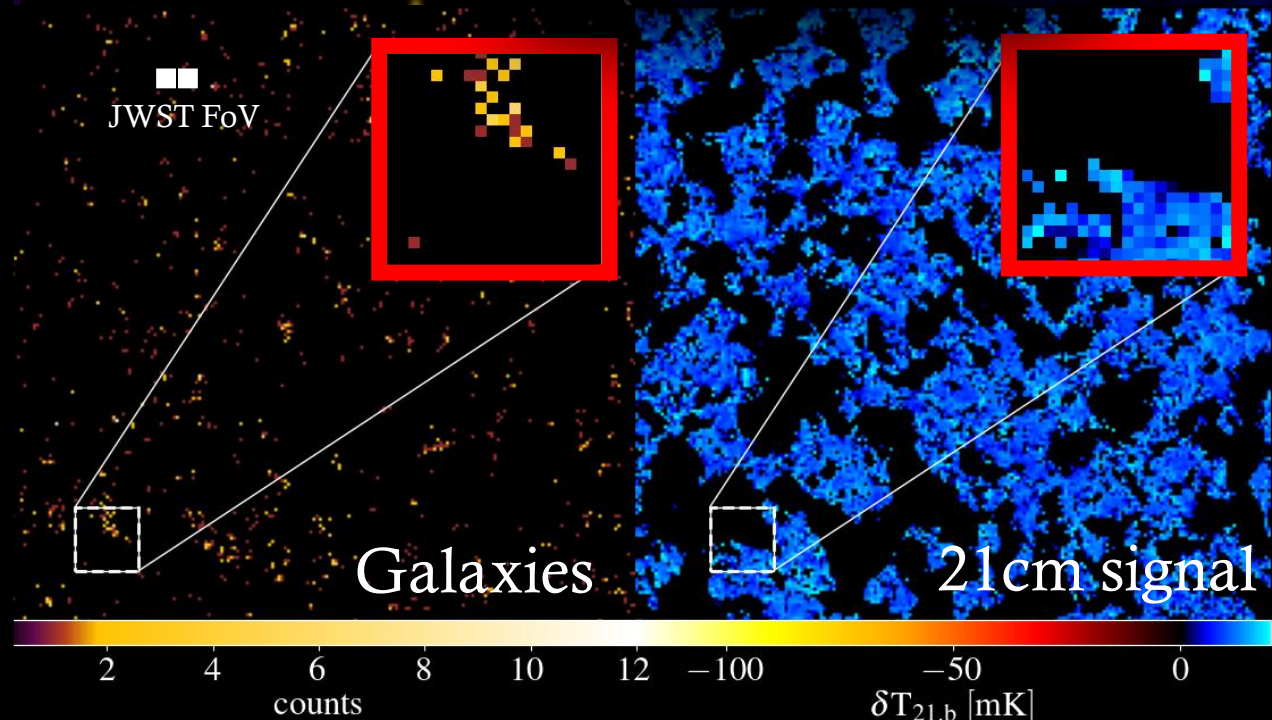
We want to leverage this complementarity

We already know they give complementary information on astronomy parameters



Observables

Highly complementary!!!



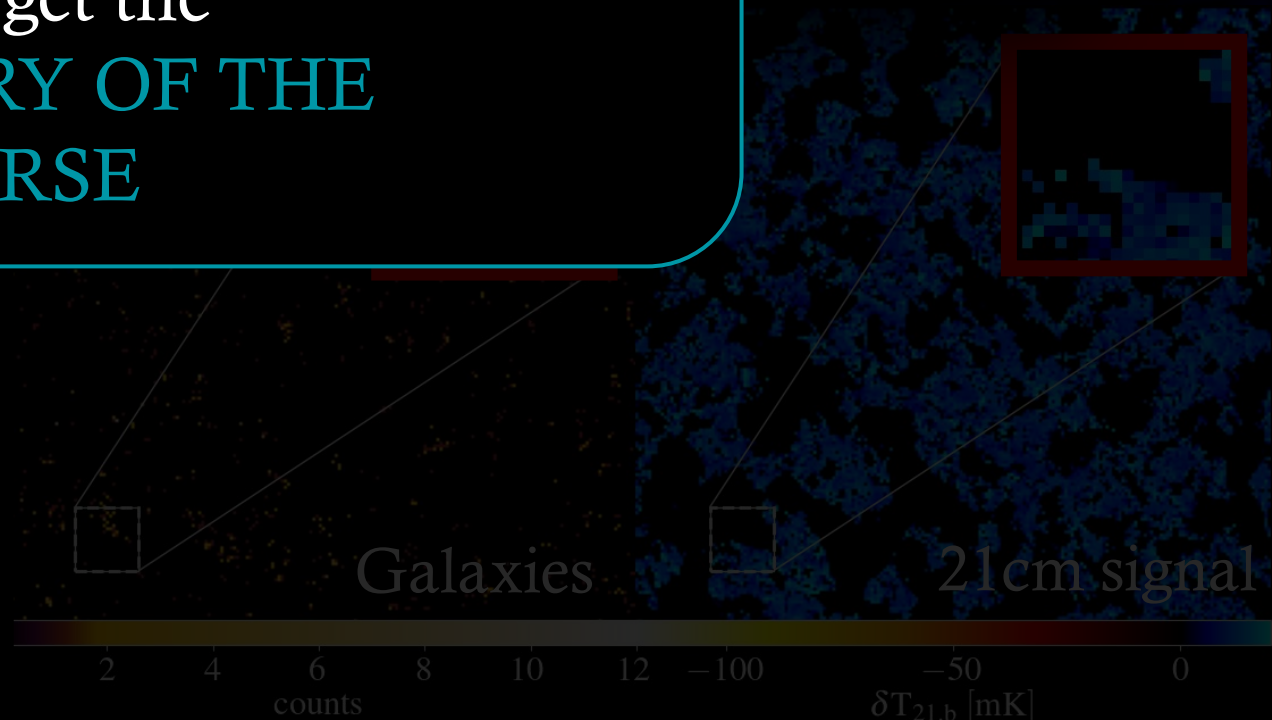
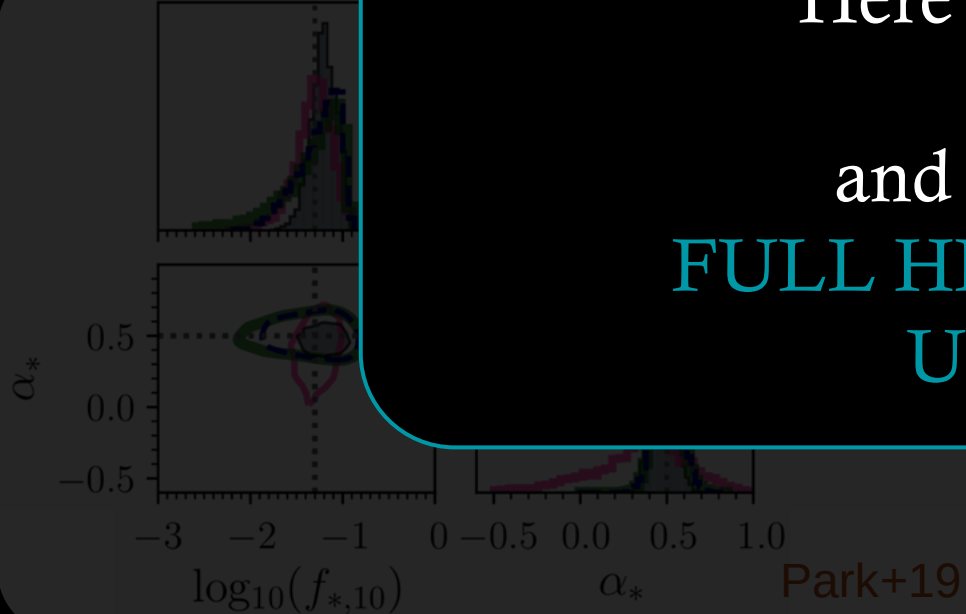
We want to leverage this complementarity

We already know they give complementary information on astronomy

Observables

Here we go beyond
and try to get the
**FULL HISTORY OF THE
UNIVERSE**

Highly
complementary!!!

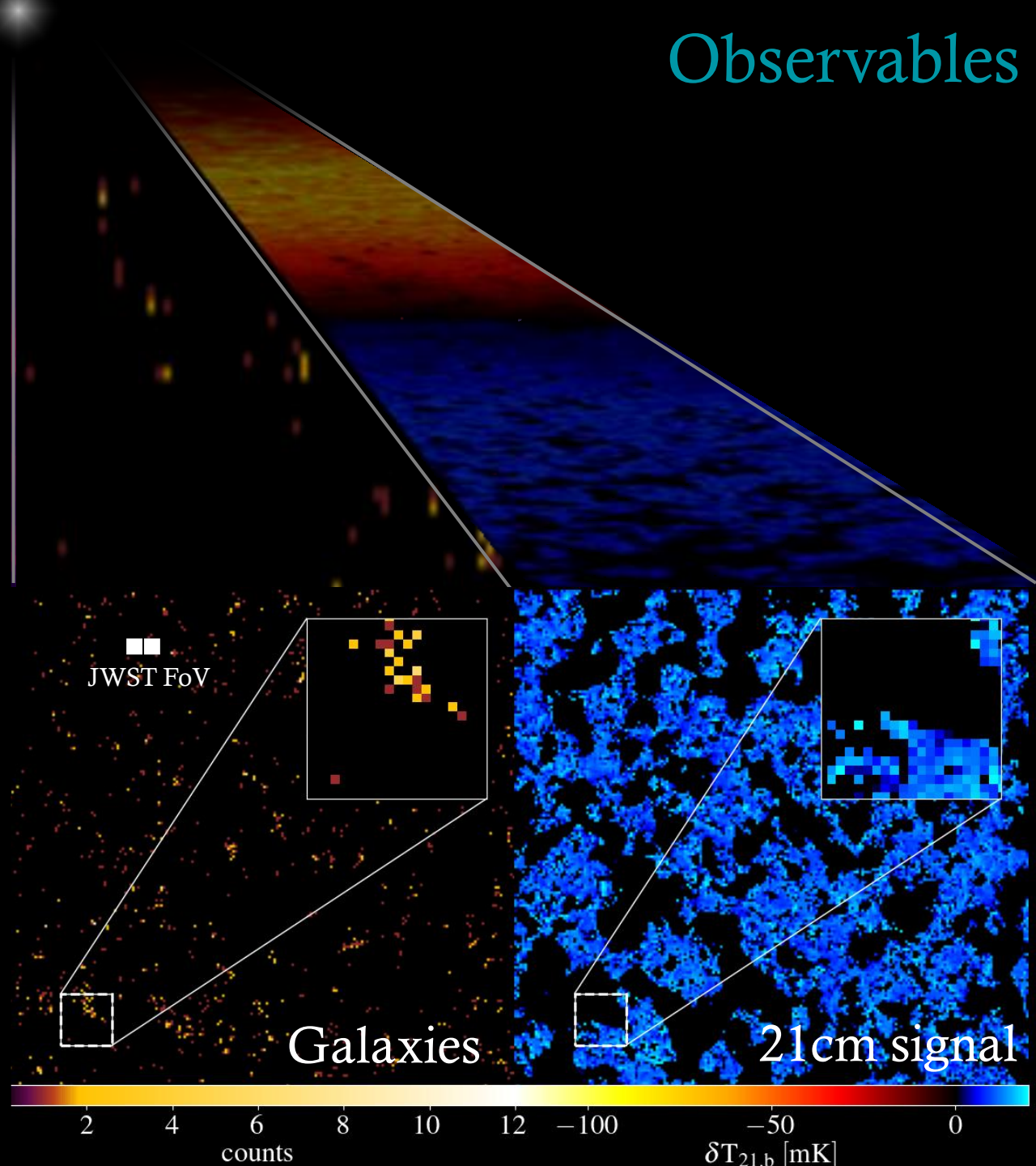


How



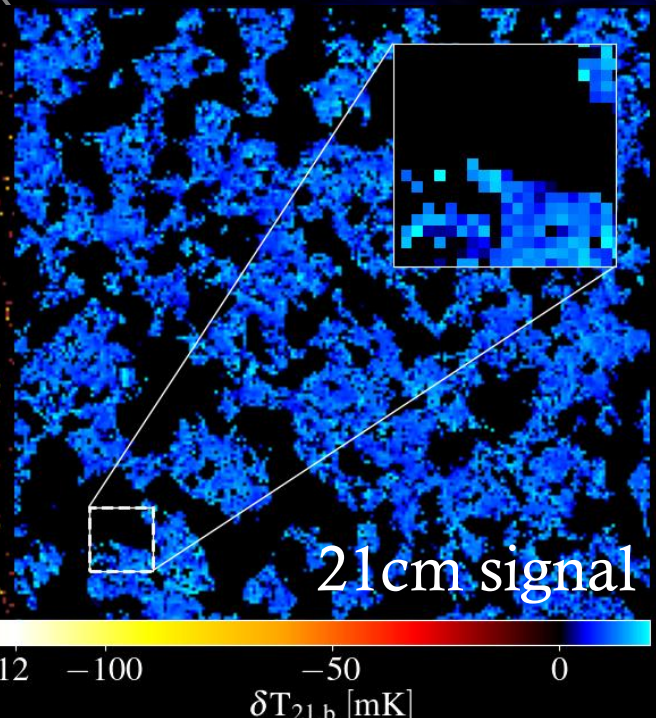
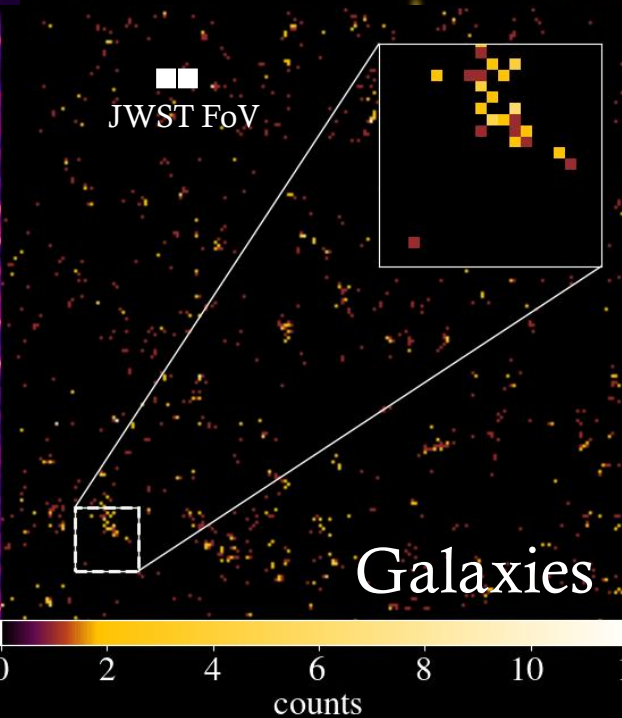
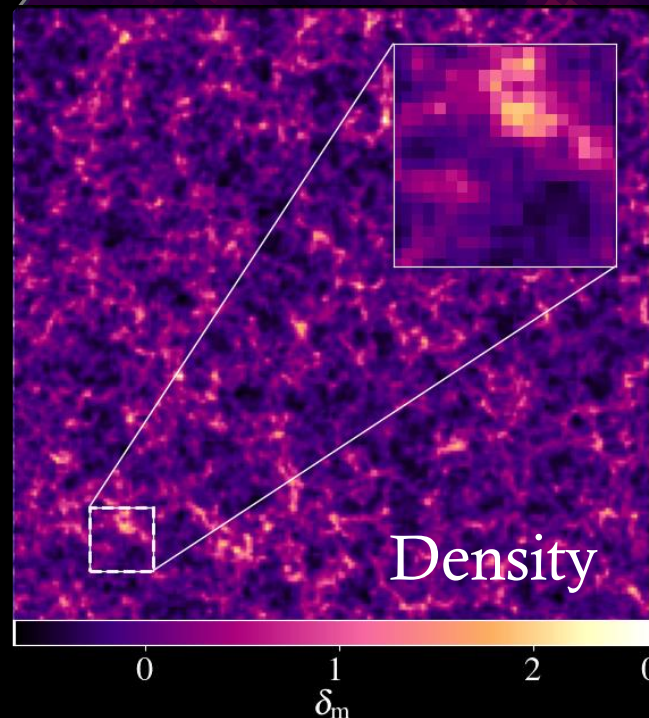
IC reconstruction

Observables



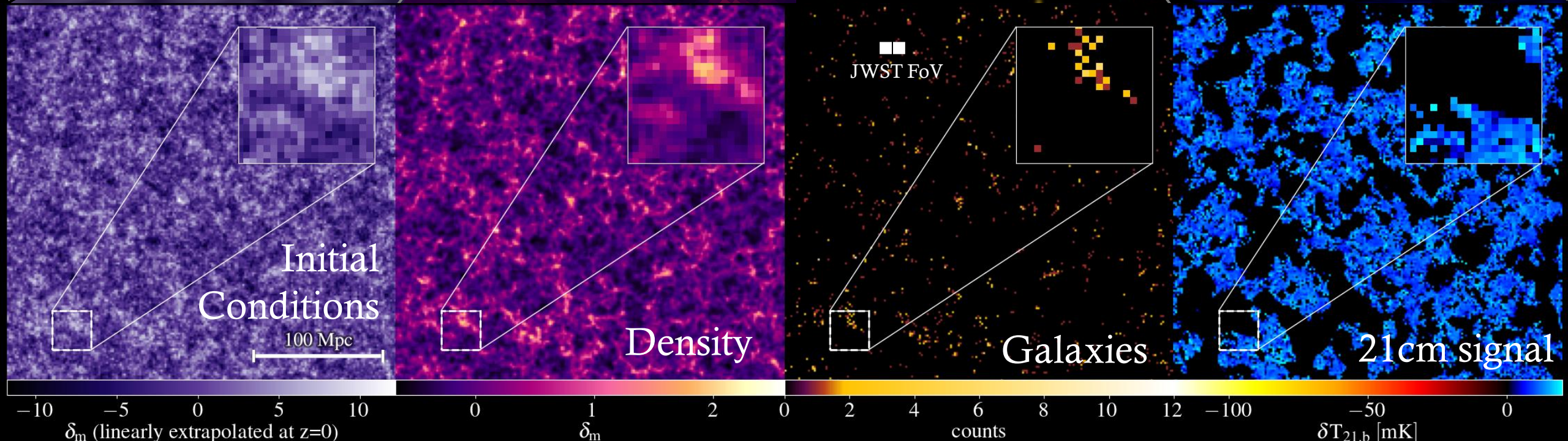
Underlying structure

Observables



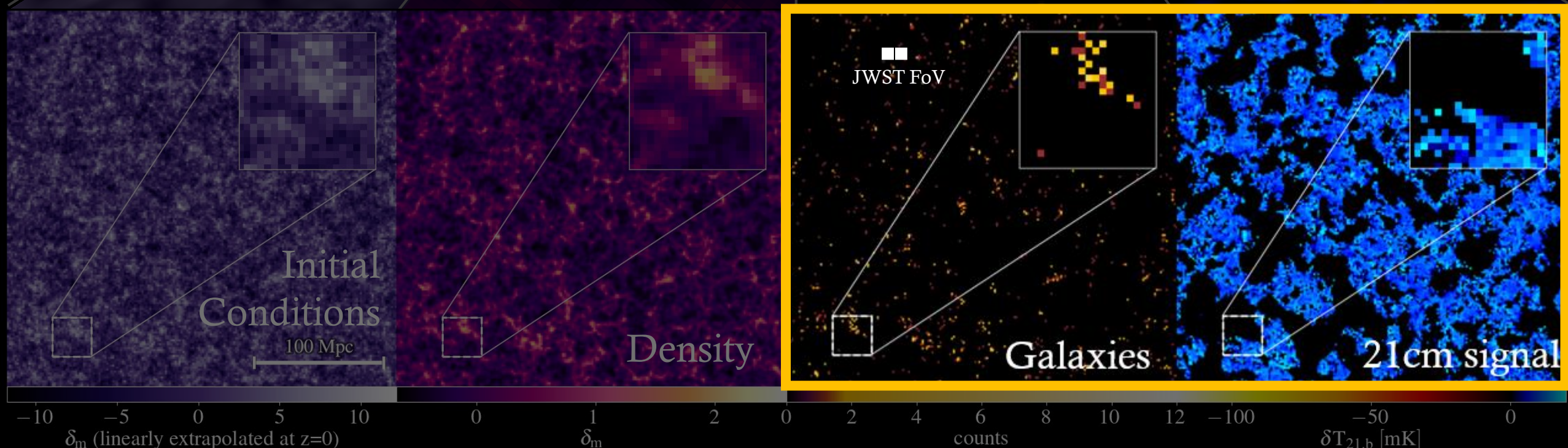
Underlying structure

Observables



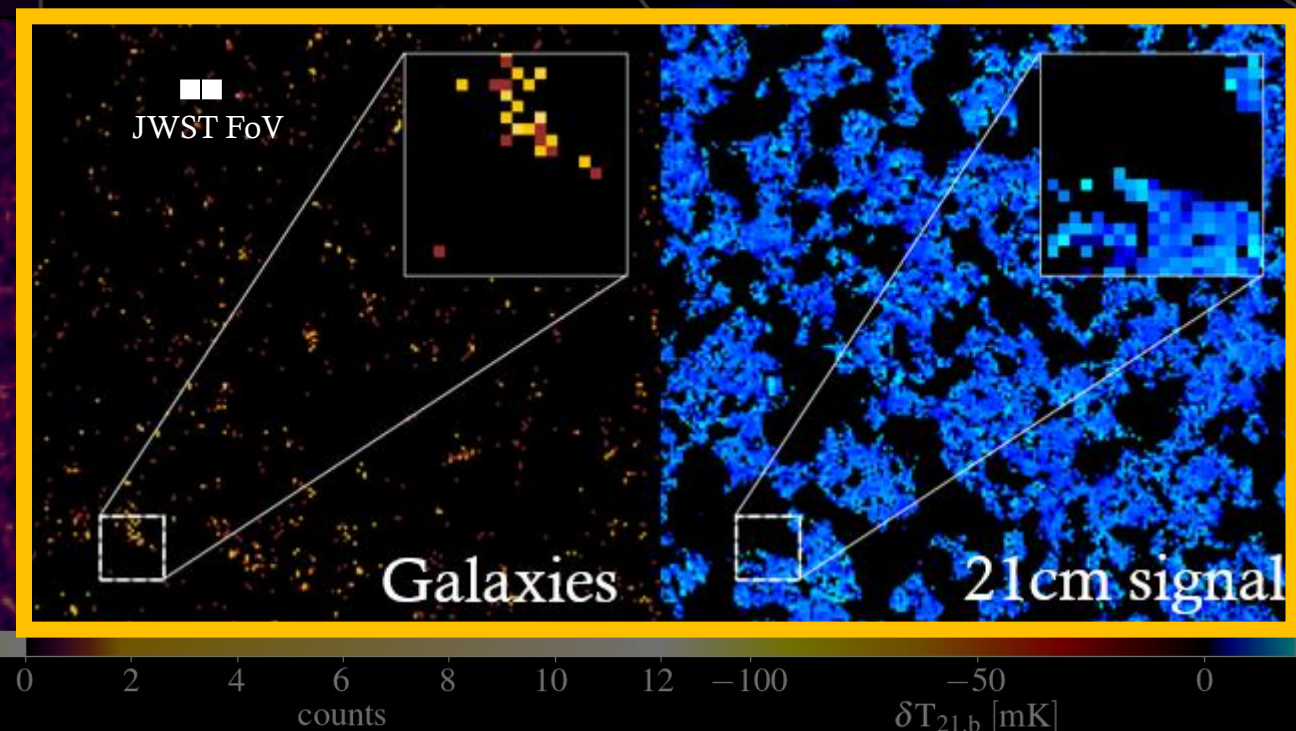
Underlying structure

Observables



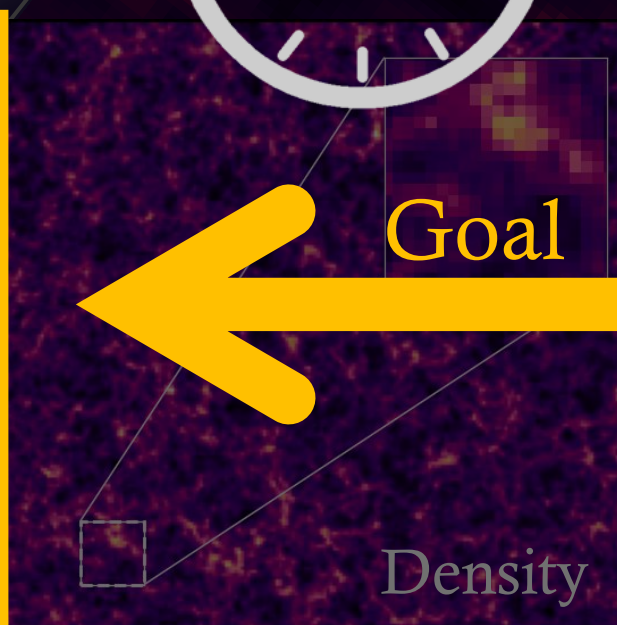
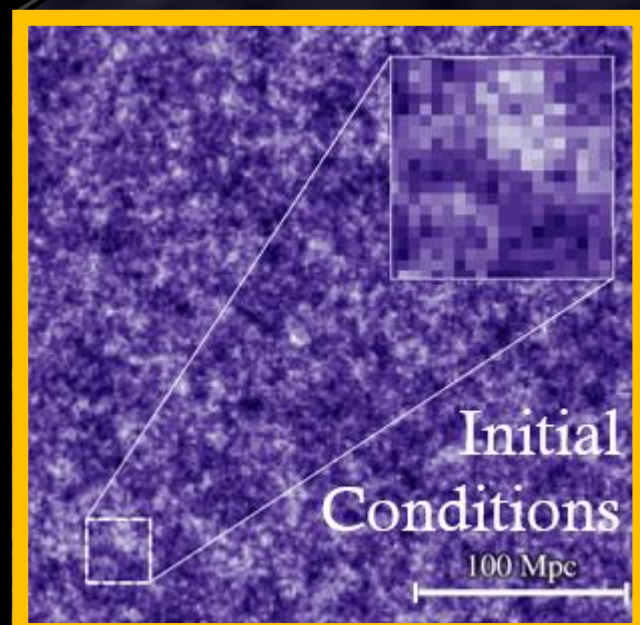
Underlying structure

Observables

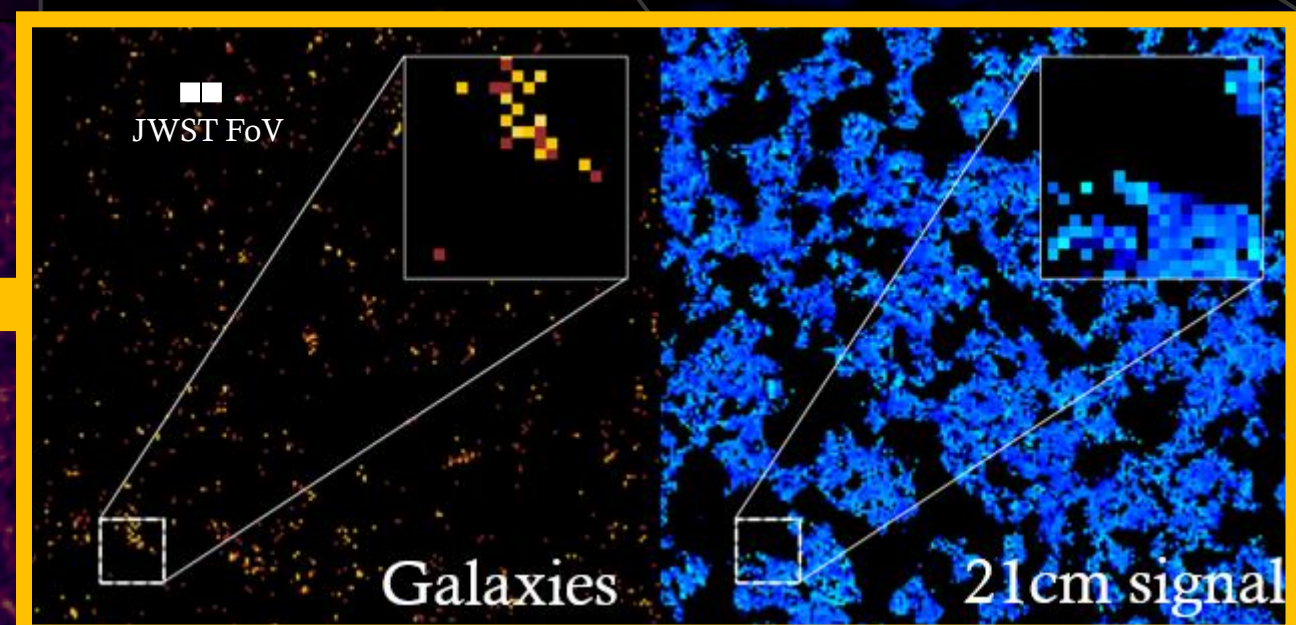


Underlying structure

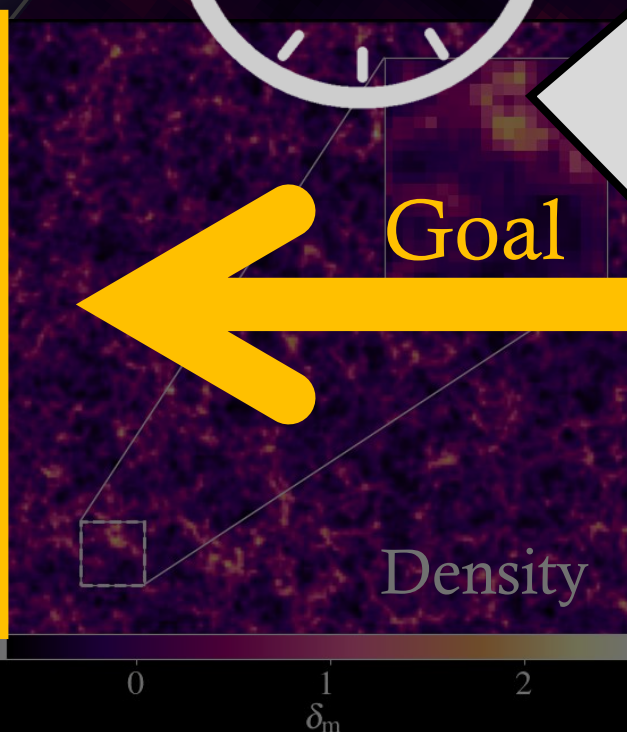
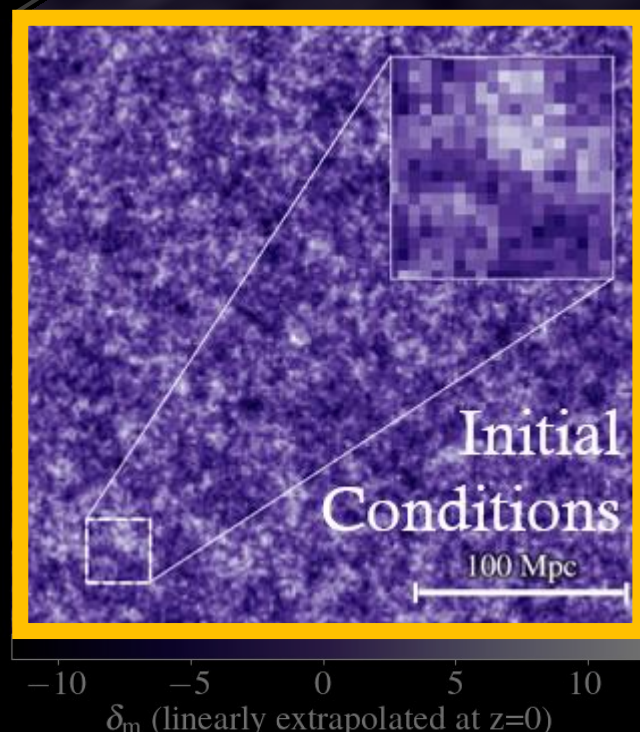
Observables



Goal



Underlying structure



Goal



This is hard

- **Millions** of parameters
(vs <10 in classical inferences)

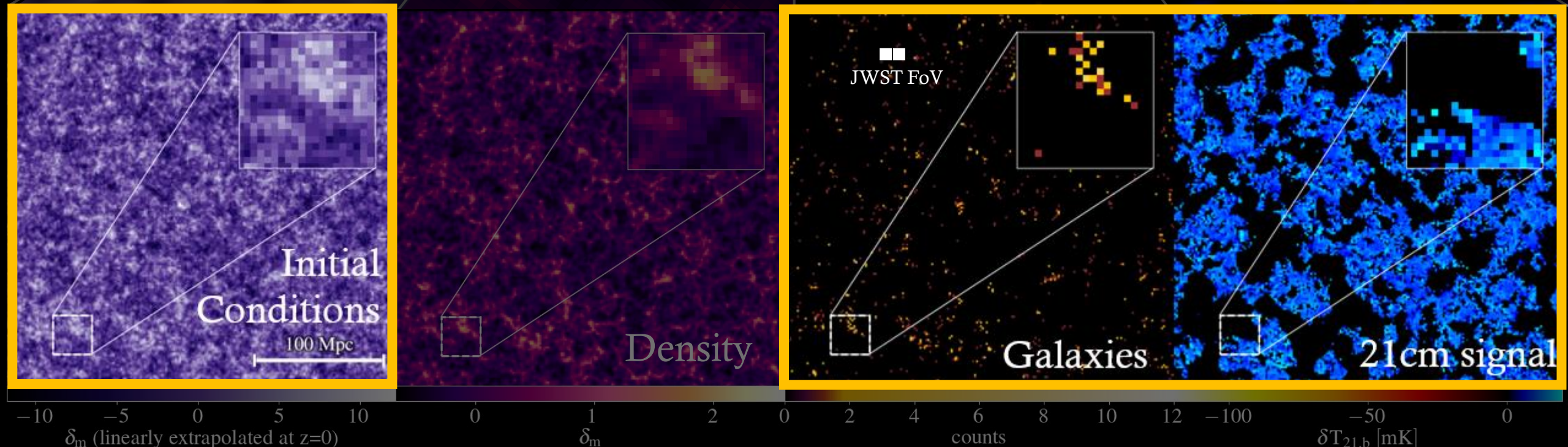
This is done at low z ,
but using just galaxy maps.
(e.g., [Doeser+24](#), [Sorce+26](#))

- At high z **observations are more challenging**

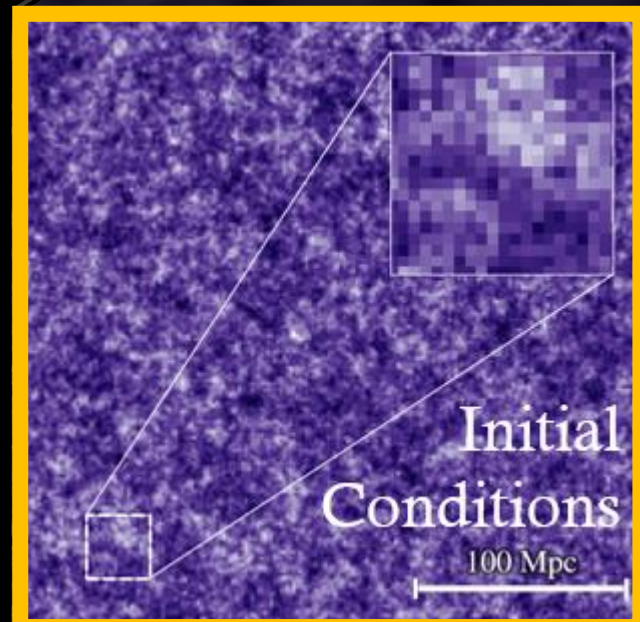
(though see [Zhang+24](#), [Chen+25](#))



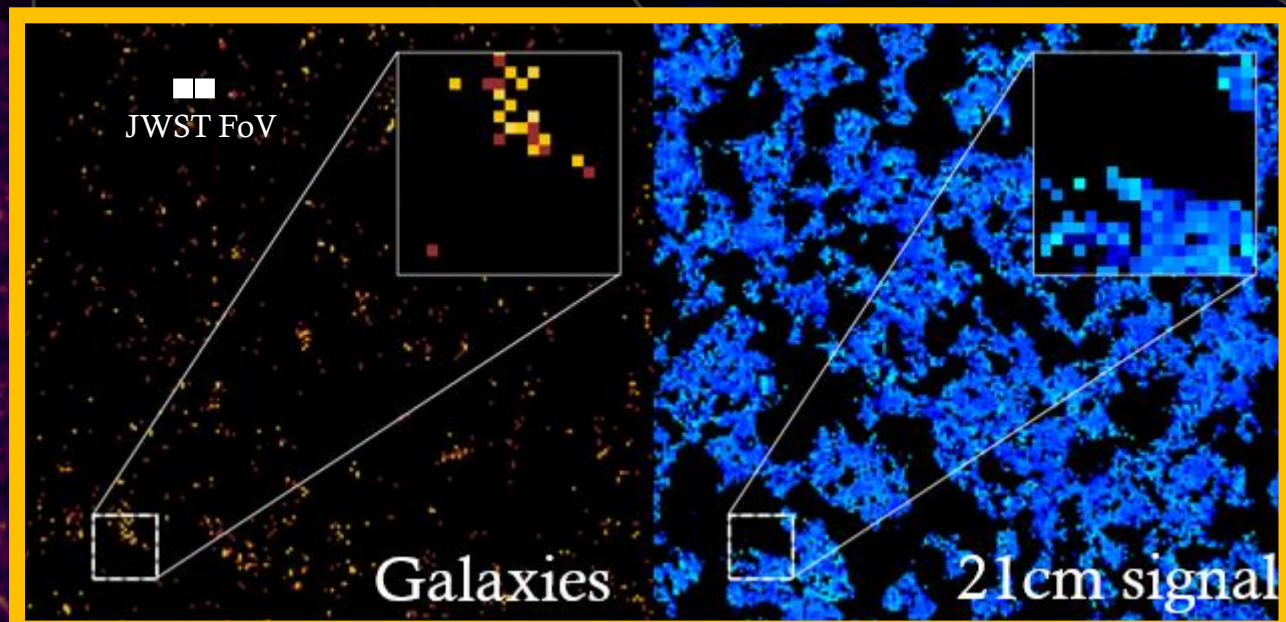
Why bother?



Why bother?



Full
evolution



δ_m (linearly extrapolated at $z=0$)

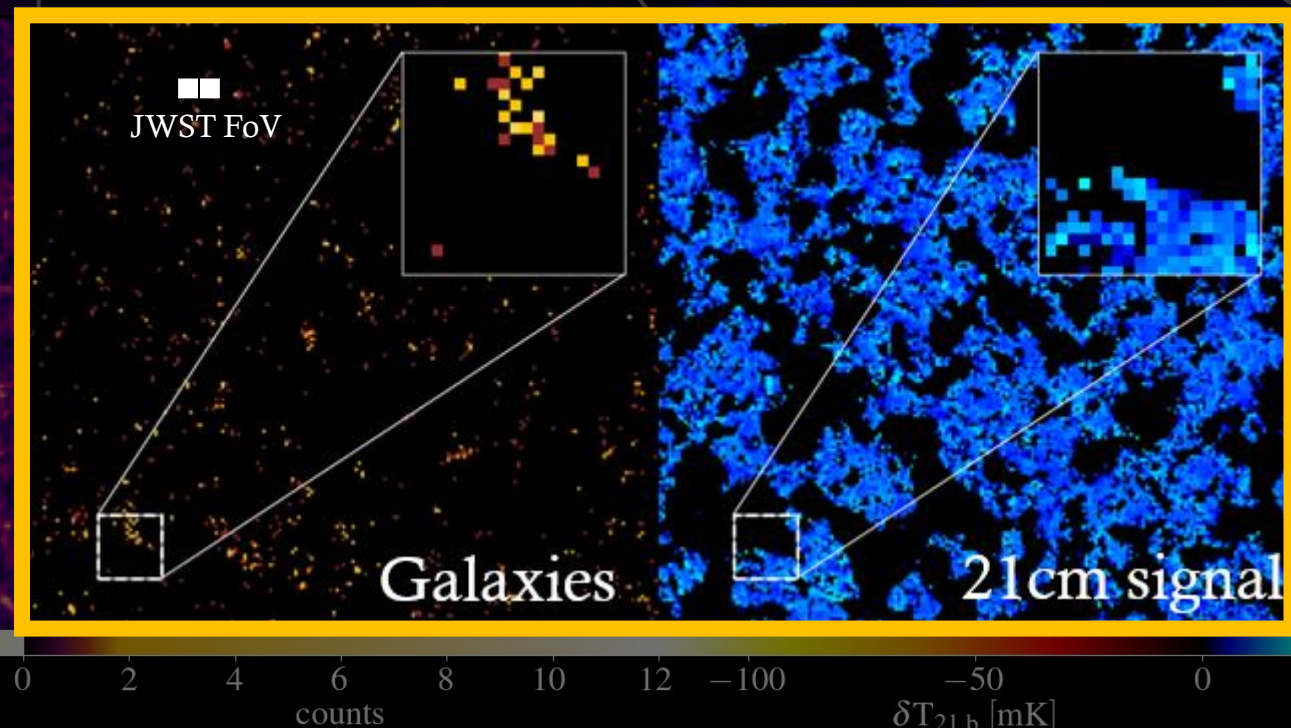
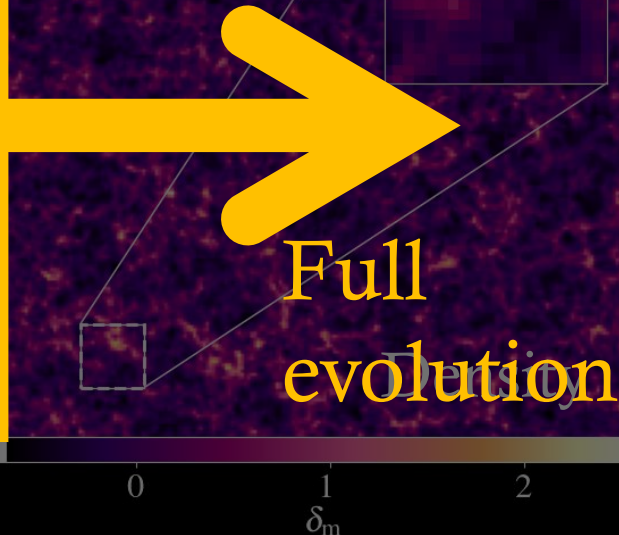
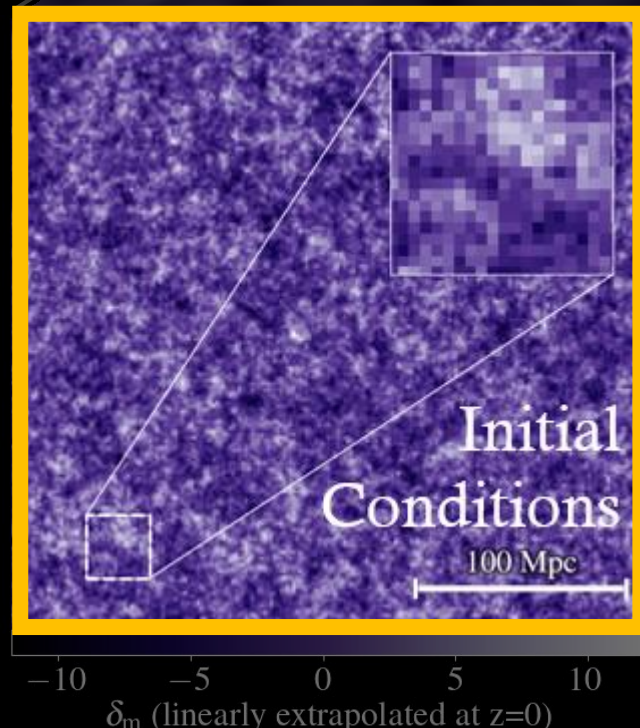
δ_m

counts

$\delta T_{21,b}$ [mK]

Why bother?

Even for the same physics we get different observations

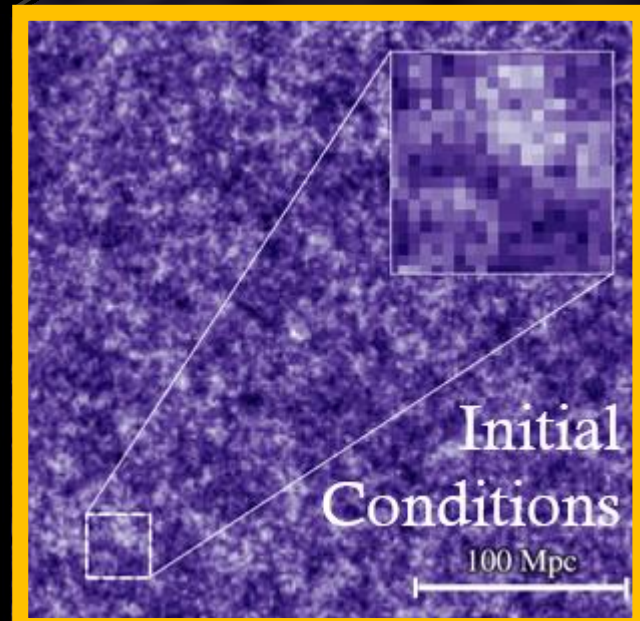


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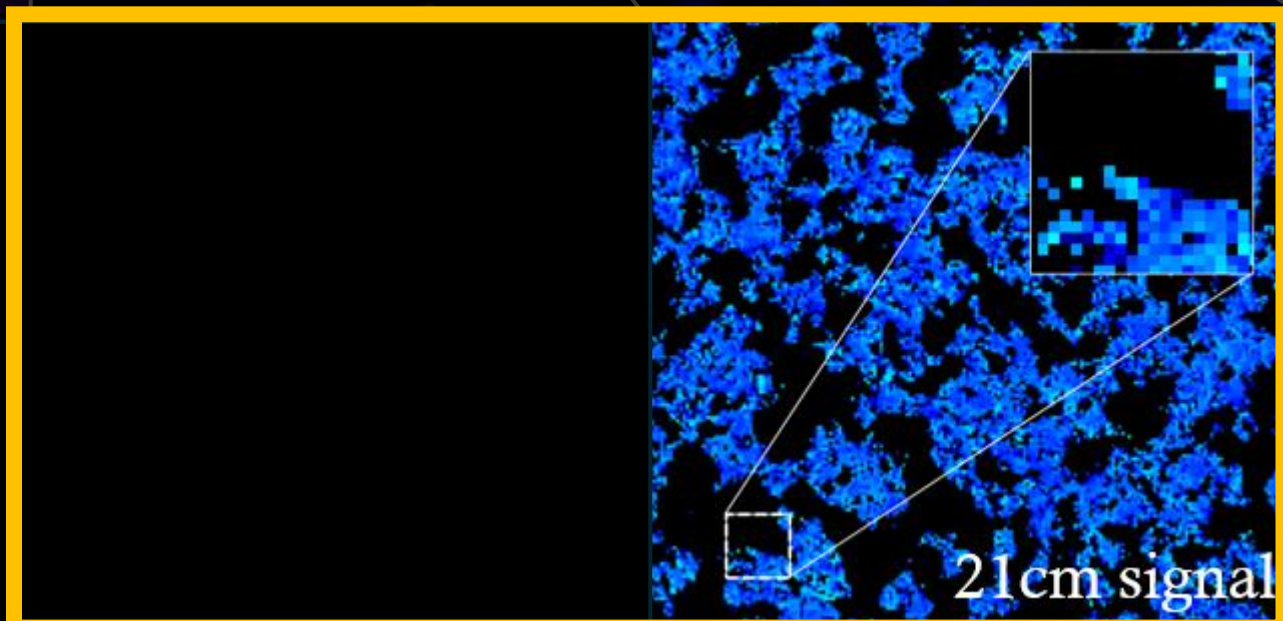
Full
evolution



Initial
Conditions

100 Mpc

δ_m (linearly extrapolated at $z=0$)



21cm signal

counts

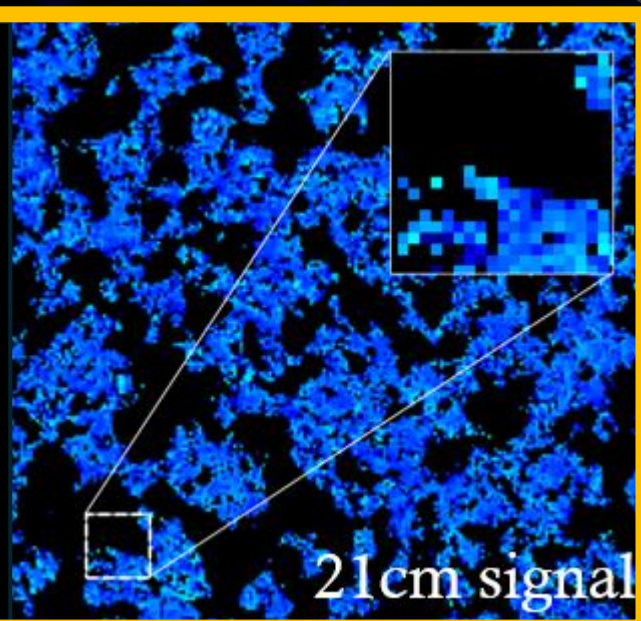
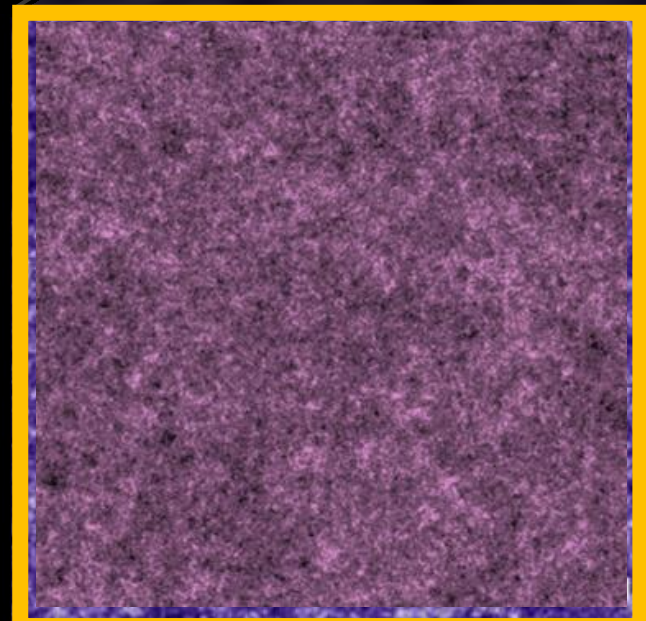
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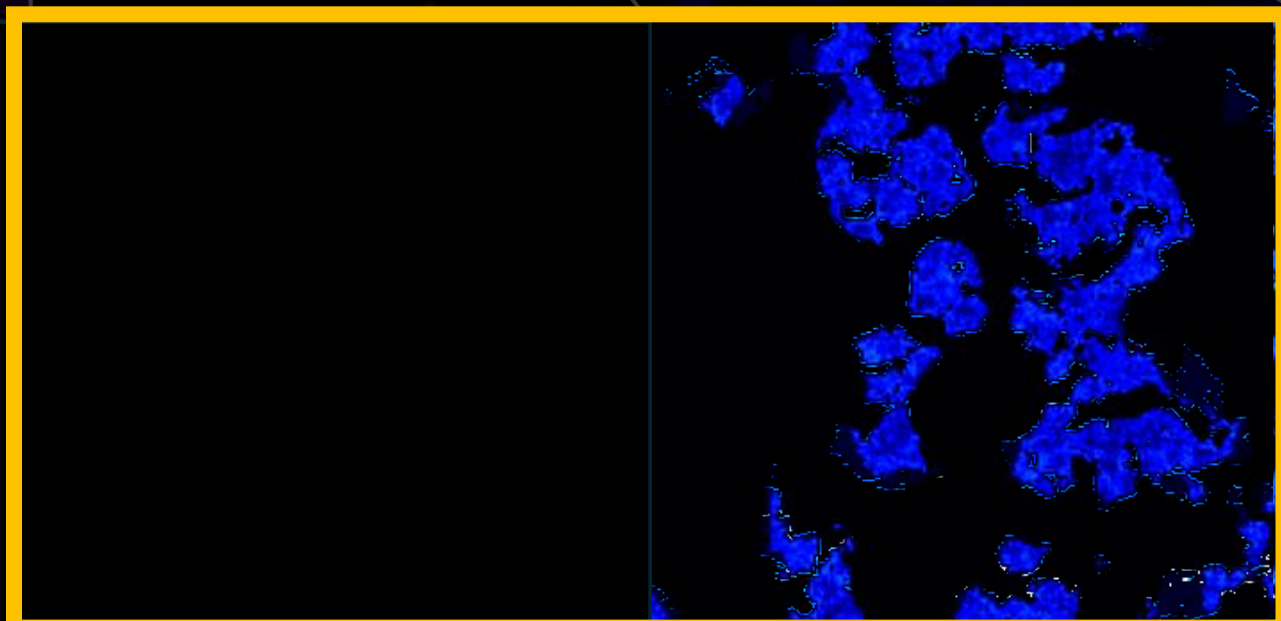
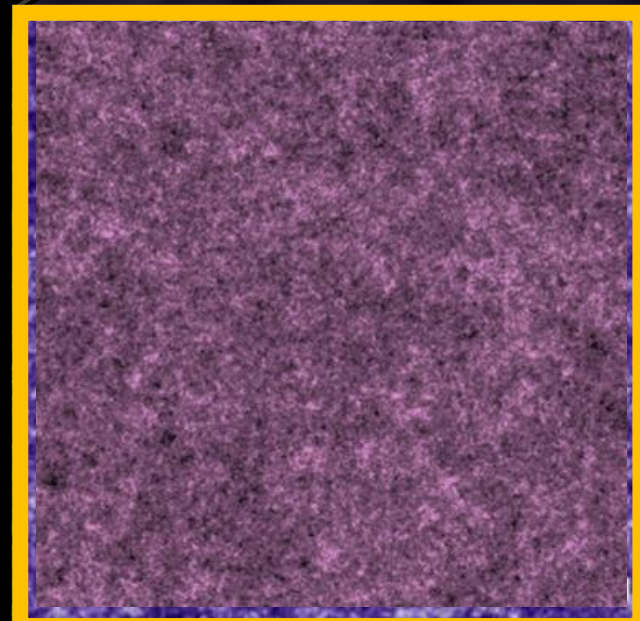


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Full
evolution



Why bother?

Why bother?

It allows us to:

- Build a **treasure map** for the Universe

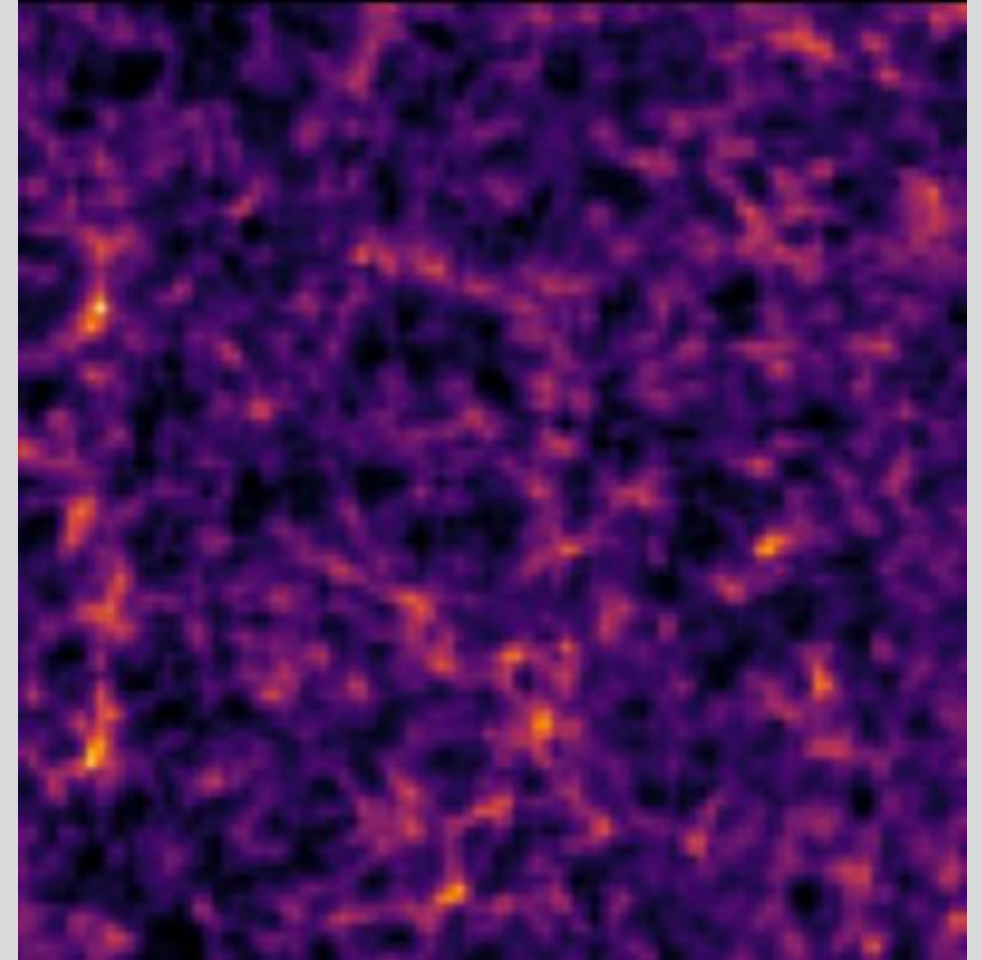


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Density field

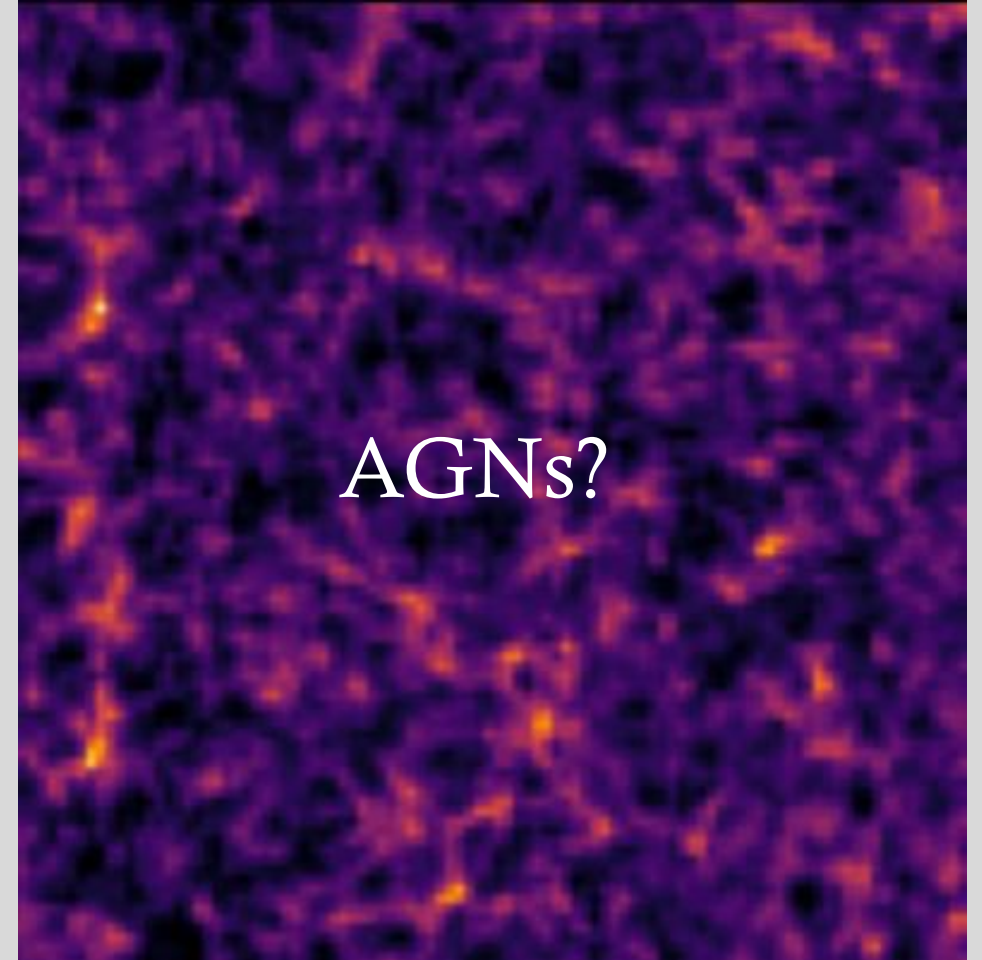


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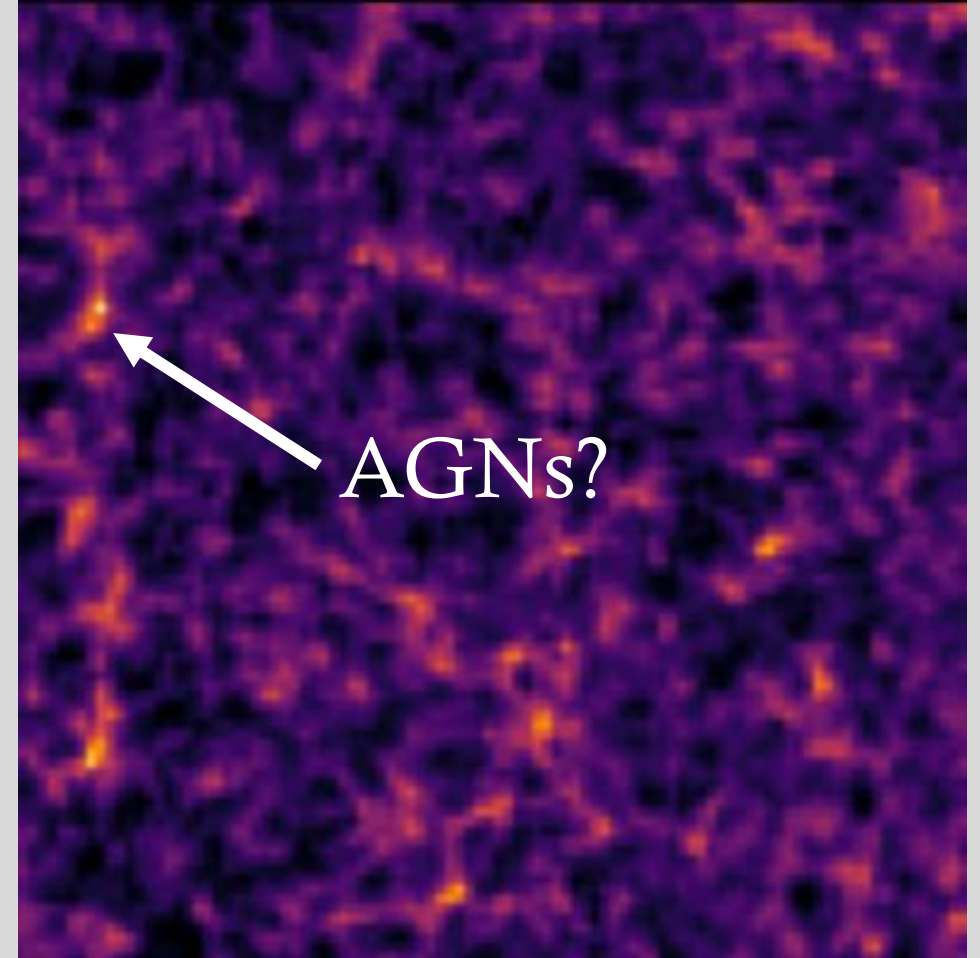


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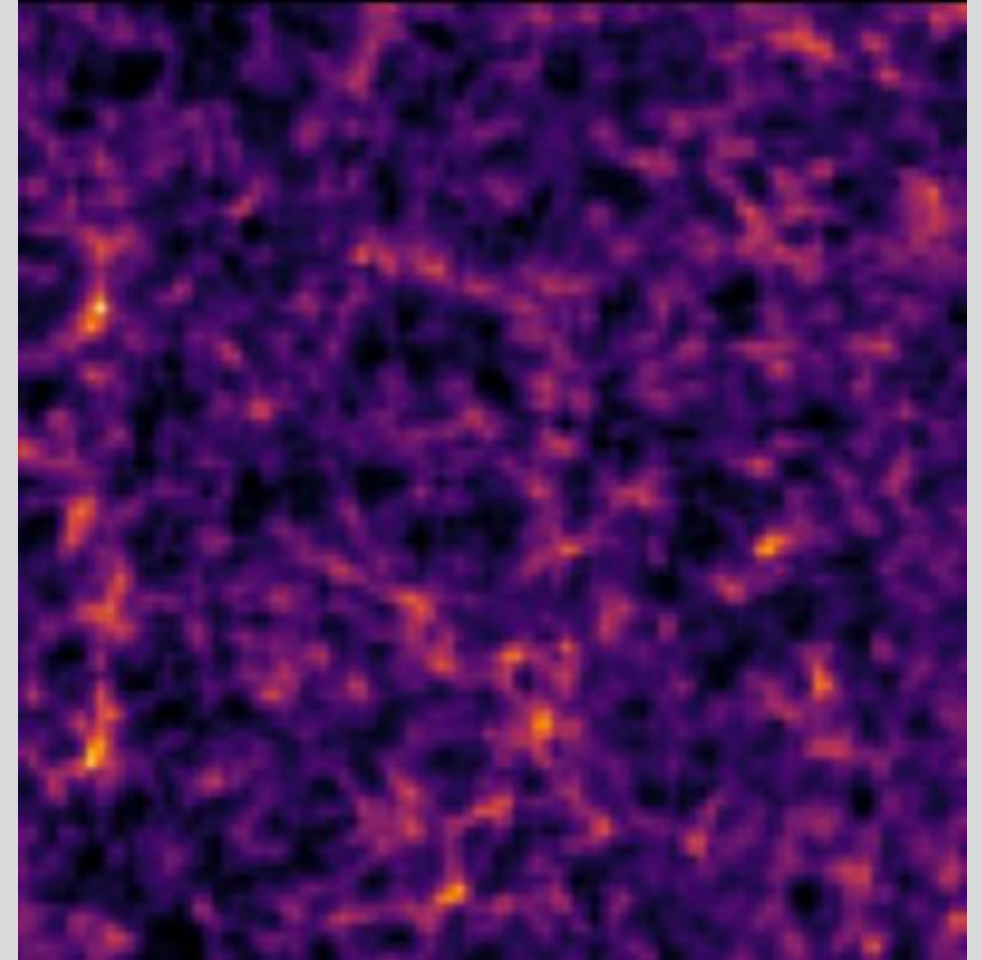


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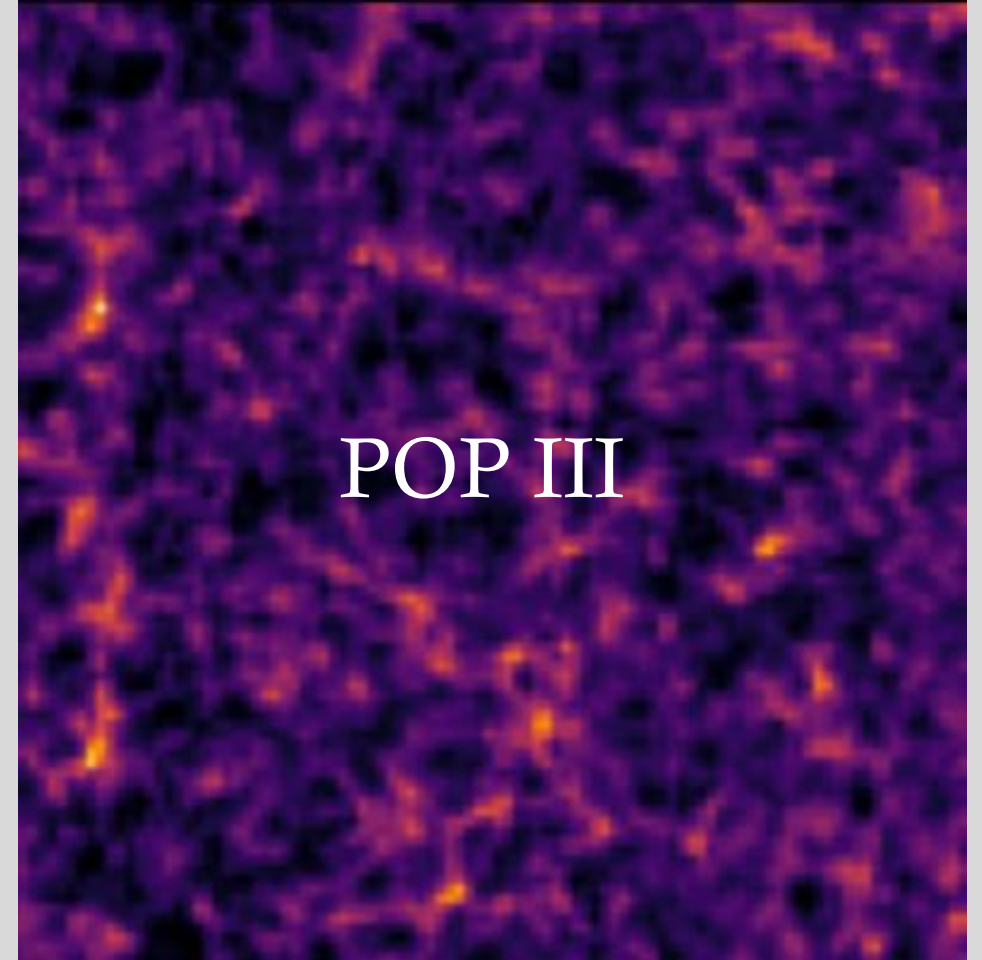


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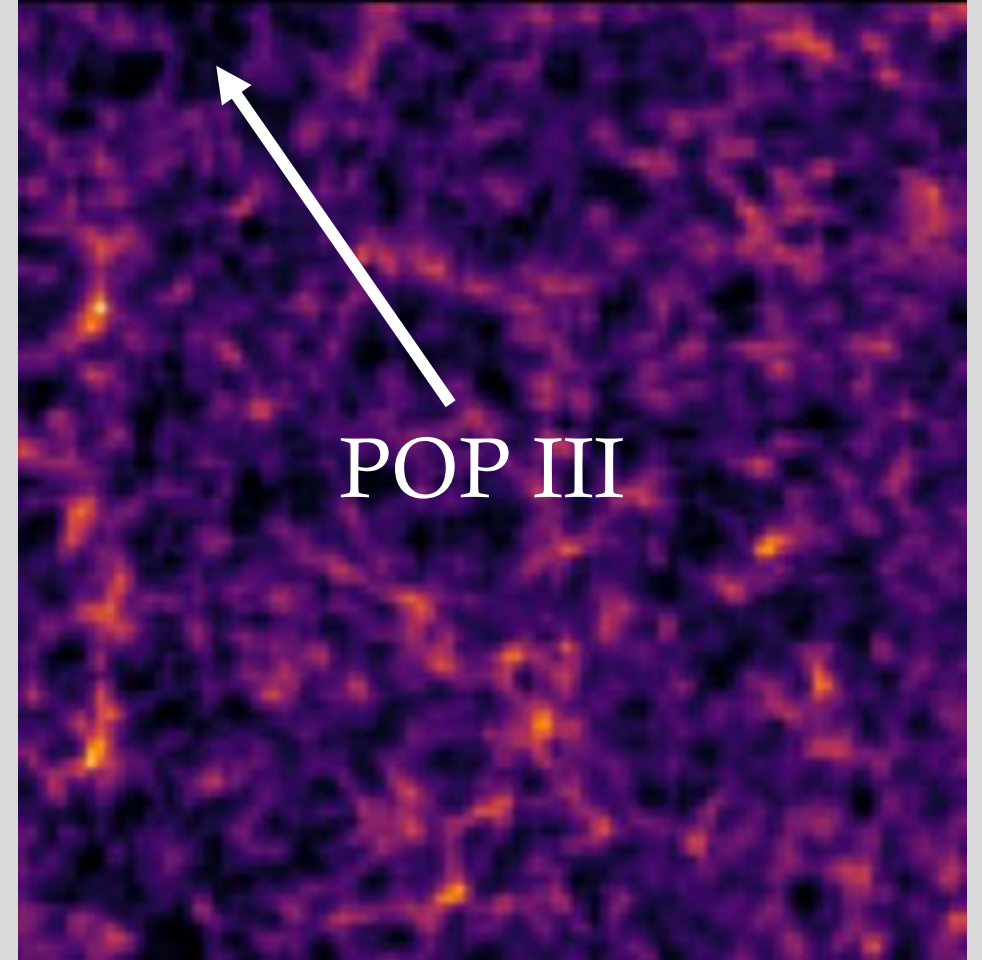


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Density field



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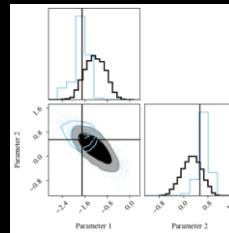
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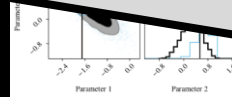
- Build a **treasure map** for the Universe
- **Improve inferences**
(e.g., by modelling cosmic variance)



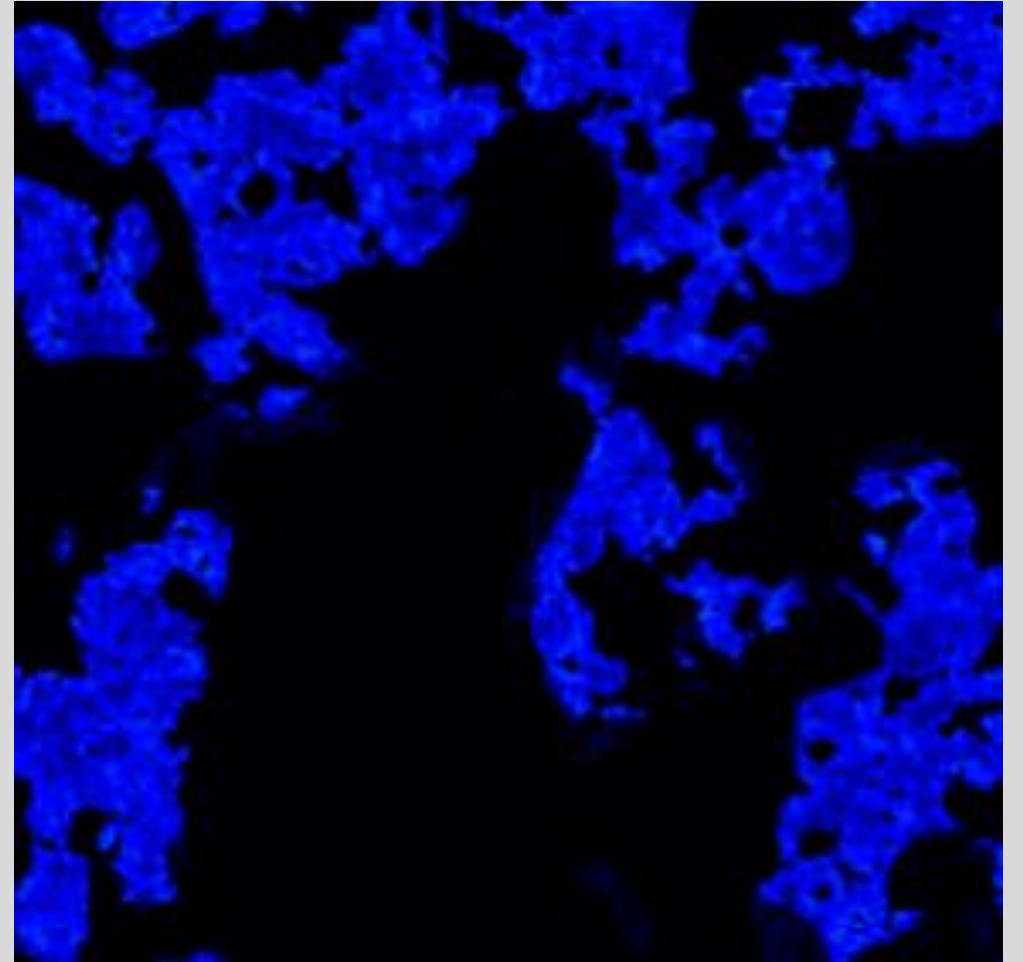
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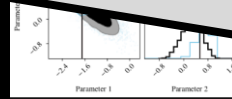
21cm signal



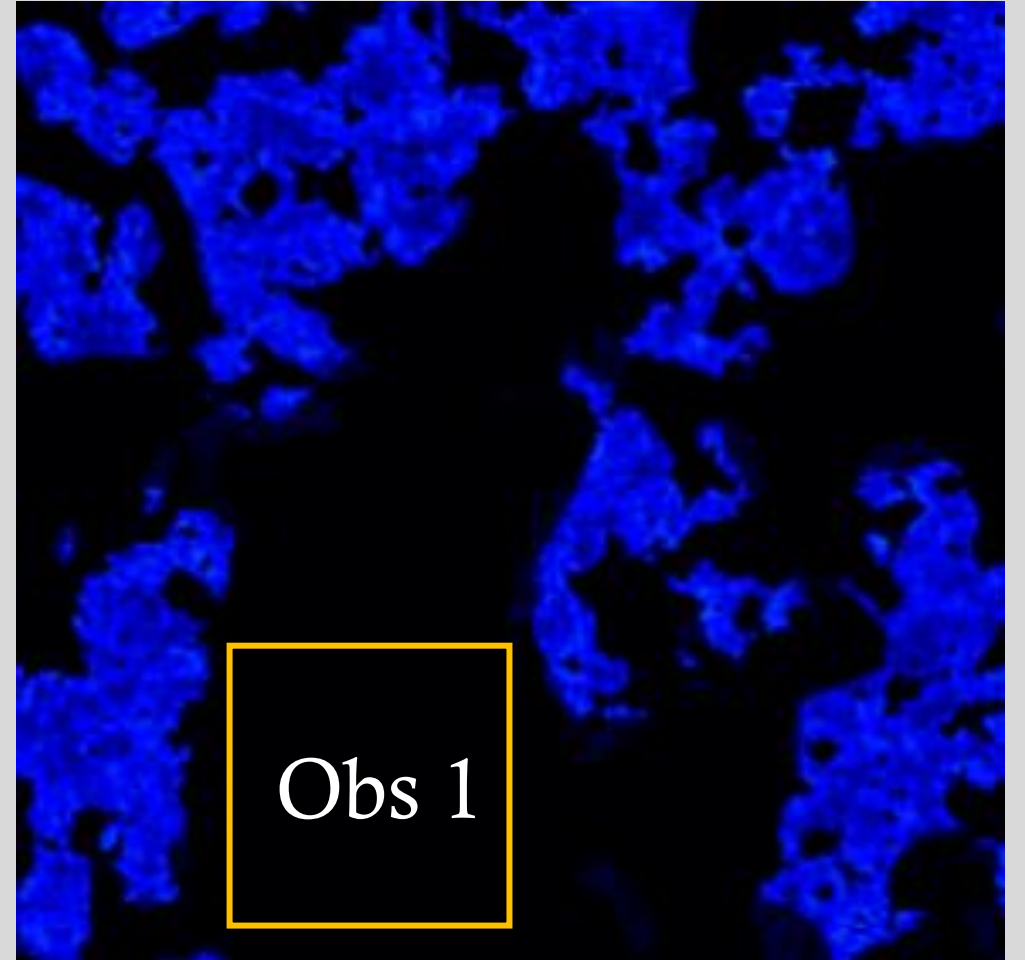
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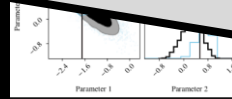
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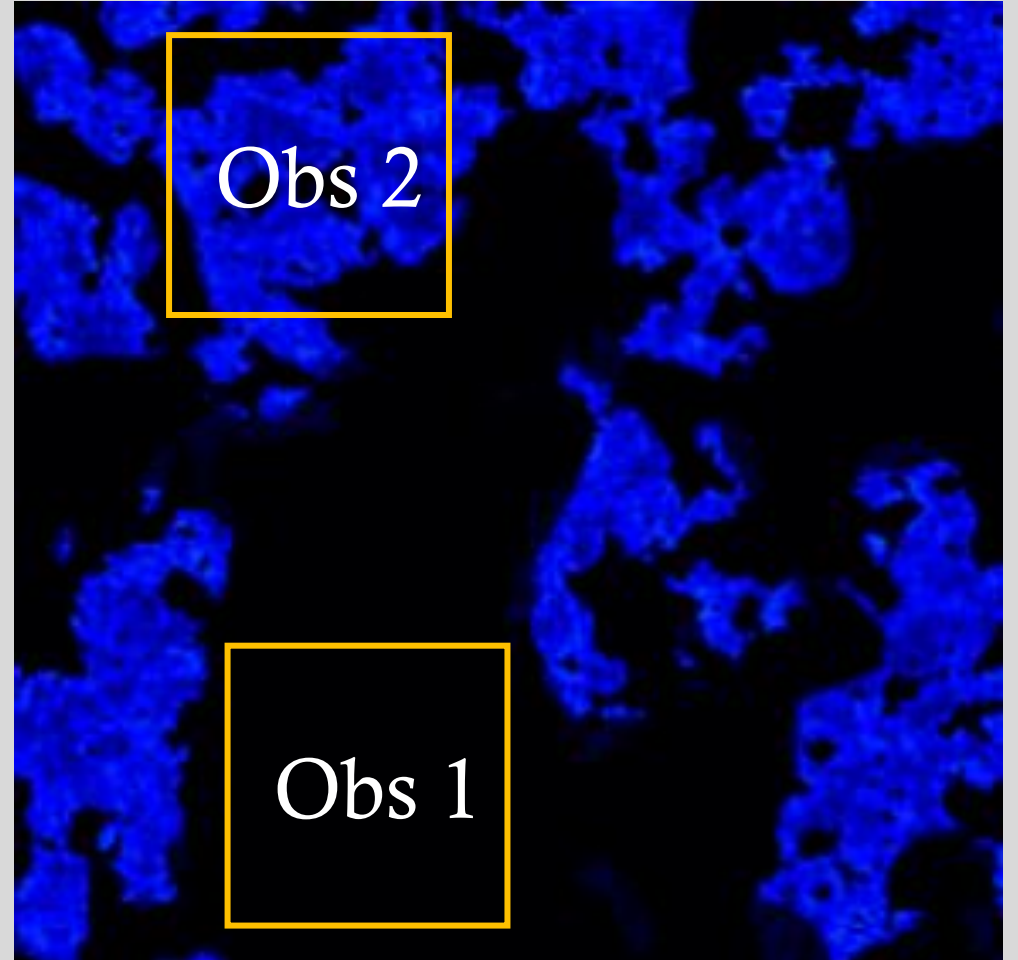
- Build a **treasure map** for the Universe
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21cm signal

Obs 2

Obs 1



Why bother?

It allows us to:

- Build a *treasure map* for the Universe
- **Improve** it
(e.g., by modelling)

If I know they are special regions, they
say the same thing

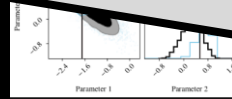
21cm signal

Obs 1

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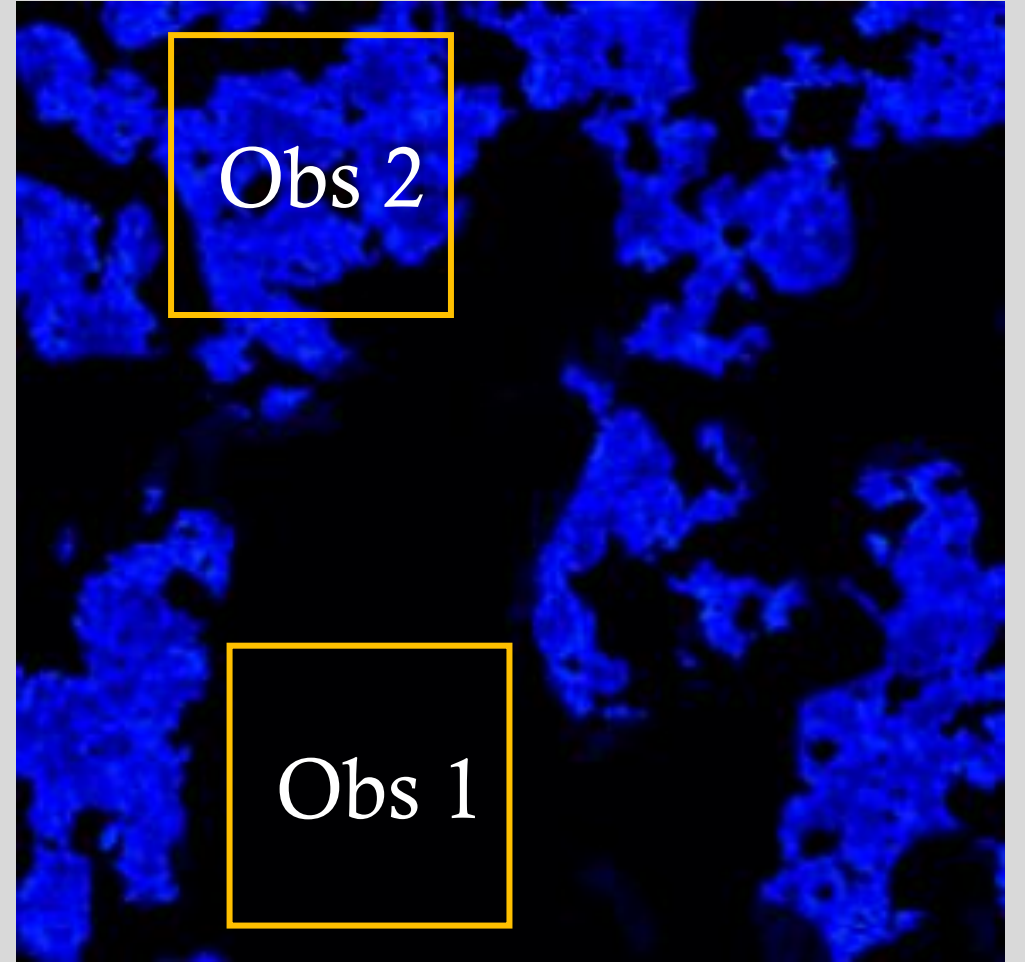
- Build a **treasure map** for the Universe
- **Improve inferences**
(e.g., by modelling cosmic variance)



21cm signal

Obs 2

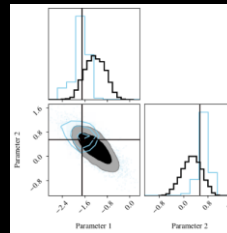
Obs 1



Why bother?

It allows us to:

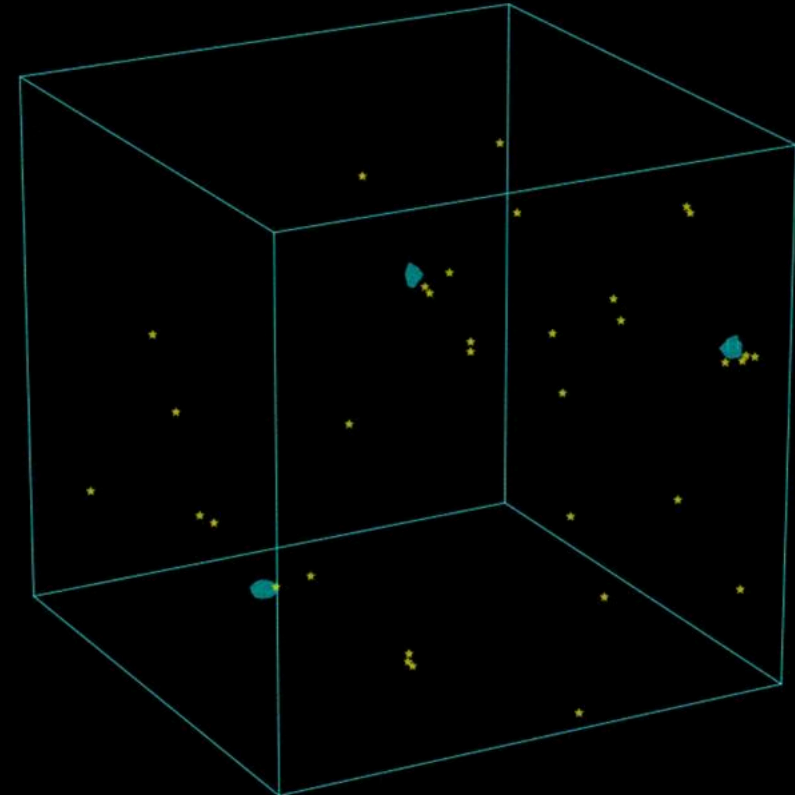
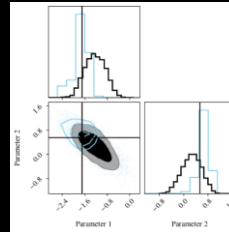
- Build a **treasure map** for the Universe
- **Improve inferences**
(e.g., by modelling cosmic variance)



Why bother?

It allows us to:

- Build a **treasure map** for the Universe
- **Improve inferences**
(e.g., by modelling cosmic variance)
- Know where **faint galaxies** are and how they interact with the IGM -> **reionization morphology**



How?

Bayesian inference

Bayesian inference

$$p(\theta|d) \propto p(d|\theta) \cdot p(\theta)$$

$\theta \rightarrow$ parameters

$d \rightarrow$ data

Bayesian inference

Posterior

$$p(\theta|d) \propto p(d|\theta) \cdot p(\theta)$$

$\theta \rightarrow$ parameters

$d \rightarrow$ data

Bayesian inference

Posterior

$$p(\theta|d) \propto p(d|\theta) \cdot p(\theta)$$

If there is no analytical form?

$\theta \rightarrow$ parameters

$d \rightarrow$ data

Bayesian inference

Posterior

$$p(\theta|d) \propto p(d|\theta) \cdot p(\theta)$$

$\theta \rightarrow$ parameters
 $d \rightarrow$ data

If there is no analytical form?

Generate θ and d pairs with a simulator and use machine learning to approximate $p(\theta|d)$

Bayesian inference

Posterior

$$p(\theta|d) \propto p(d|\theta) \cdot p(\theta)$$



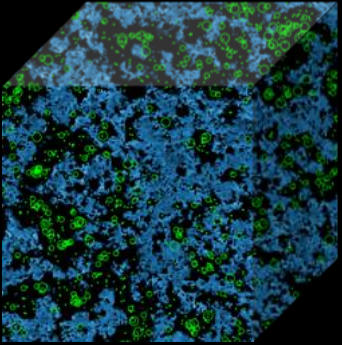
Davies, Murray & Mesinger +25

If there is no analytical form?

Generate θ and d pairs with a **simulator** and use machine learning to approximate $p(\theta|d)$

$\theta \rightarrow$ parameters
 $d \rightarrow$ data

Workflow outline

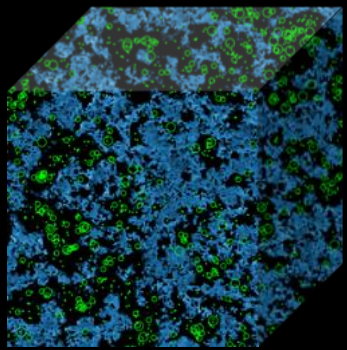


Simulated data

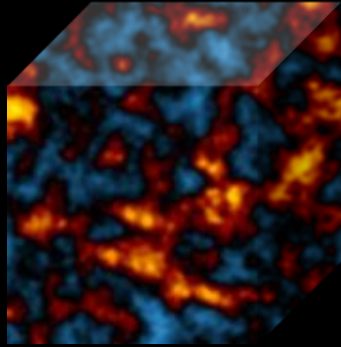
(galaxy and 21cm
 200^3 pix, 300^3 Mpc
fields)

2,000 simulations at $z=8$
under default astrophysics

Workflow outline



Observational
pipeline



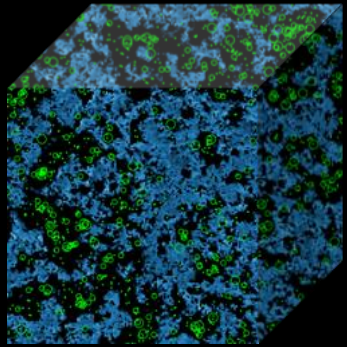
Simulated data

(galaxy and 21cm
 200^3 pix, 300^3 Mpc
fields)

Mock
observations

2,000 simulations at $z=8$
under default astrophysics

Workflow outline



Observational
pipeline



Includes:

- RSDs
- Muv cuts
- SKA noise
- foreground contamination

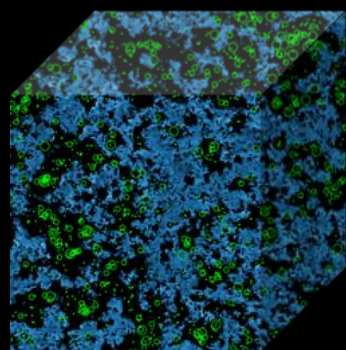
Simulated data

(galaxy and 21cm
 200^3 pix, 300^3 Mpc
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Mock
observations

2,000 simulations at $z=8$
under default astrophysics

Workflow outline



Observational
pipeline



Includes:

- RSDs
- Muv cuts
- SKA noise
- foreground contamination

Resembling observations
of SKA and galaxy
surveys

e.g,

MOONRISE; Maiolino+20,
COSMOS-3D; Kakiichi+24

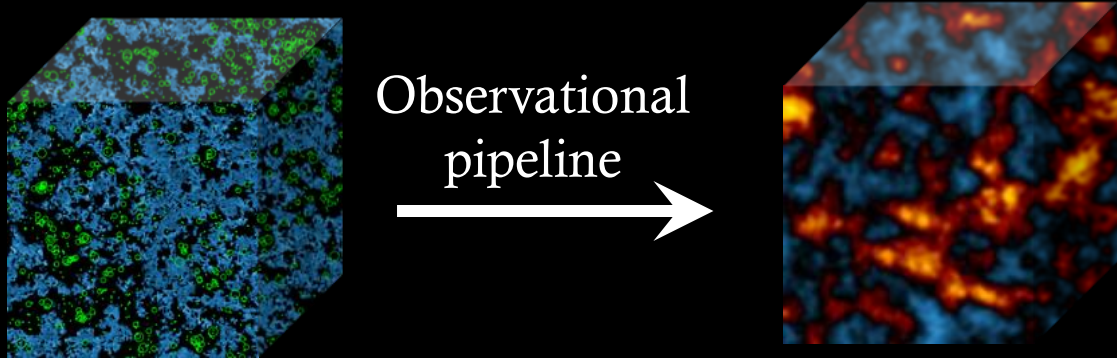
Simulated data

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Mock
observations

2,000 simulations at $z=8$
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Workflow outline



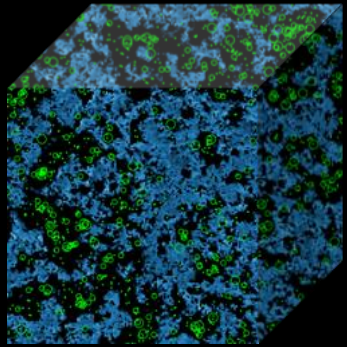
Simulated data

(galaxy and 21cm
 200^3 pix, 300^3 Mpc
fields)

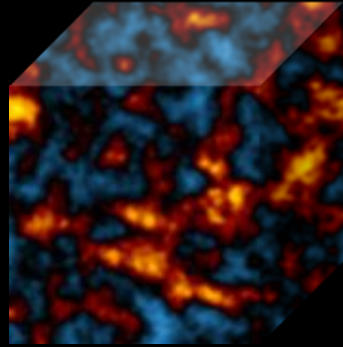
Mock
observations

2,000 simulations at $z=8$
under default astrophysics

Workflow outline



Observational
pipeline



Simulated data

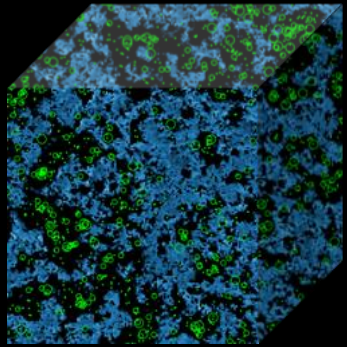
(galaxy and 21cm
 200^3 pix, 300^3 Mpc
fields)

Mock
observations

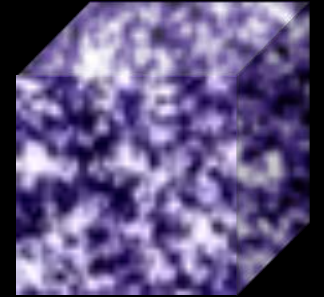
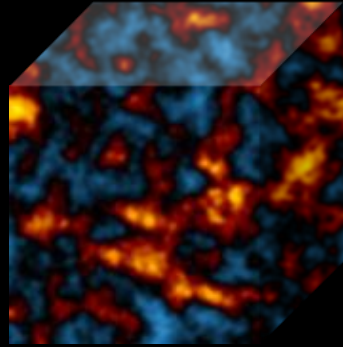
ML model

2,000 simulations at $z=8$
under default astrophysics

Workflow outline



Observational
pipeline



Simulated data

(galaxy and 21cm
 200^3 pix, 300^3 Mpc
fields)

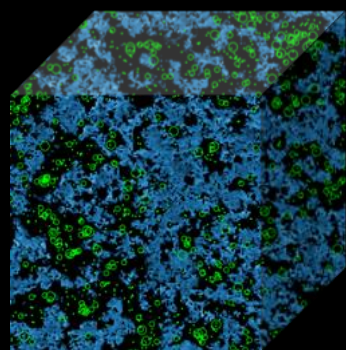
Mock
observations

ML model

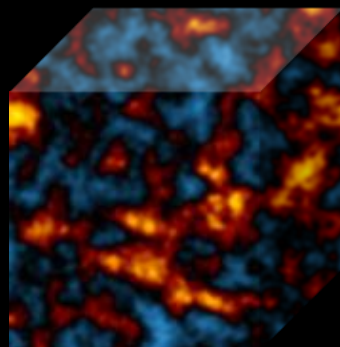
ICs samples

2,000 simulations at $z=8$
under default astrophysics

Workflow outline



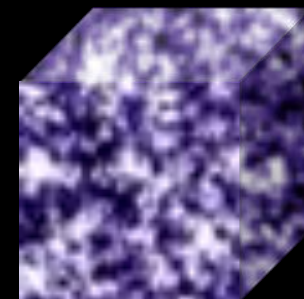
Observational
pipeline



Mock
observations



ML model



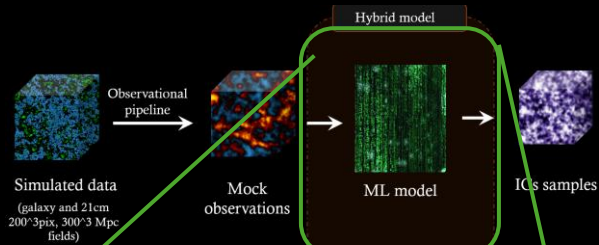
ICs samples

TILING

Simulated data
(galaxy and 21cm
 200^3 pix, 300^3 Mpc
fields)

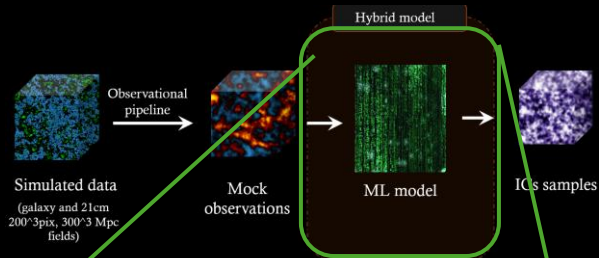
2,000 simulations at $z=8$
under default astrophysics

Workflow outline

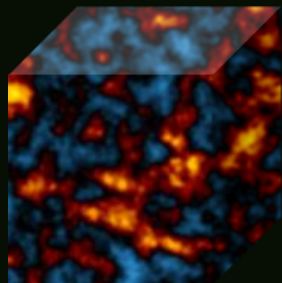


TILING

Workflow outline

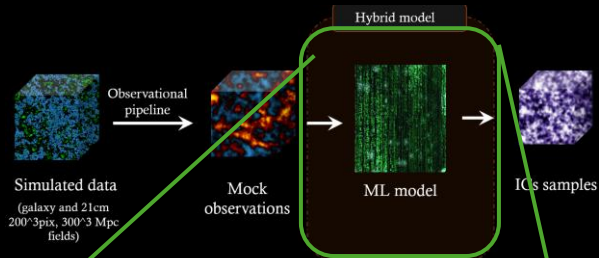


TILING

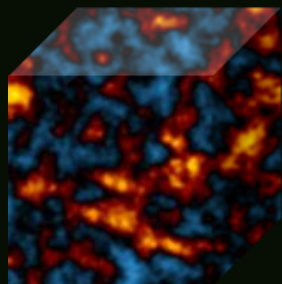


Simulated mock
observations

Workflow outline



TILING

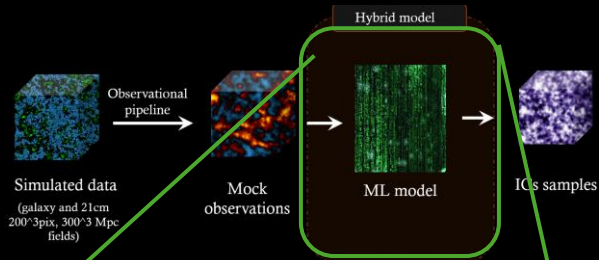


ICs MAP estimator U-net



trained with 50% of the dataset

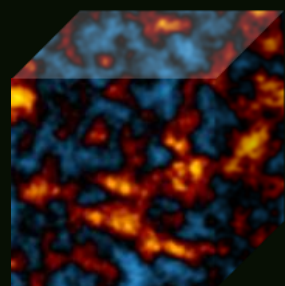
Simulated mock
observations



Workflow outline

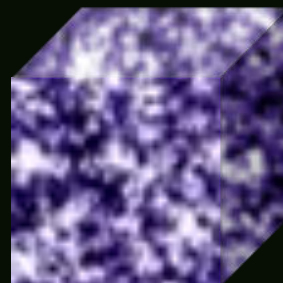
TILING

ICs MAP predictions
for the rest 50%

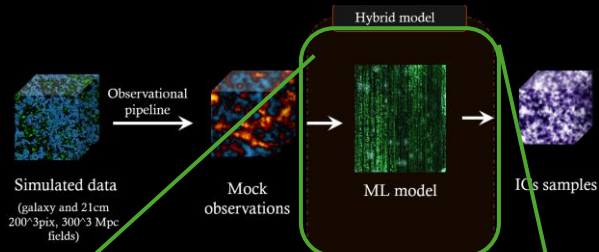


ICs MAP estimator U-net

trained with 50% of the dataset

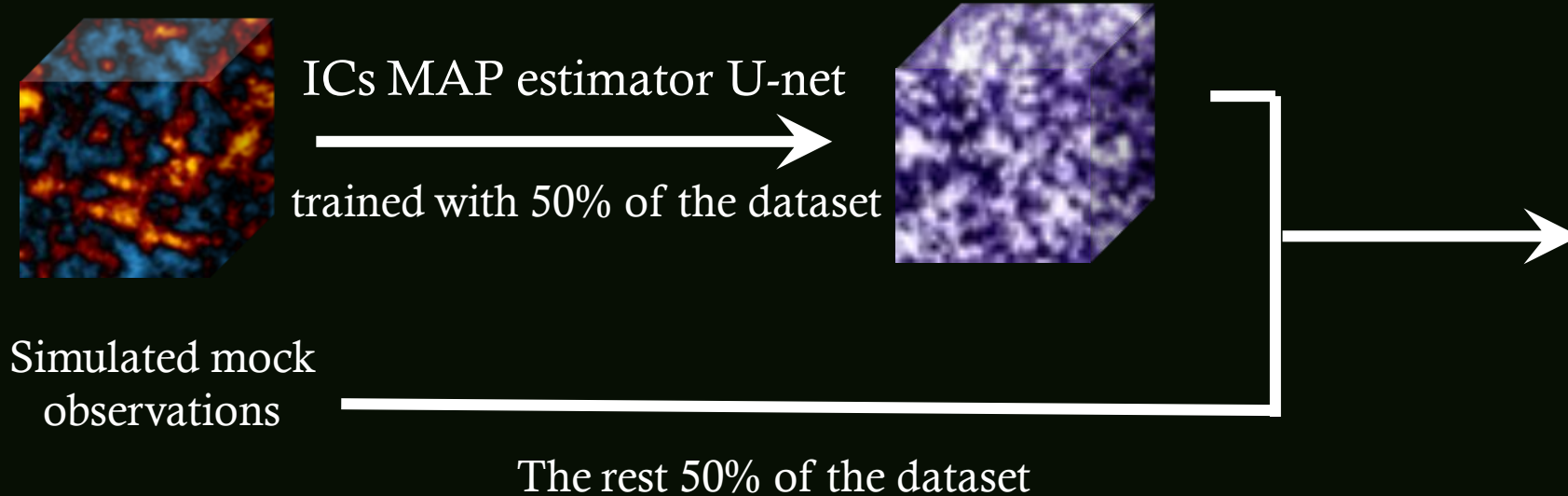


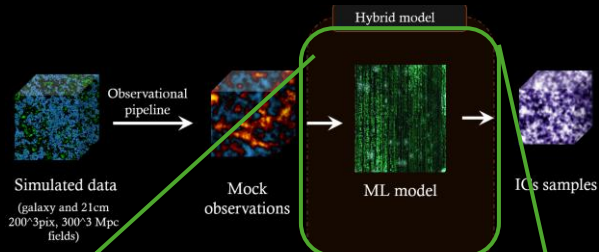
Simulated mock
observations



Workflow outline

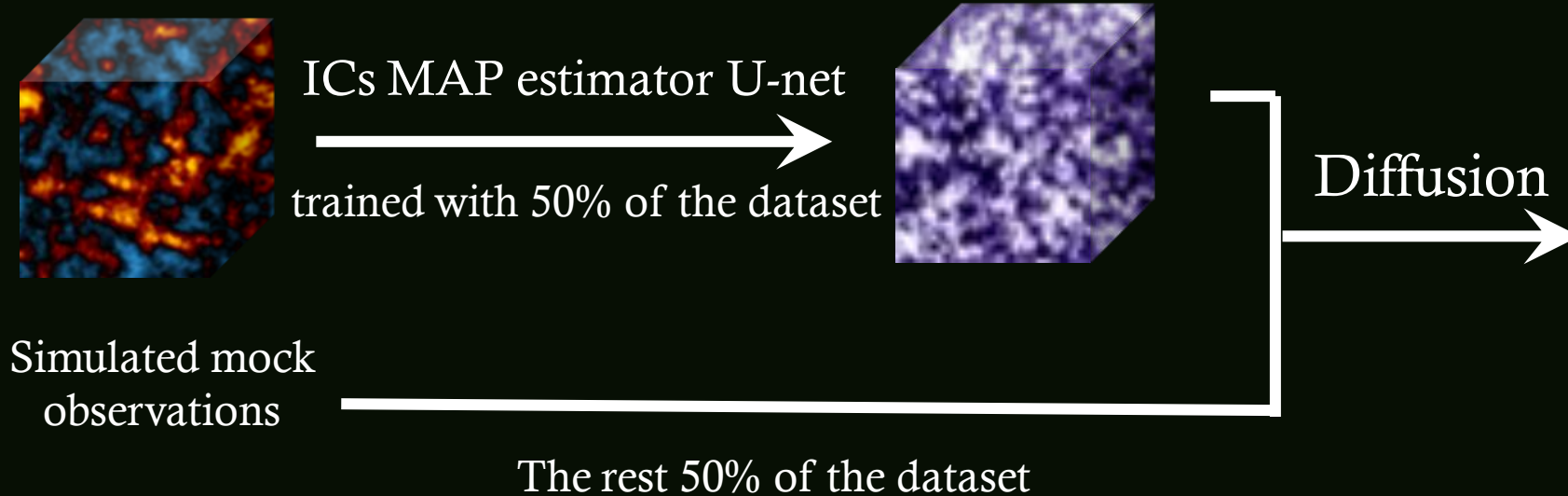
TILING

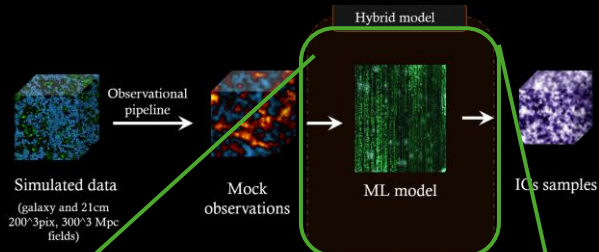




Workflow outline

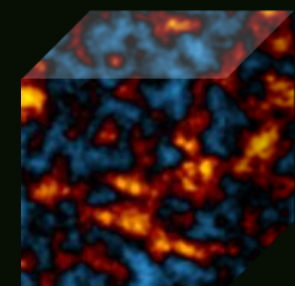
TILING





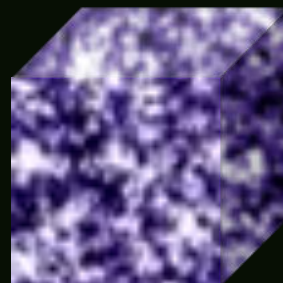
Workflow outline

TILING



Simulated mock
observations

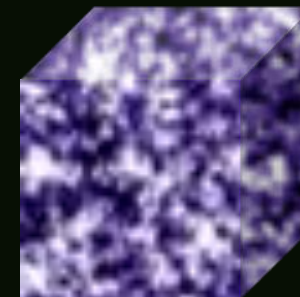
ICs MAP estimator U-net
trained with 50% of the dataset



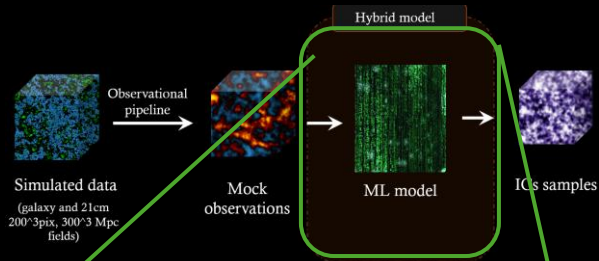
ICs MAP predictions
for the rest 50%

The rest 50% of the dataset

Diffusion

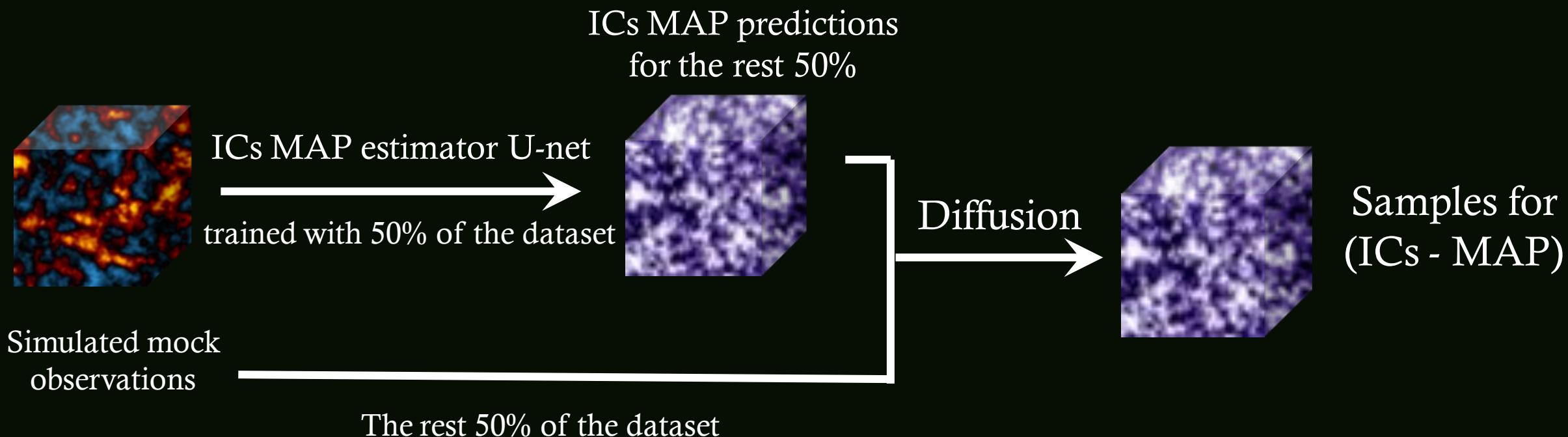


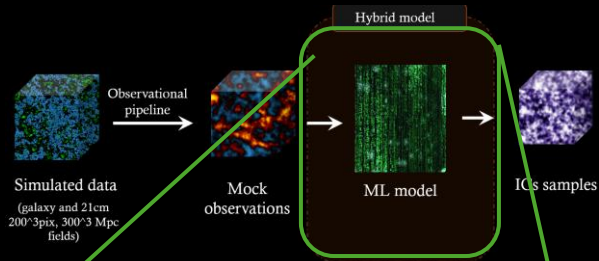
Samples for
(ICs - MAP)



Workflow outline

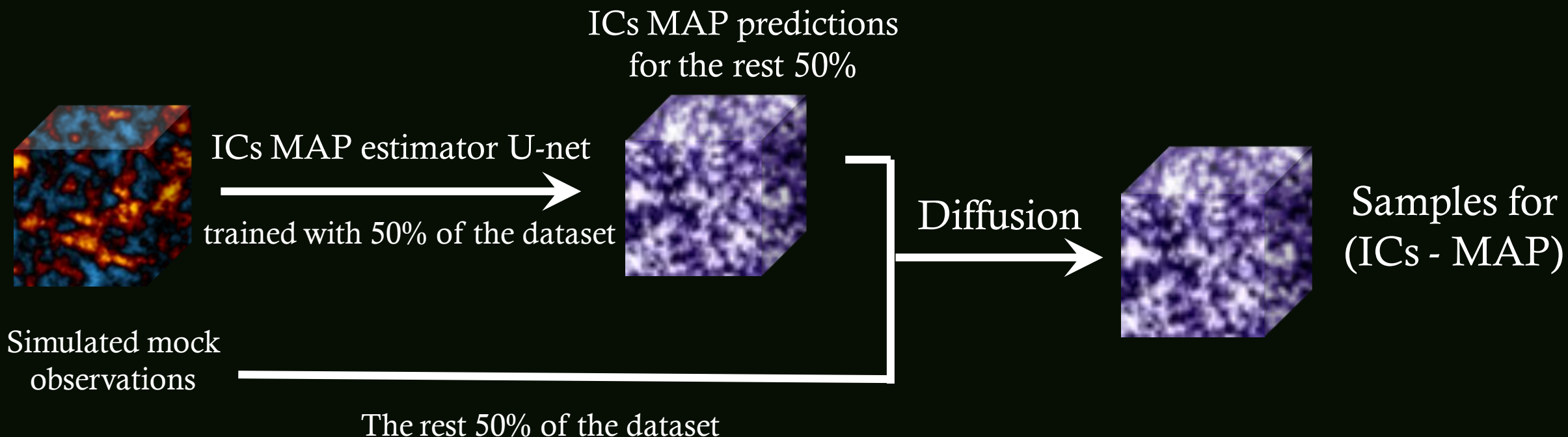
TILING

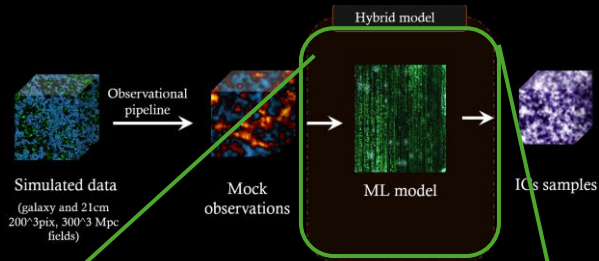




Workflow outline

TILING : Tomographic Inference of Linear ICs via Network Grafting





Workflow outline

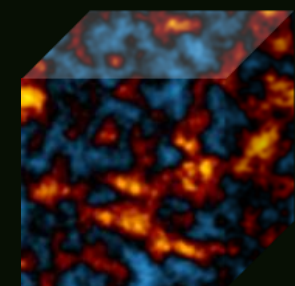
TILING: Tomographic Inference of Linear ICs via Network Grafting

Grafting in Biology:



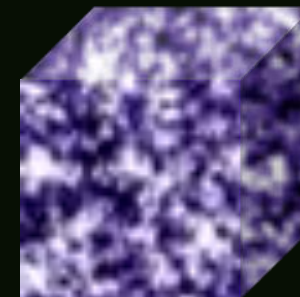
Diffusion

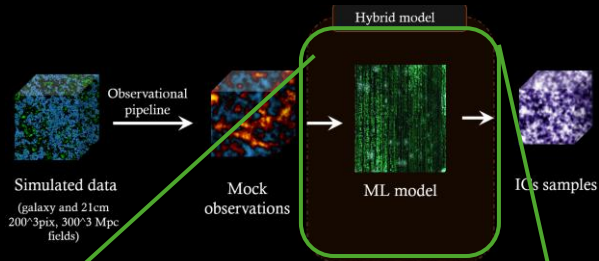
Samples for
(ICs - MAP)



ICs MAP
trained with

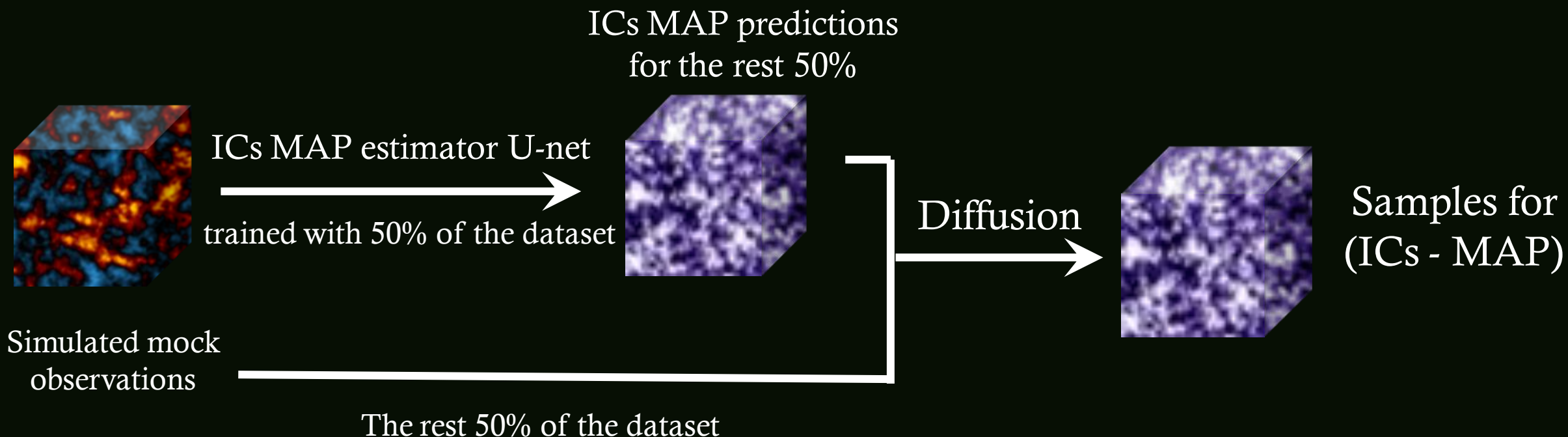
Simulated mock
observations





Workflow outline

TILING : Tomographic Inference of Linear ICs via Network Grafting



Results

Results

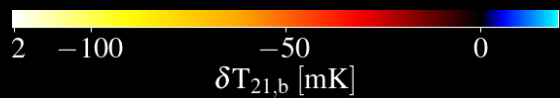
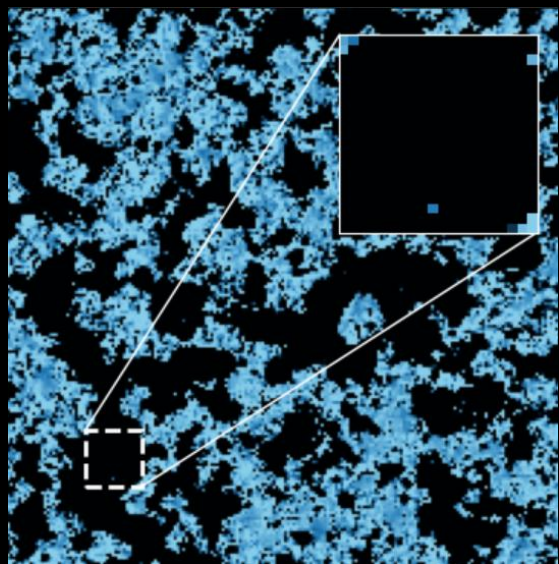
Only Muv cut = -17

Results

Only Muv cut = -17

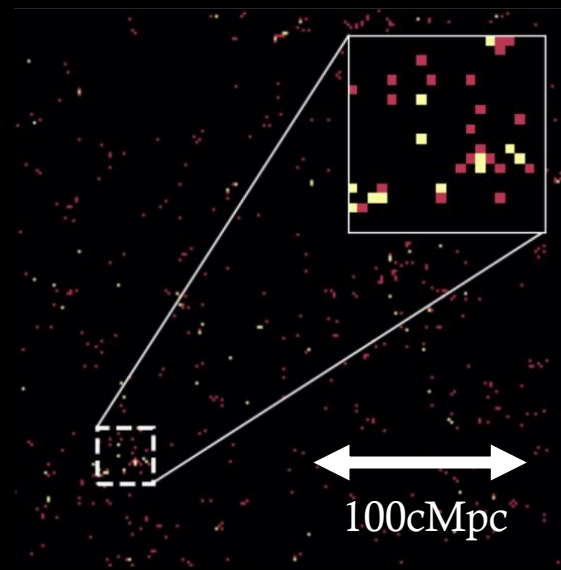
$z=8$, 200^3 voxel cubes

21cm



$\delta T_{21,b}$ [mK]

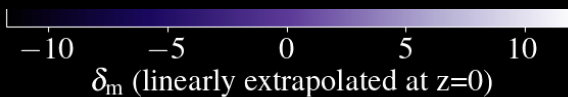
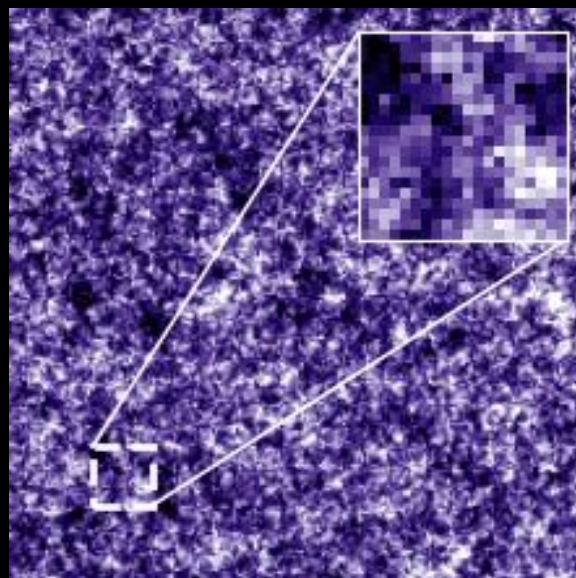
galaxies



counts

100cMpc

True ICs



δ_m (linearly extrapolated at $z=0$)

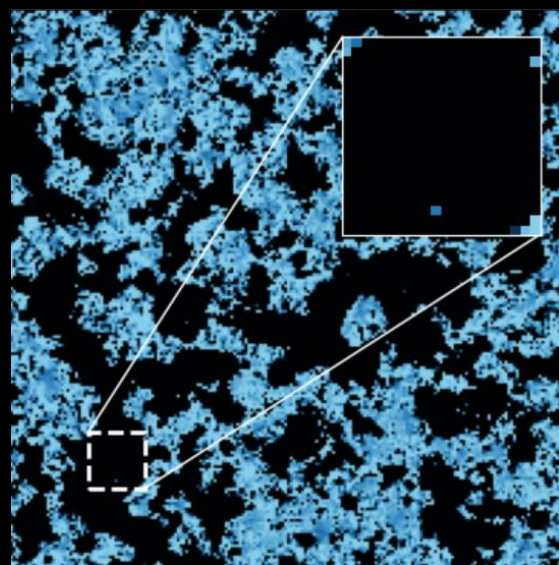
Results

Only Muv cut = -17

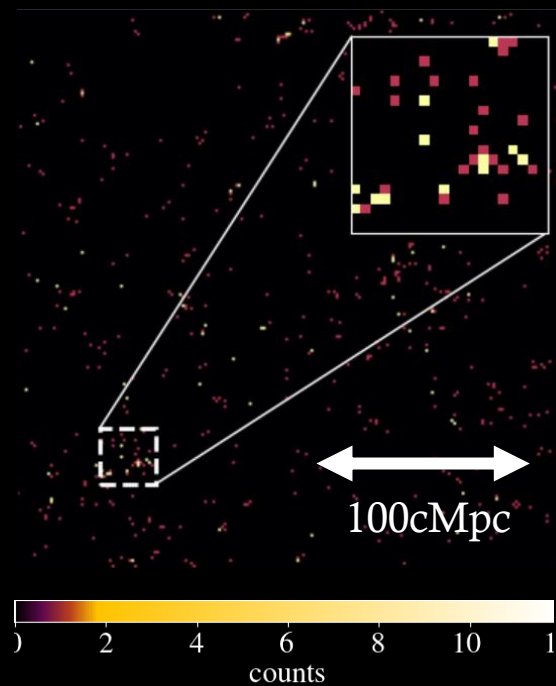
8 million parameter reconstruction

$z=8$, 200^3 voxel cubes

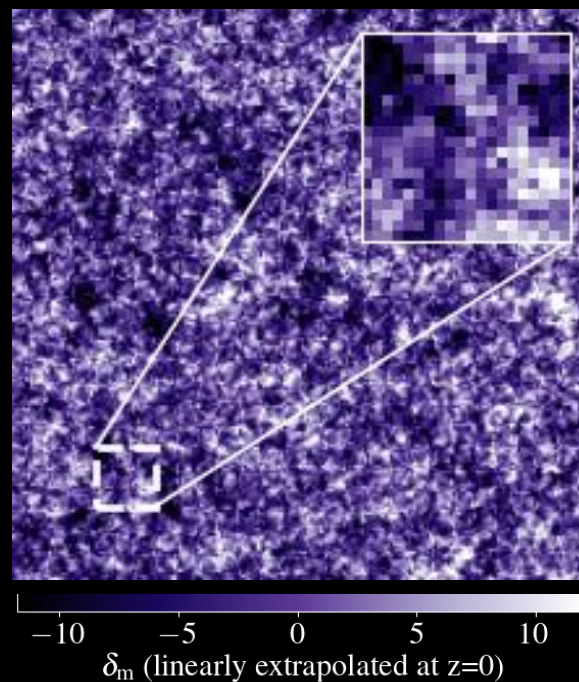
21cm



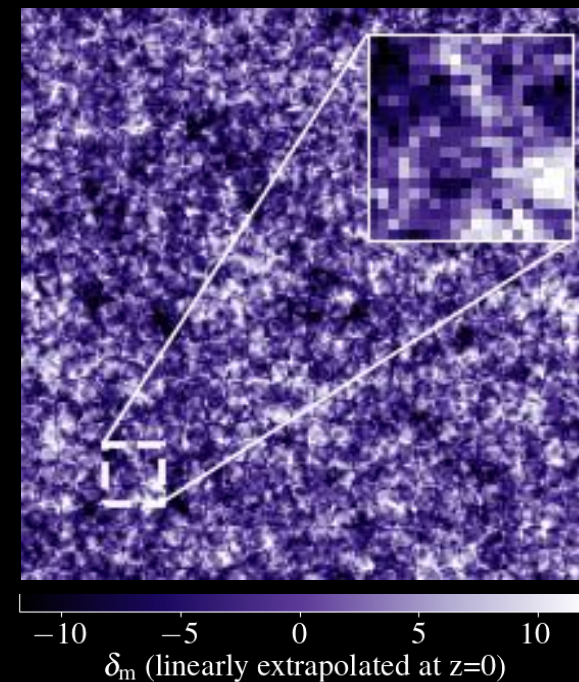
galaxies



True ICs



Sample



Results

Only Muv cut = -17

8 million parameter reconstruction

$z=8$, 200^3 voxel cubes

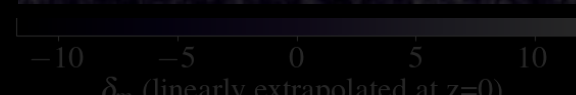
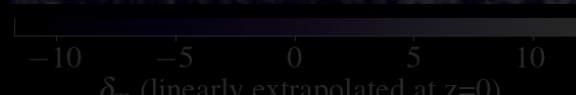
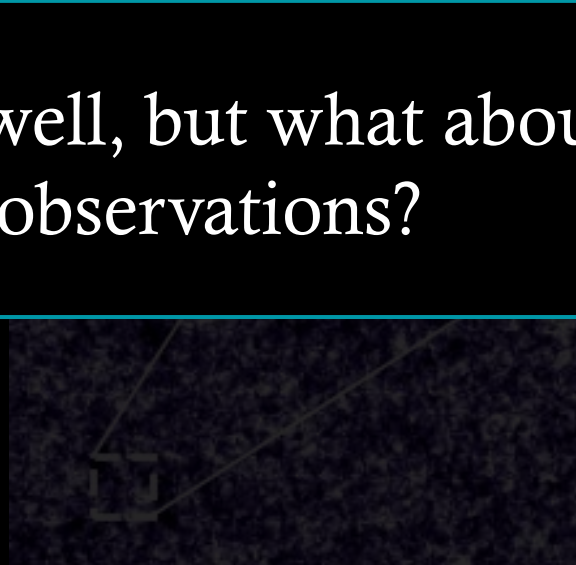
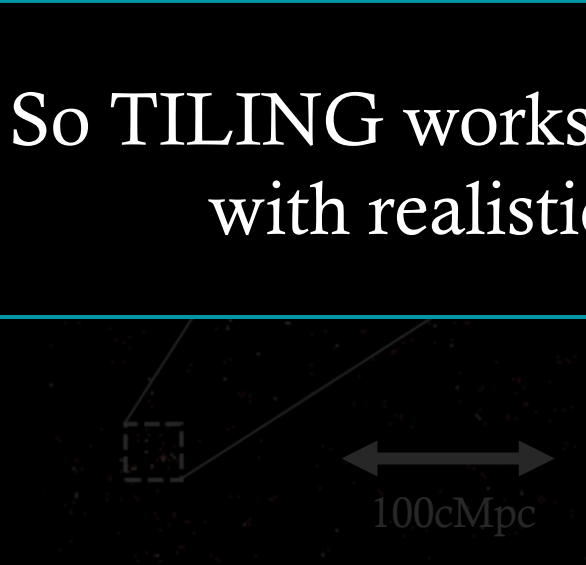
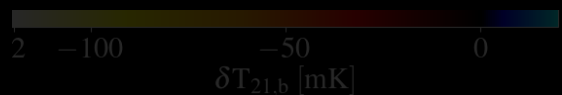
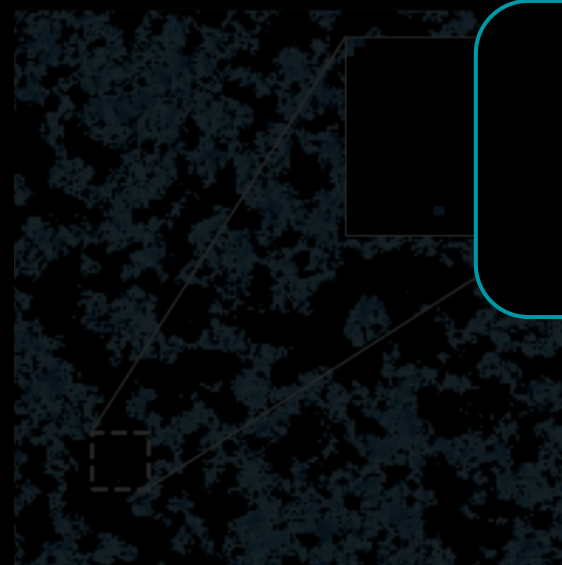
21cm

galaxies

True ICs

Sample

So TILING works well, but what about
with realistic observations?



Results

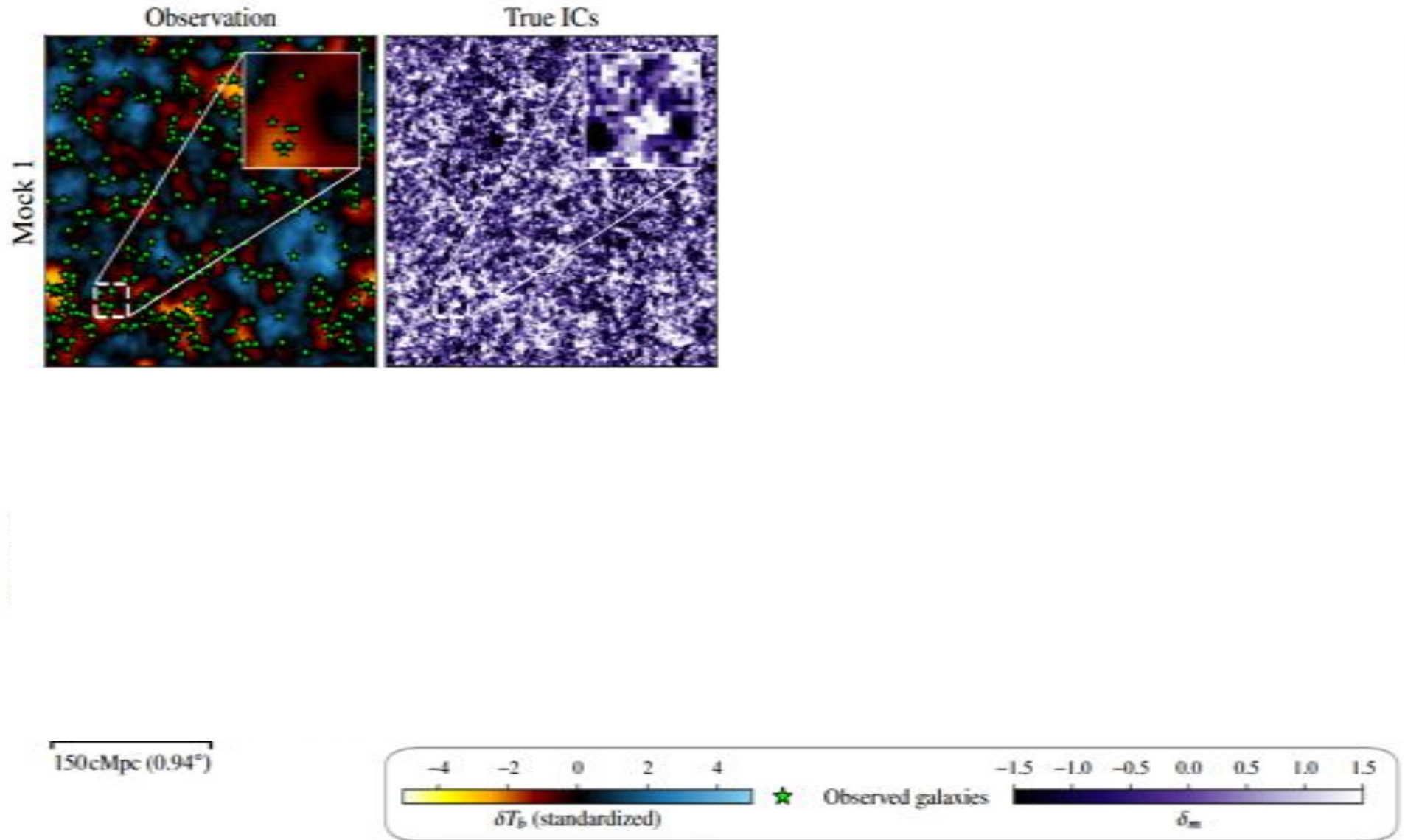
Fiducial

- 1000h SKA w/
foregrounds
- $M_{\text{UV}} < -18$

Results

Fiducial

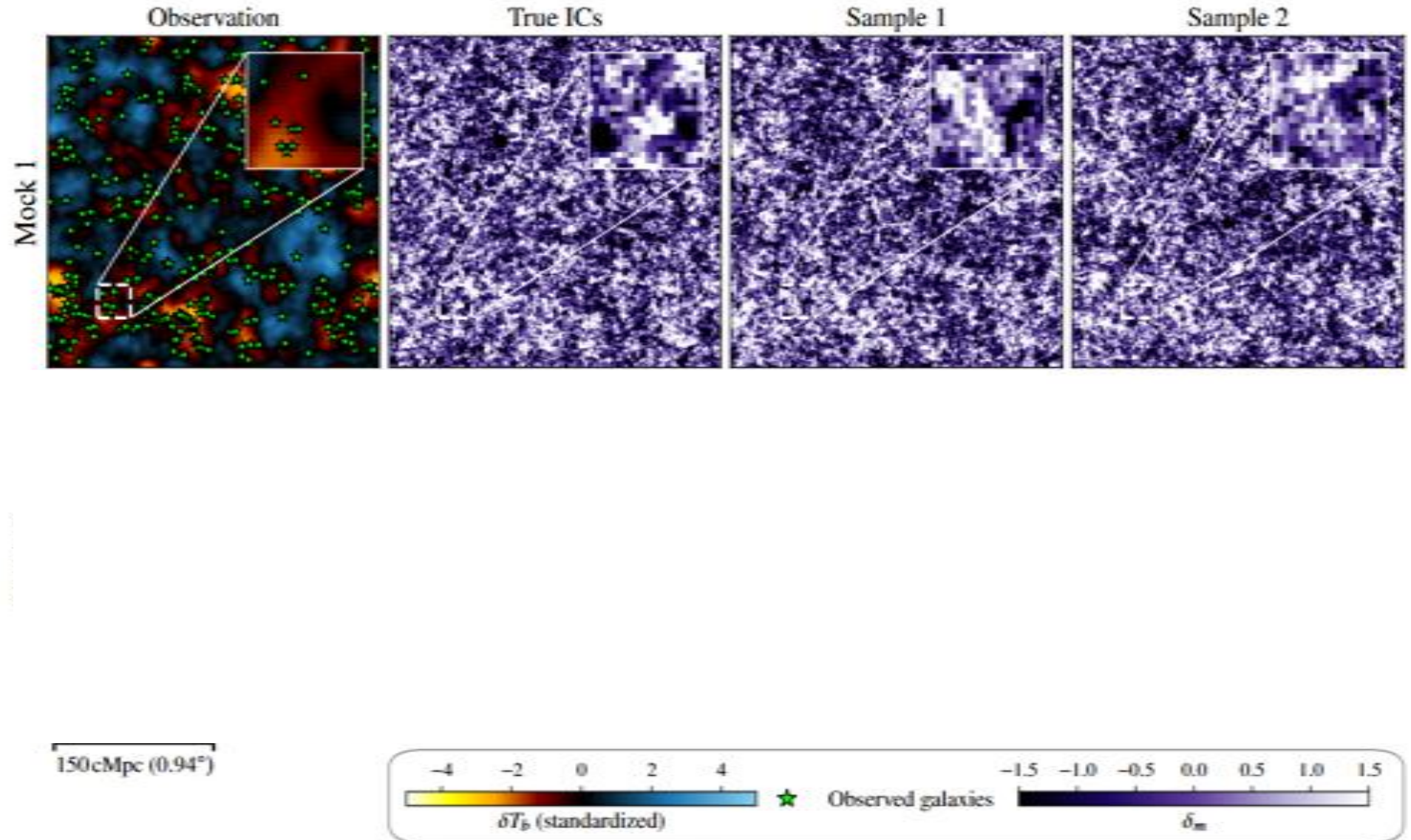
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

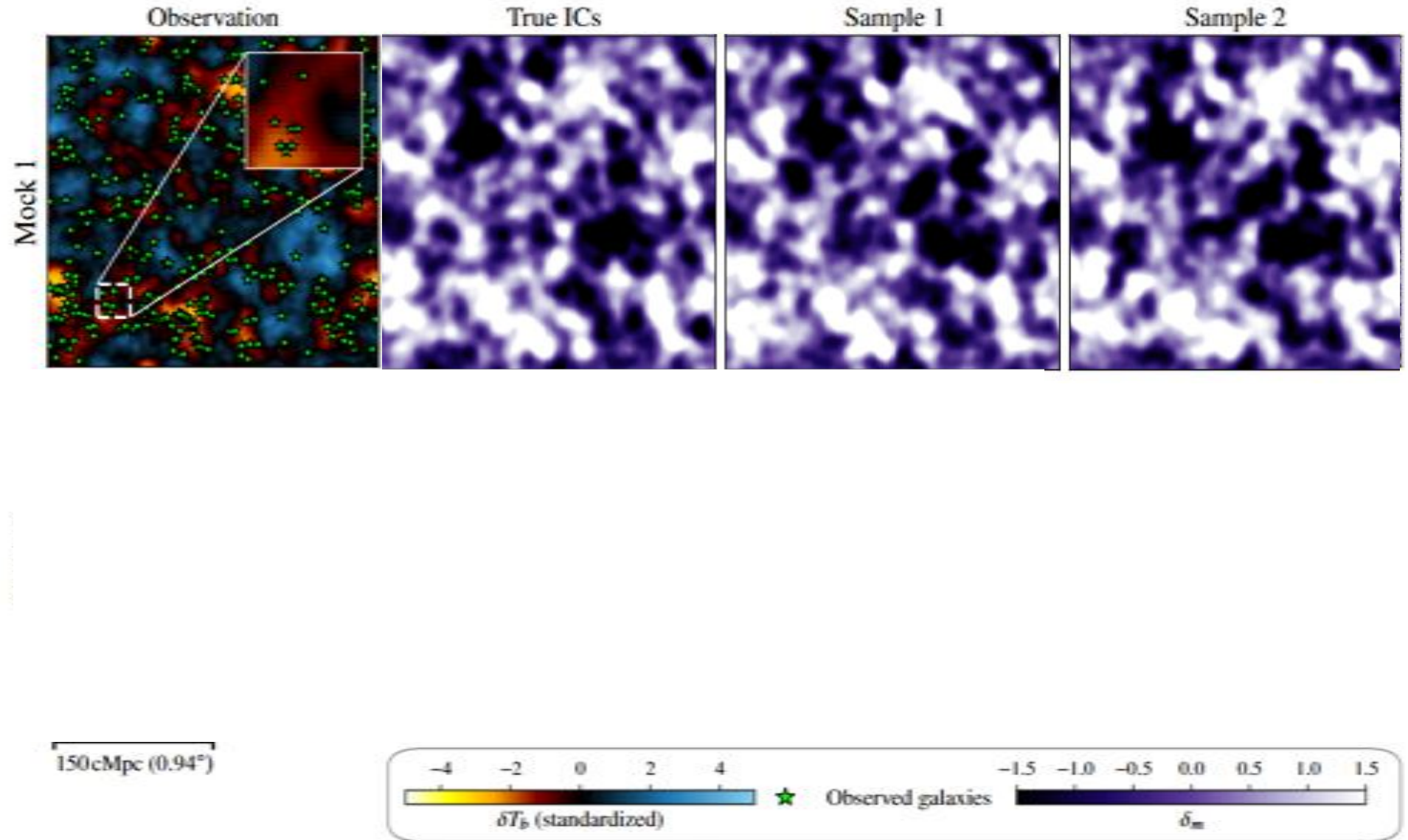
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

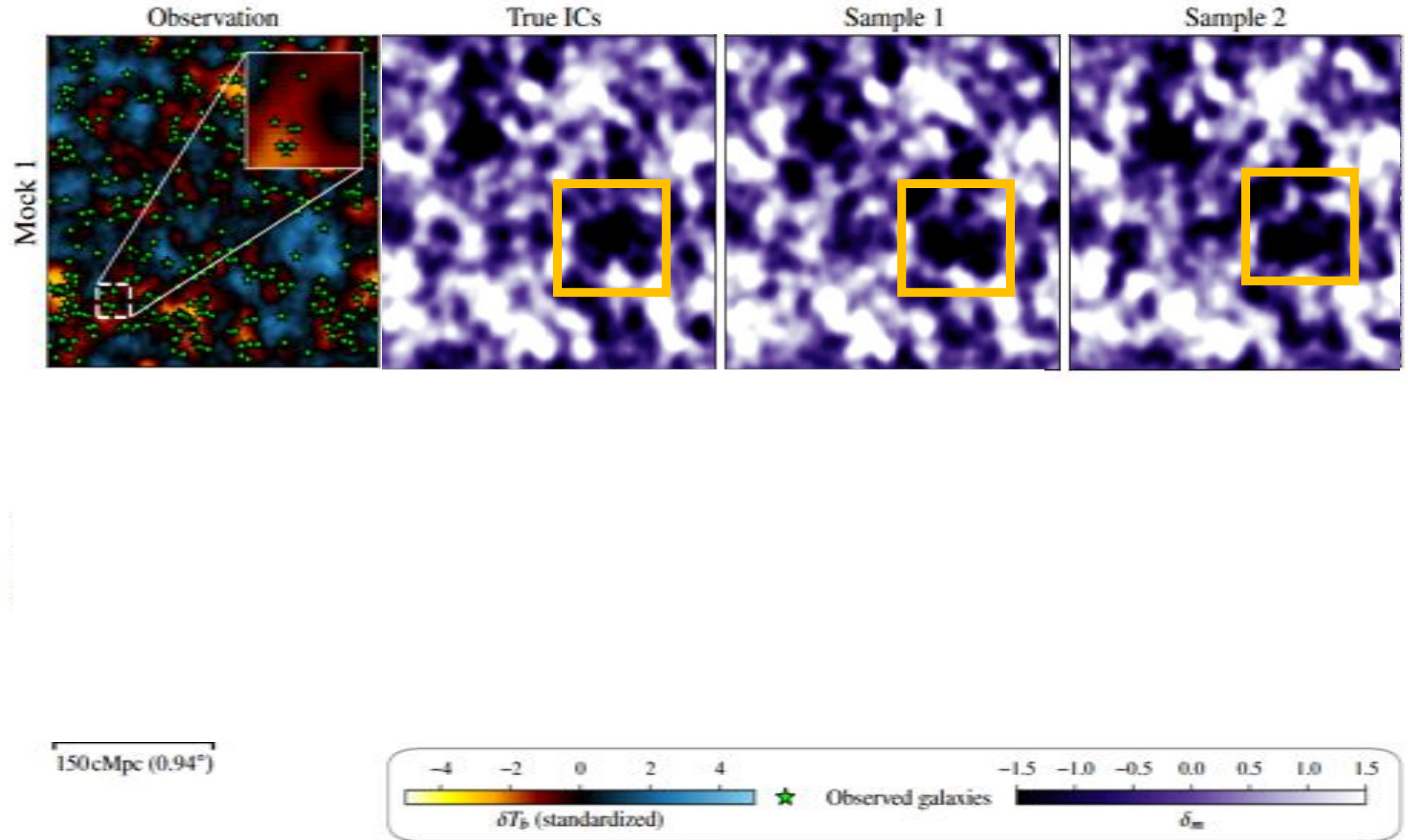
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

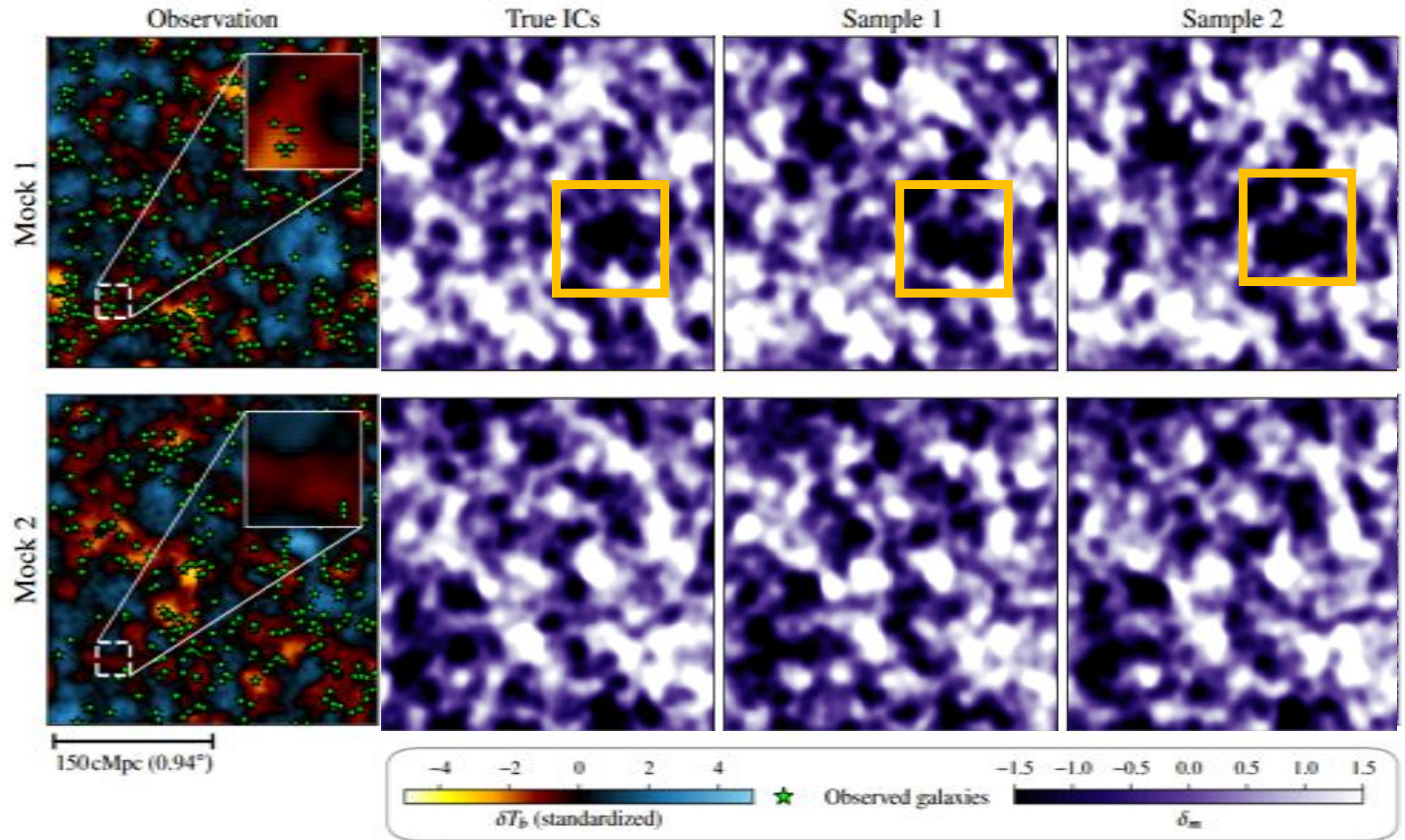
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

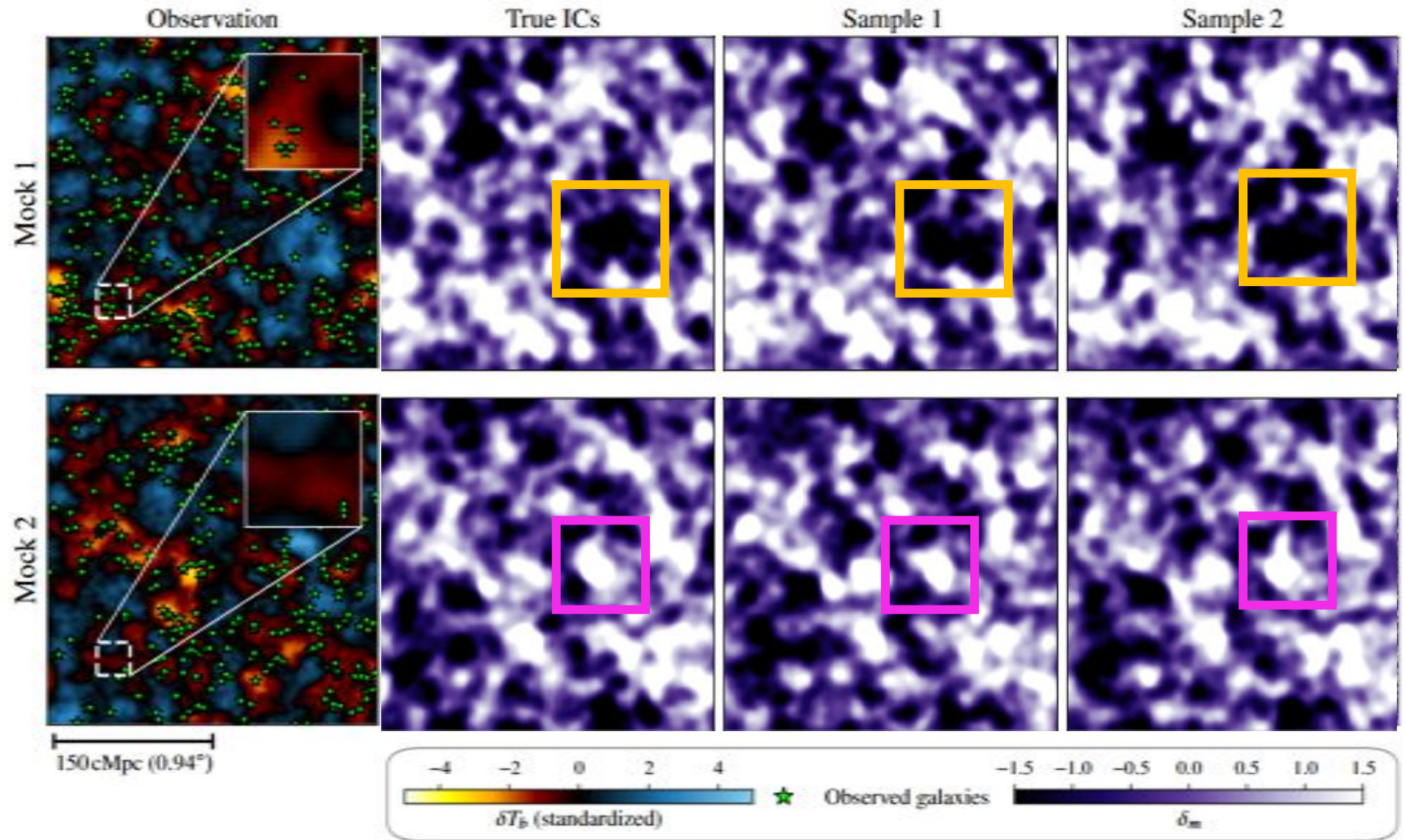
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

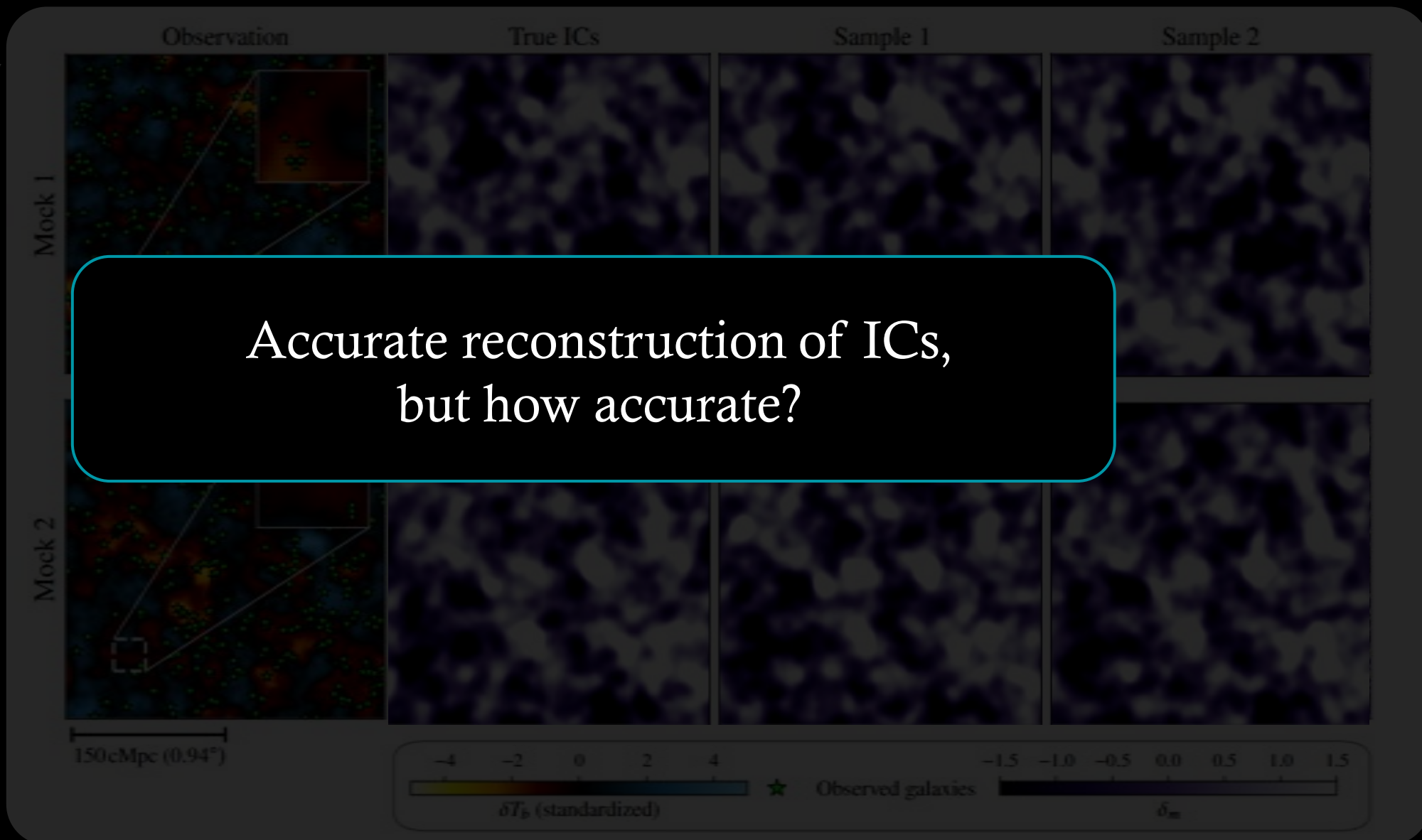
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

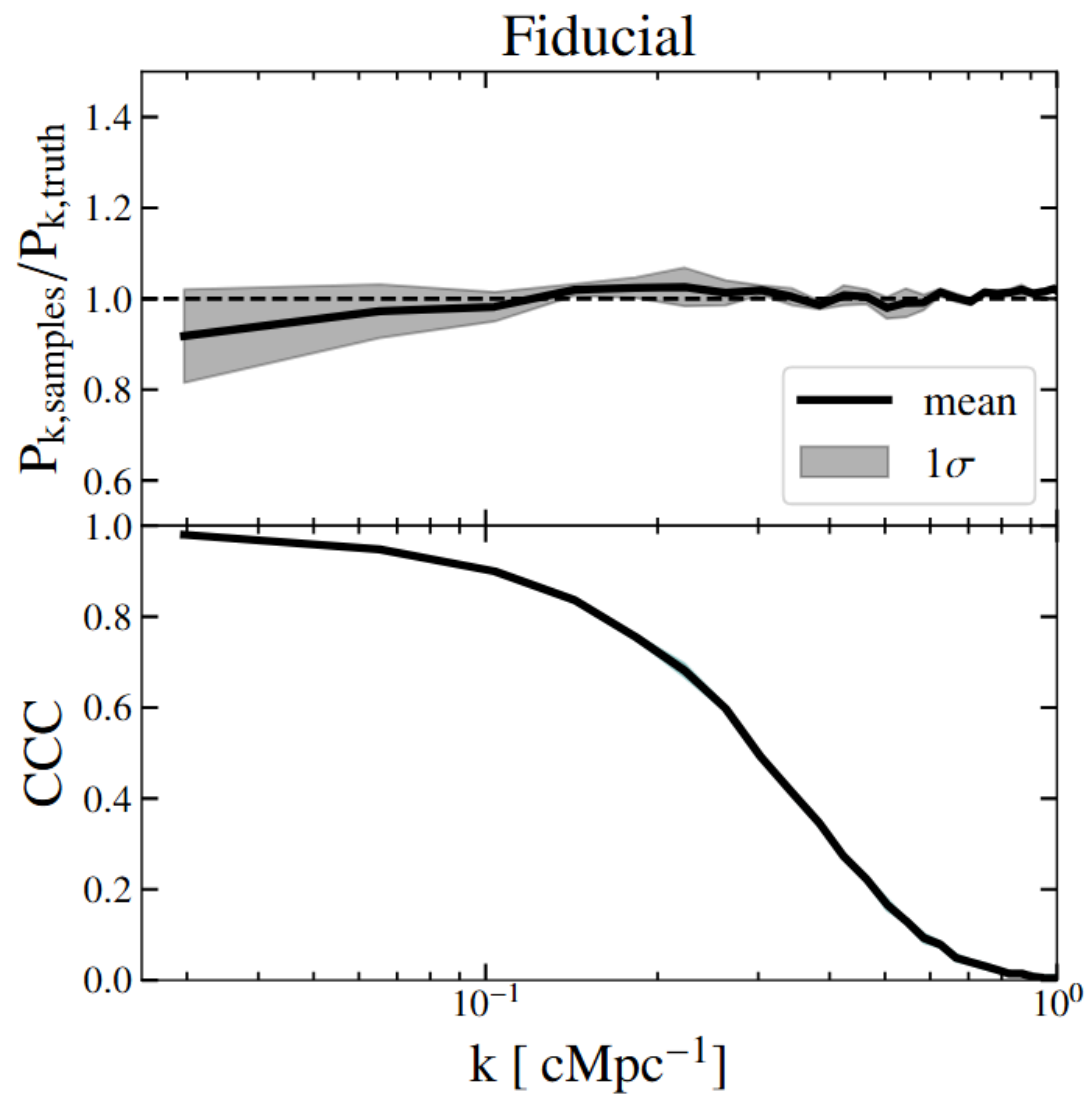
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

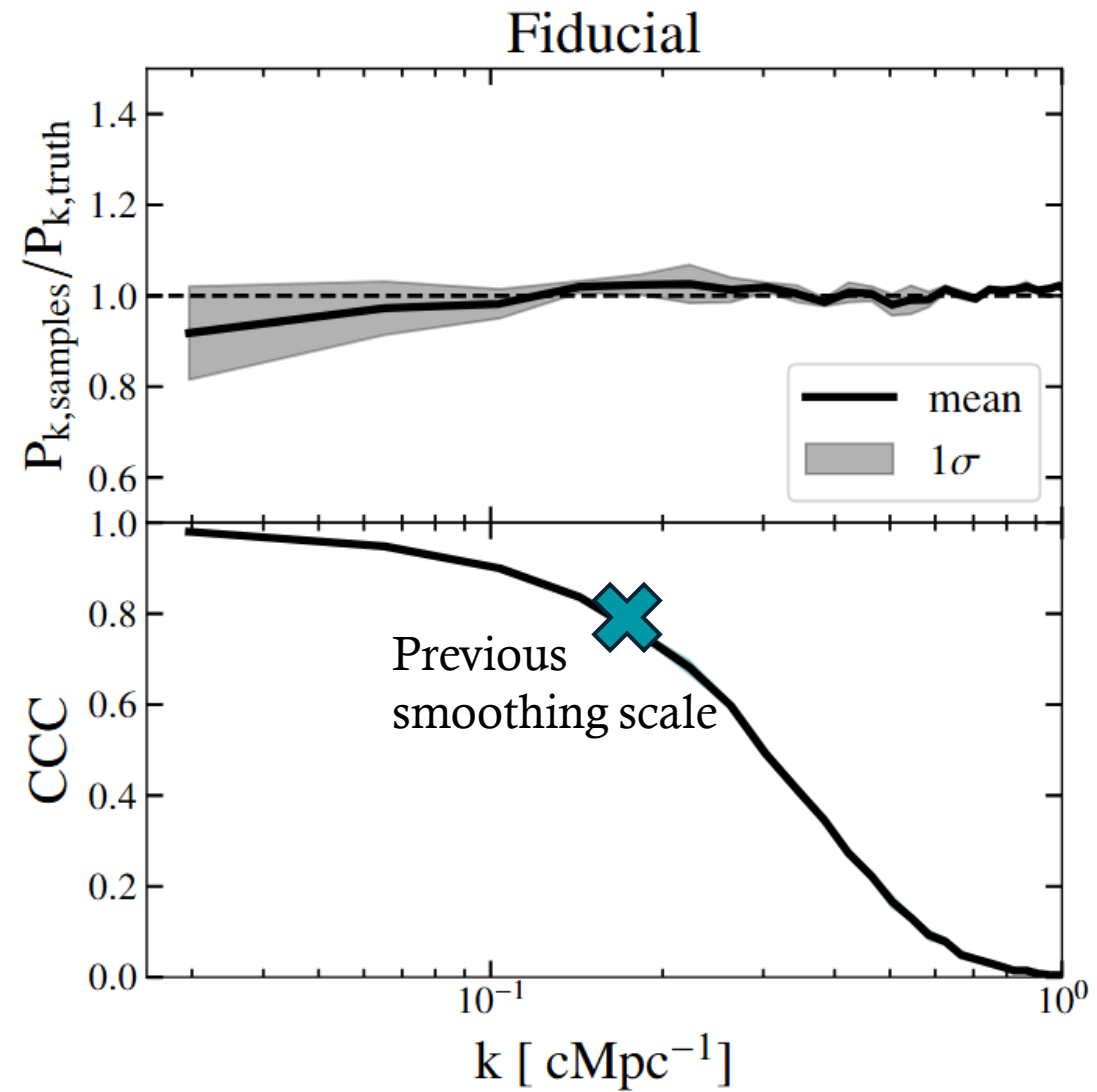
- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$



Results

Fiducial

- 1000h SKA w/ foregrounds
- $M_{\text{uv}} < -18$

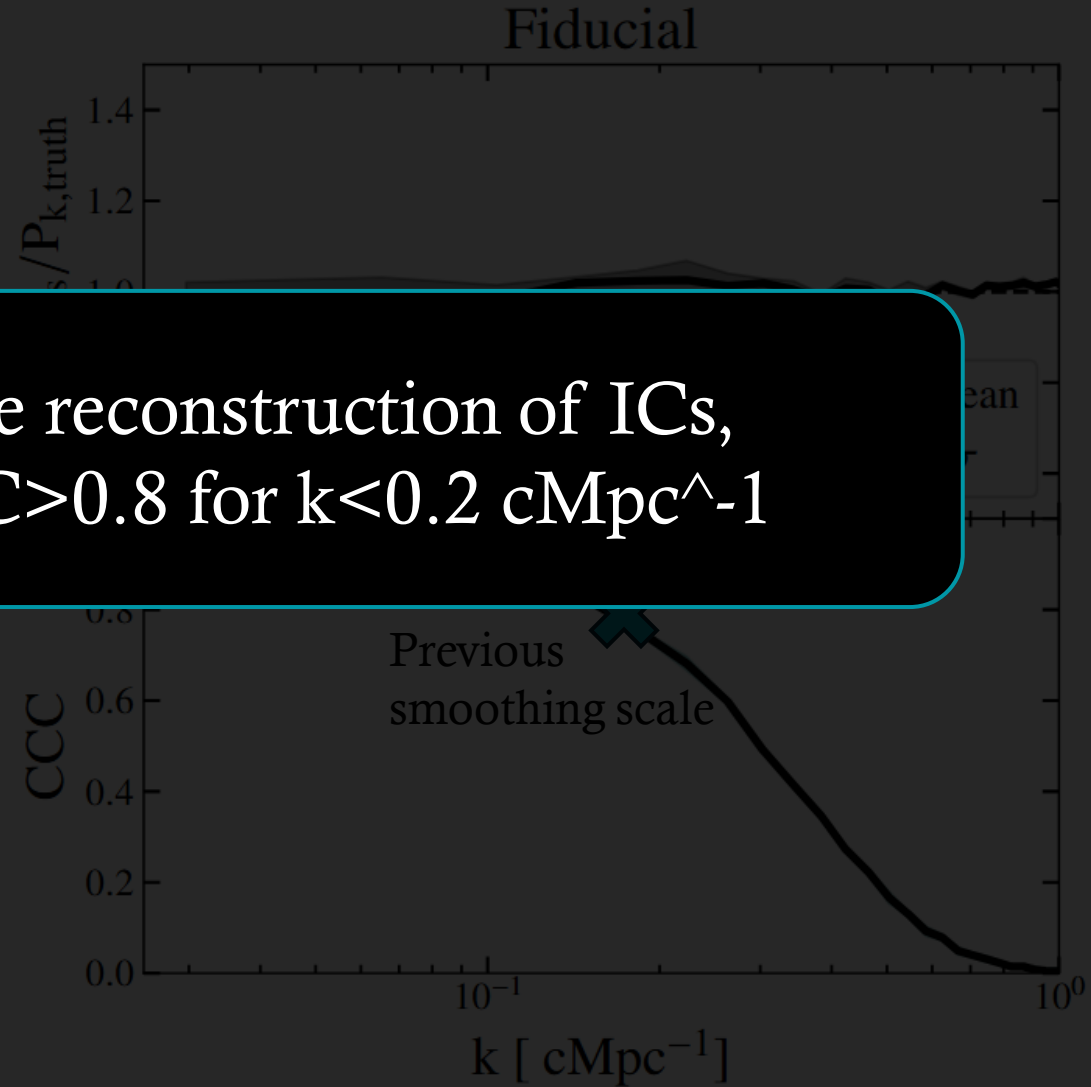


Results

Fiducial

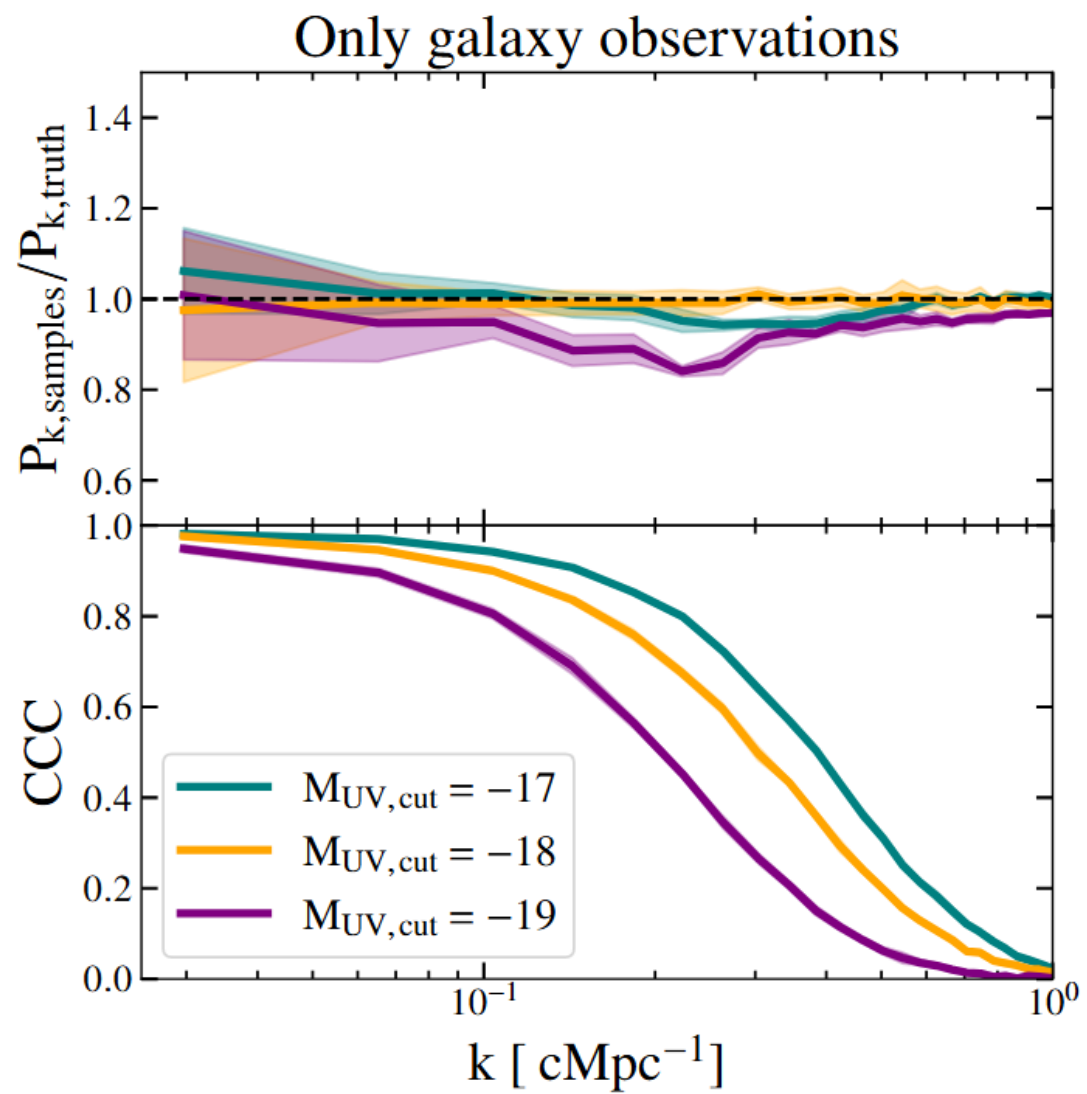
- 1000h SKA w/ foregrounds
- $M_{UV} < -18$

Accurate reconstruction of ICs,
with $CCC > 0.8$ for $k < 0.2 \text{ cMpc}^{-1}$



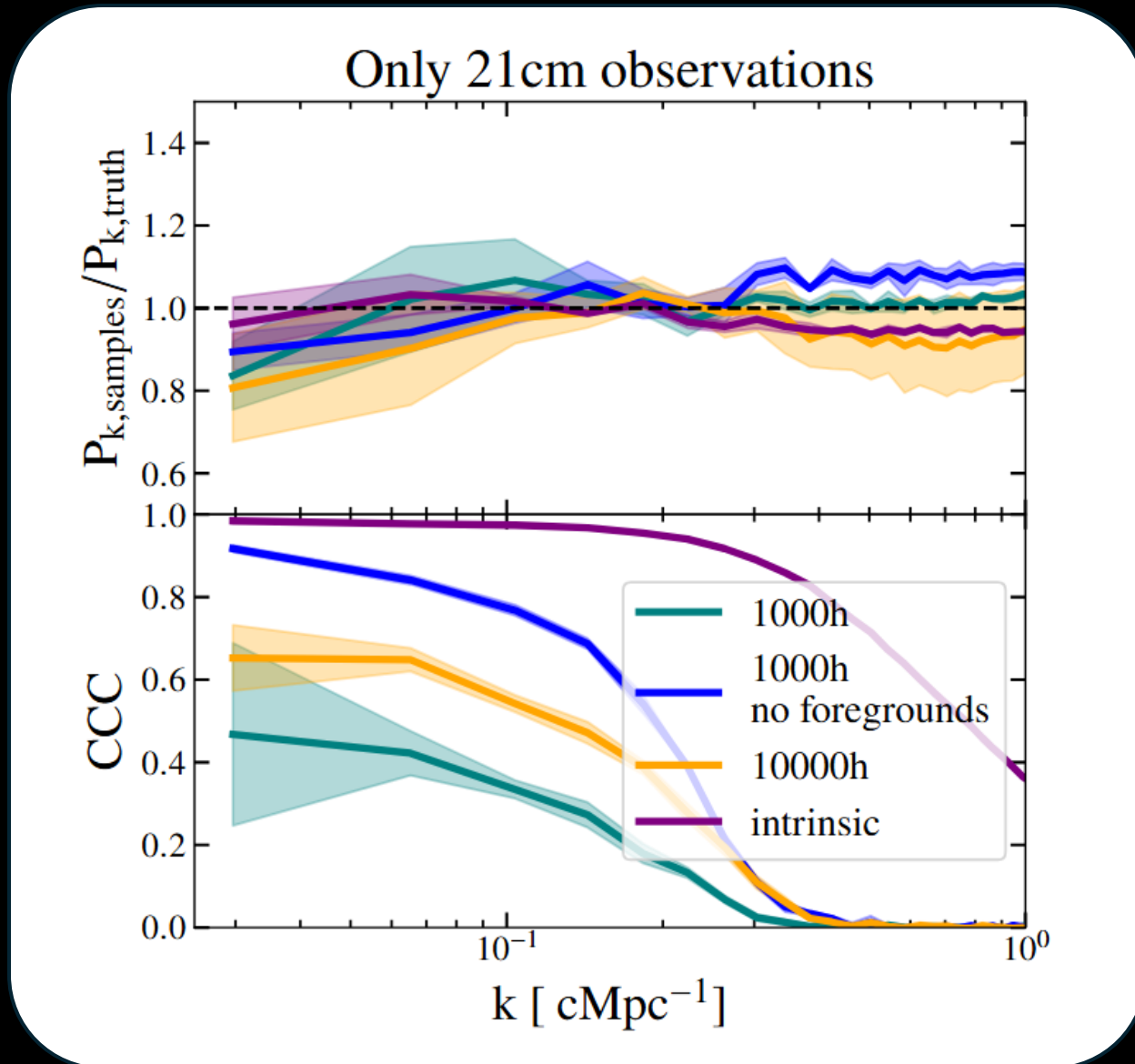
Results

Only observing one tracer



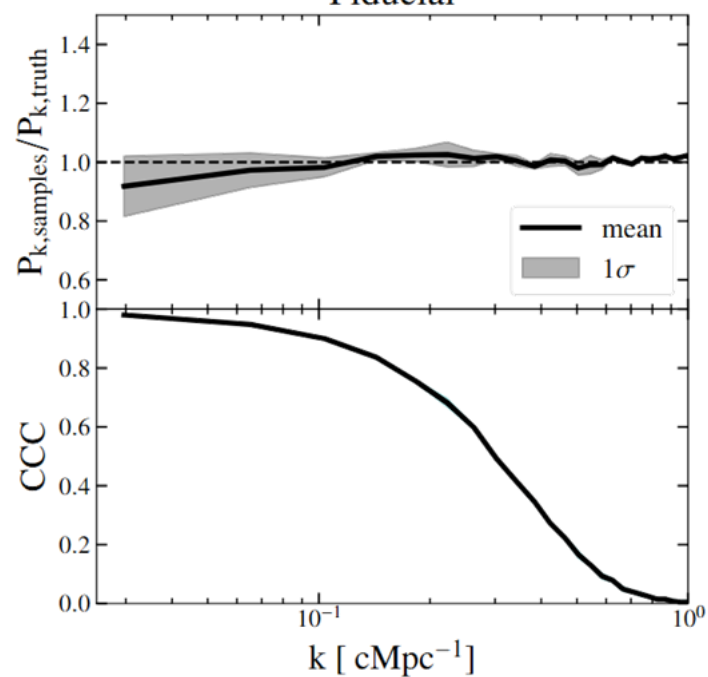
Results

Only observing one tracer

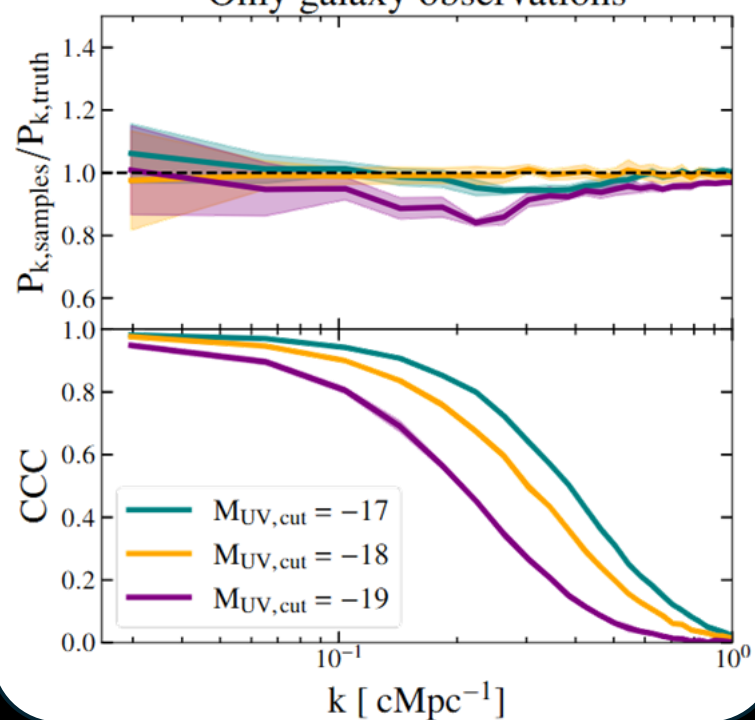


Results

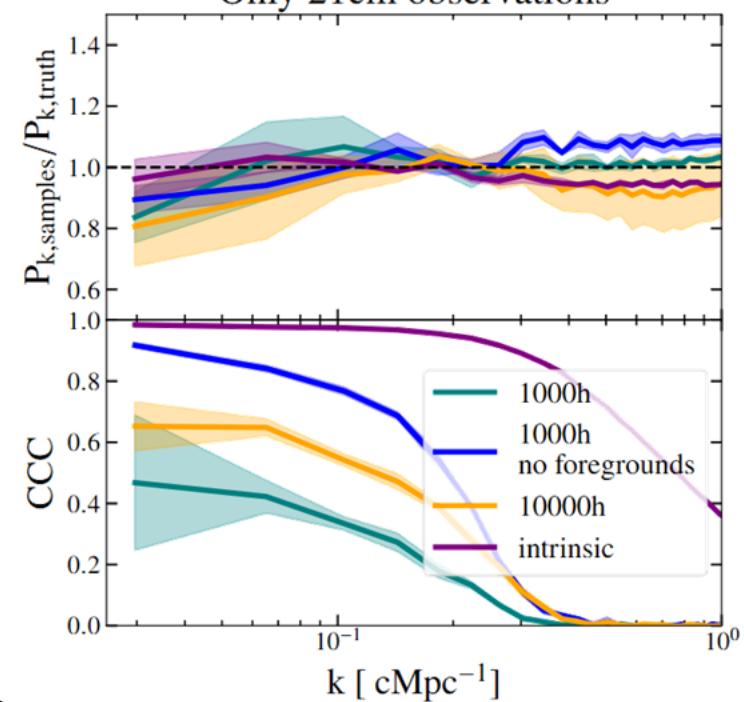
Fiducial



Only galaxy observations



Only 21cm observations



Consequences

Getting the full history of the Universe

• The *cosmological principle* is the assumption that the Universe is homogeneous and isotropic on large scales

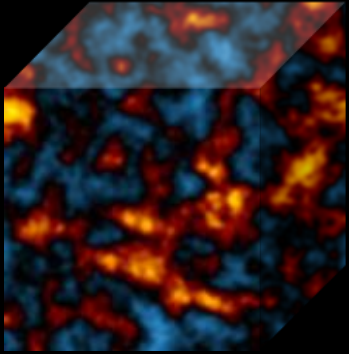
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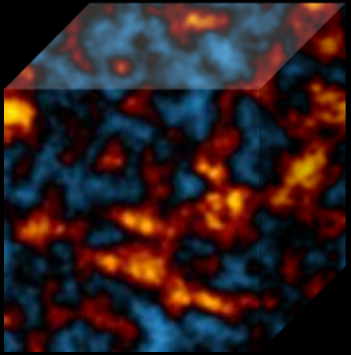
Getting the full history of the Universe

Mock
observation

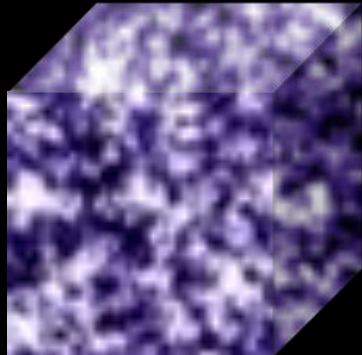


Getting the full history of the Universe

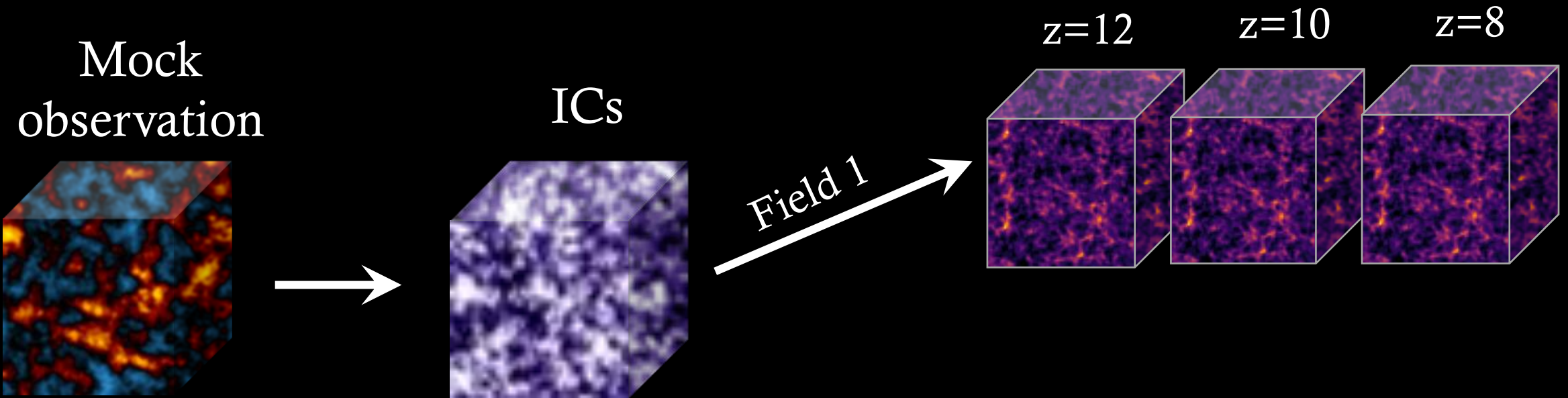
Mock
observation



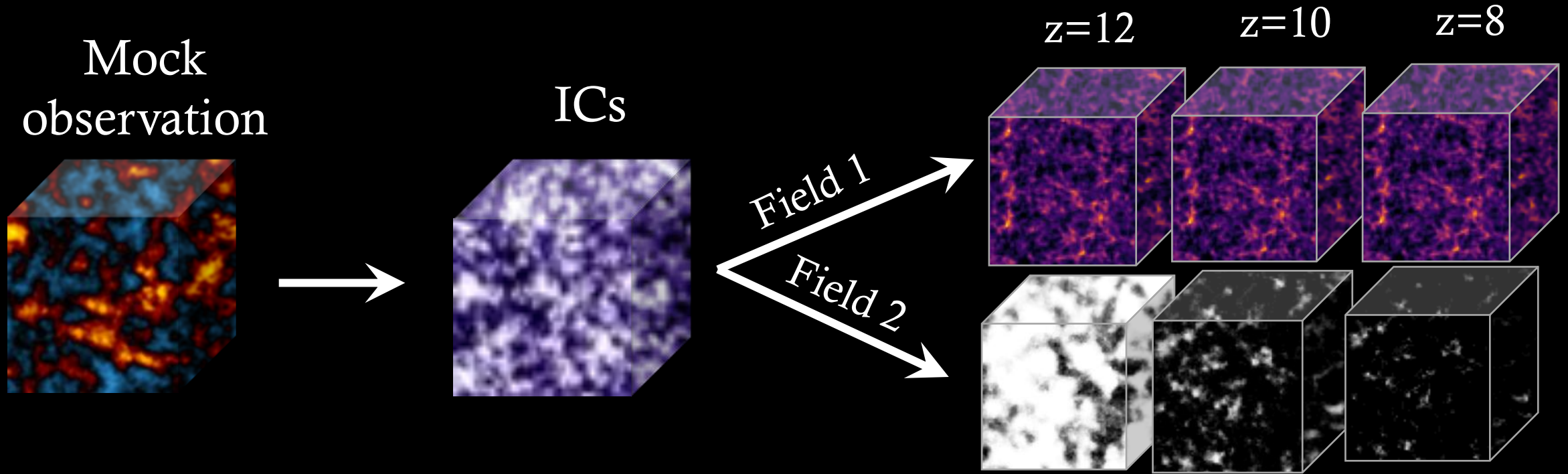
ICs



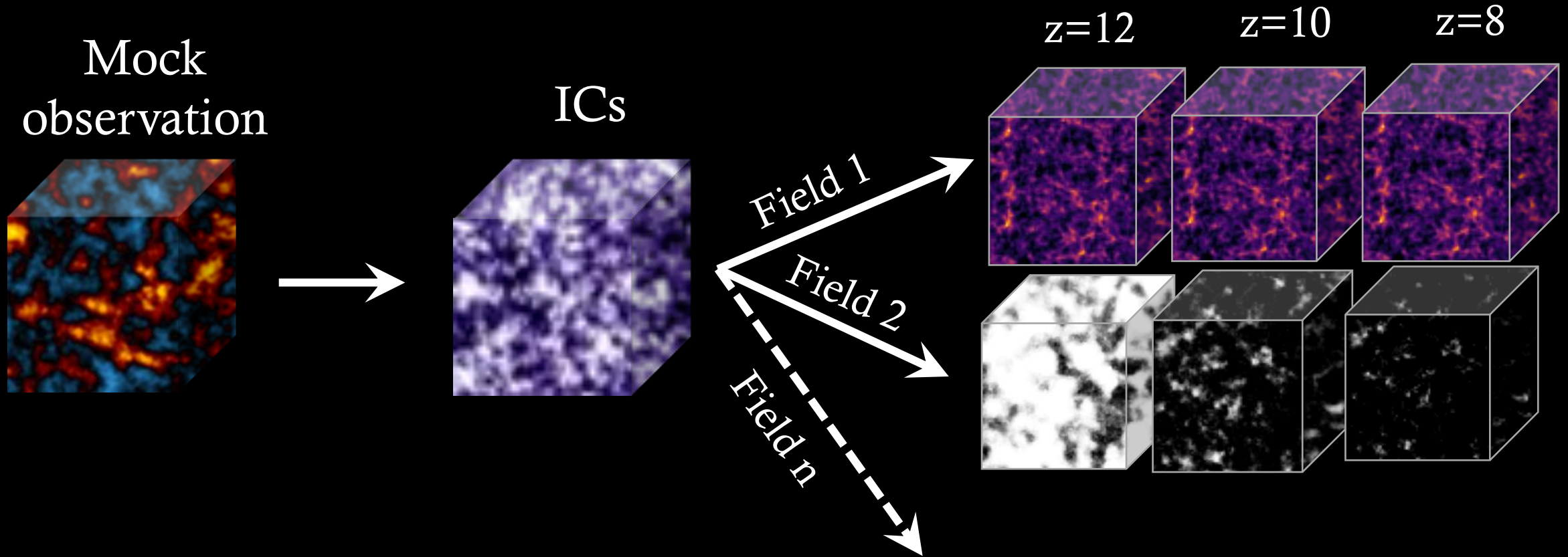
Getting the full history of the Universe



Getting the full history of the Universe



Getting the full history of the Universe



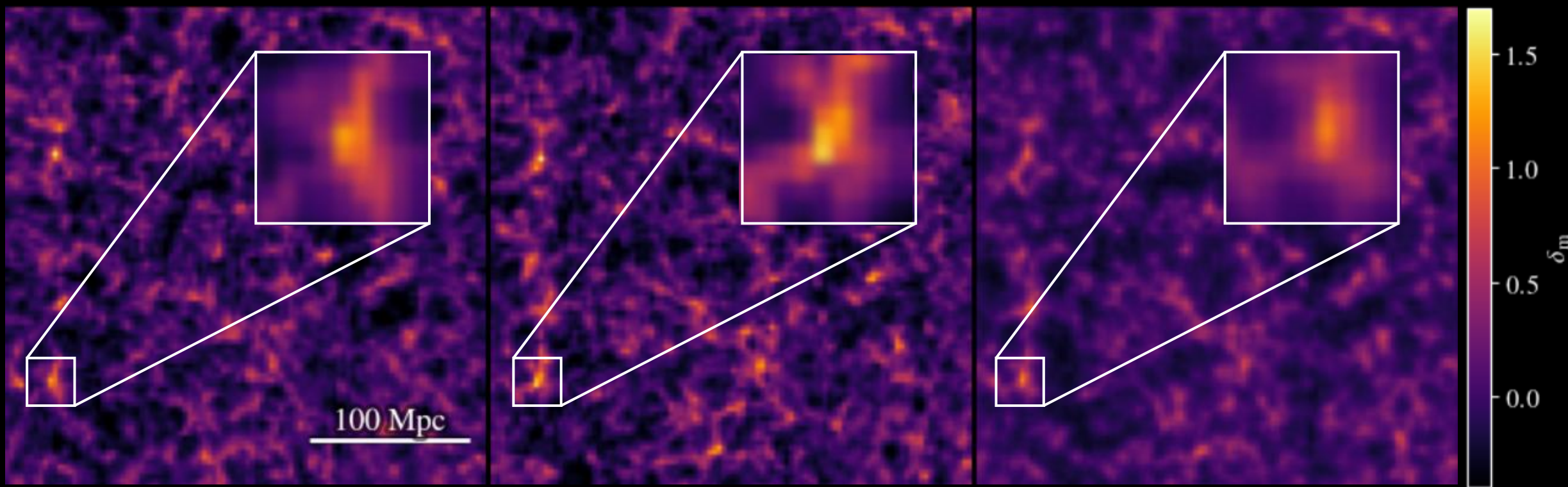
Reconstructed density field

$z=8$

True

Sample

Mean of samples



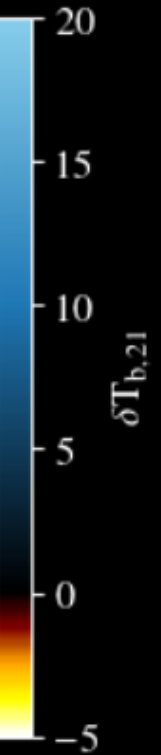
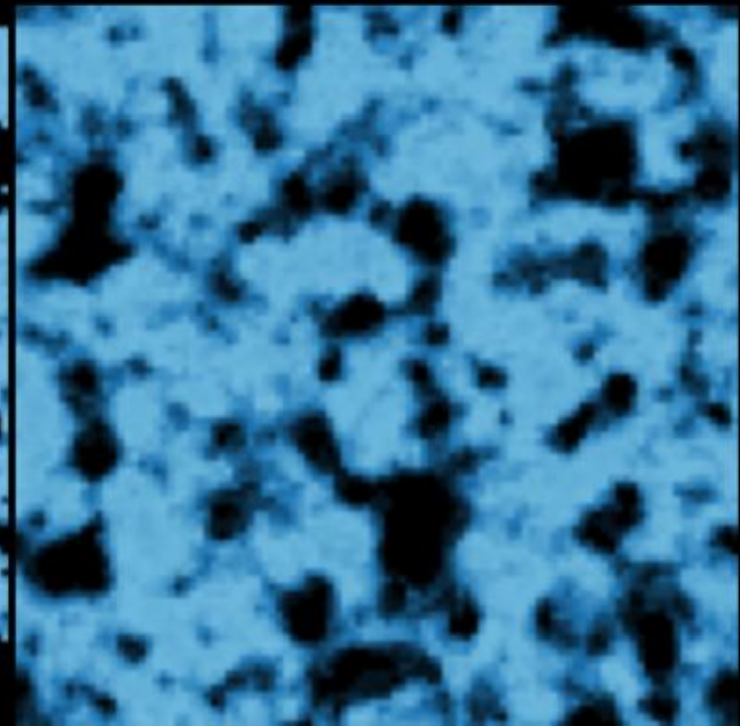
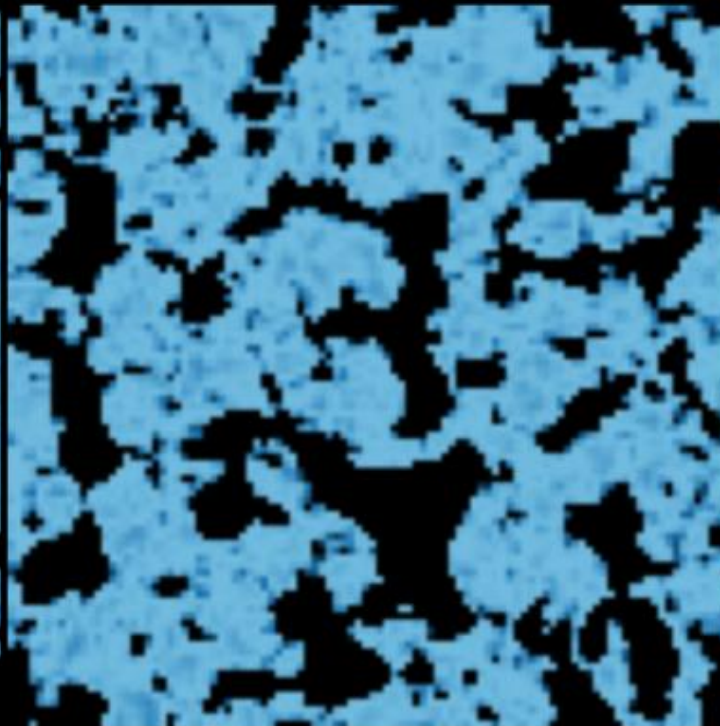
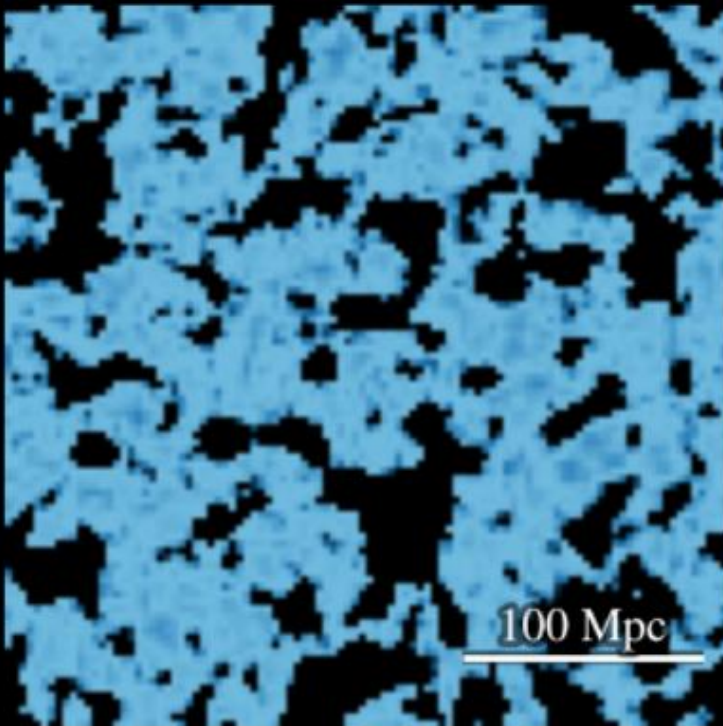
Reconstructed 21cm signal

$z=8$

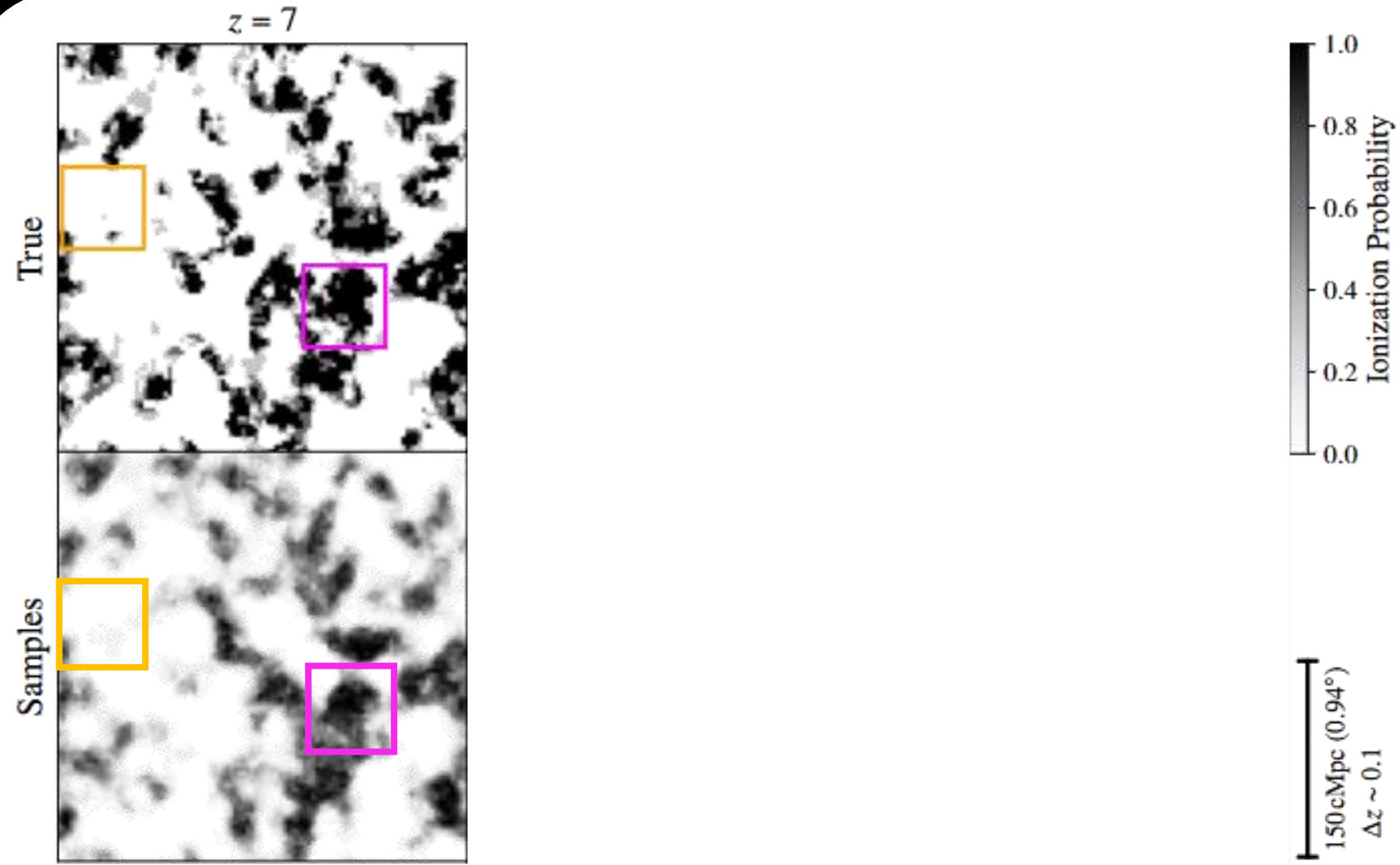
True

Sample

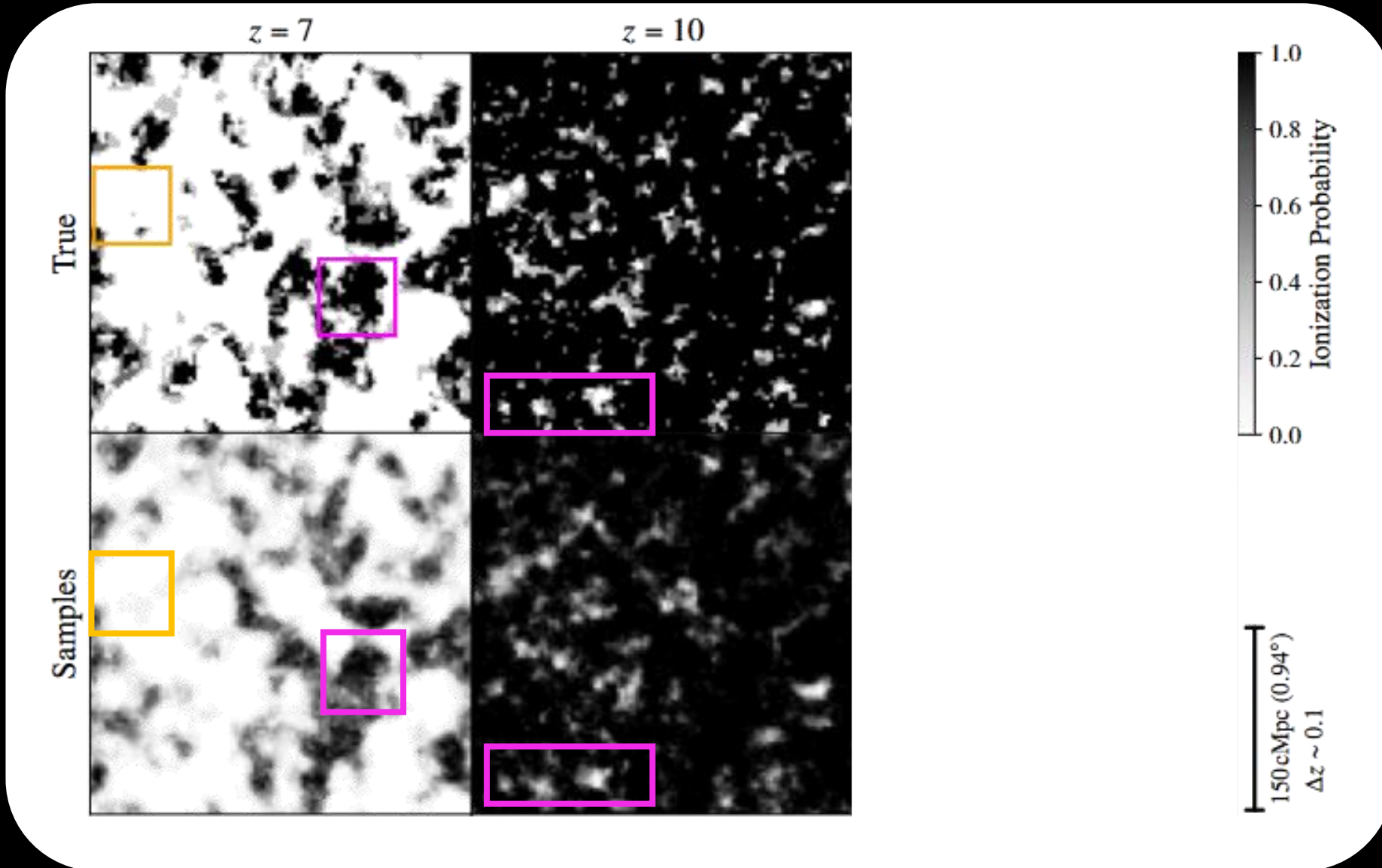
Mean of samples



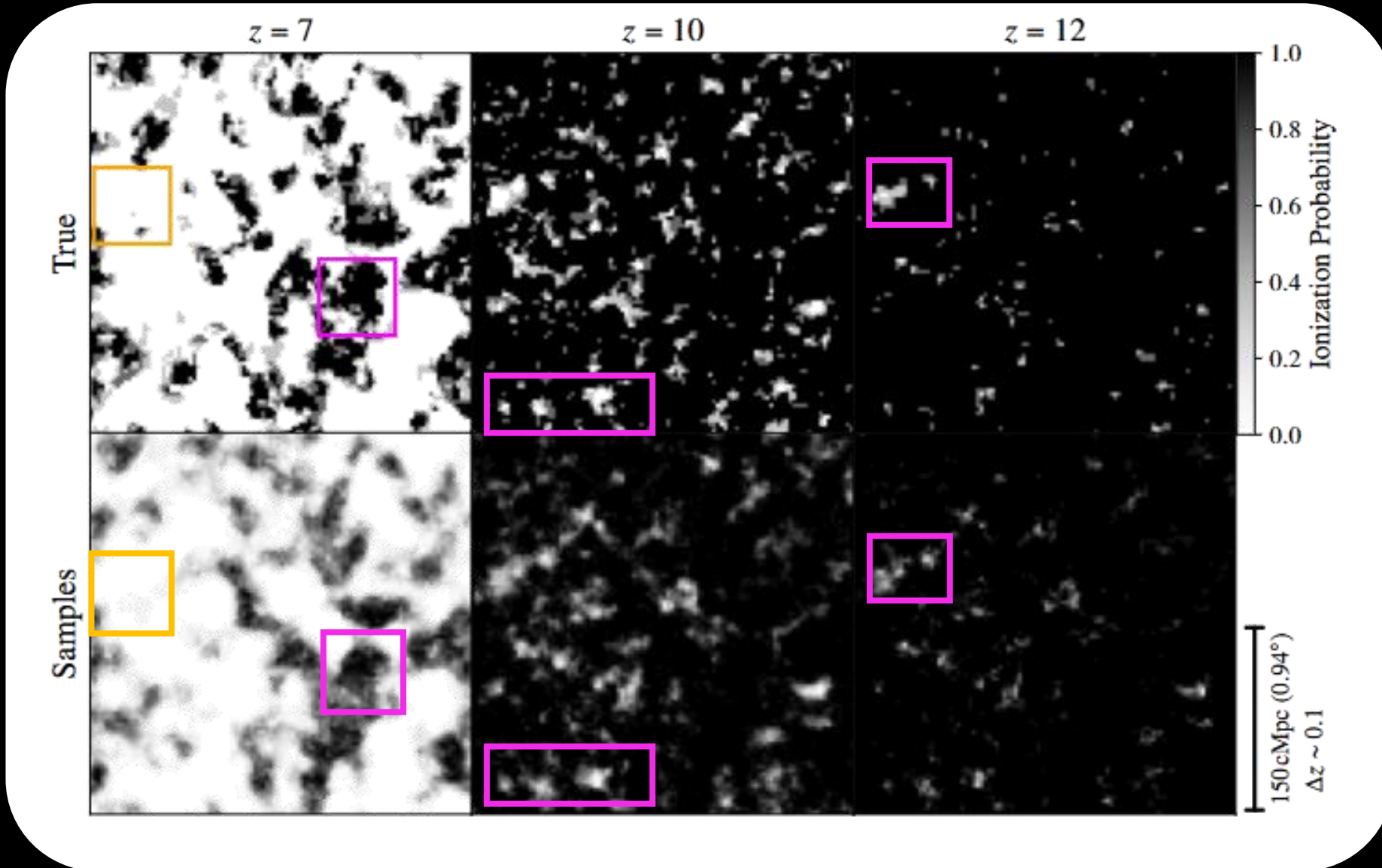
Ionization history



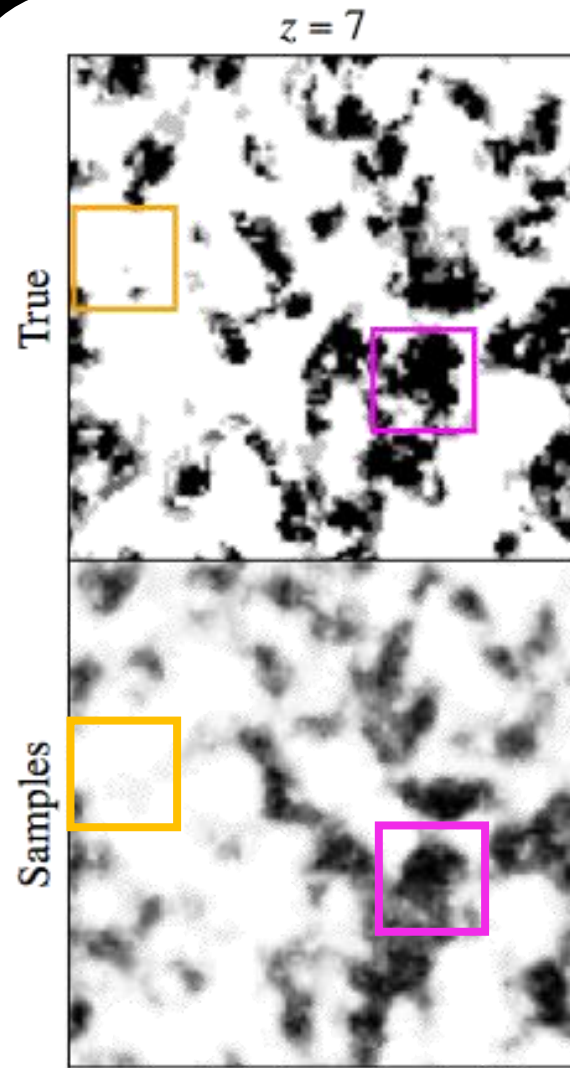
Ionization history



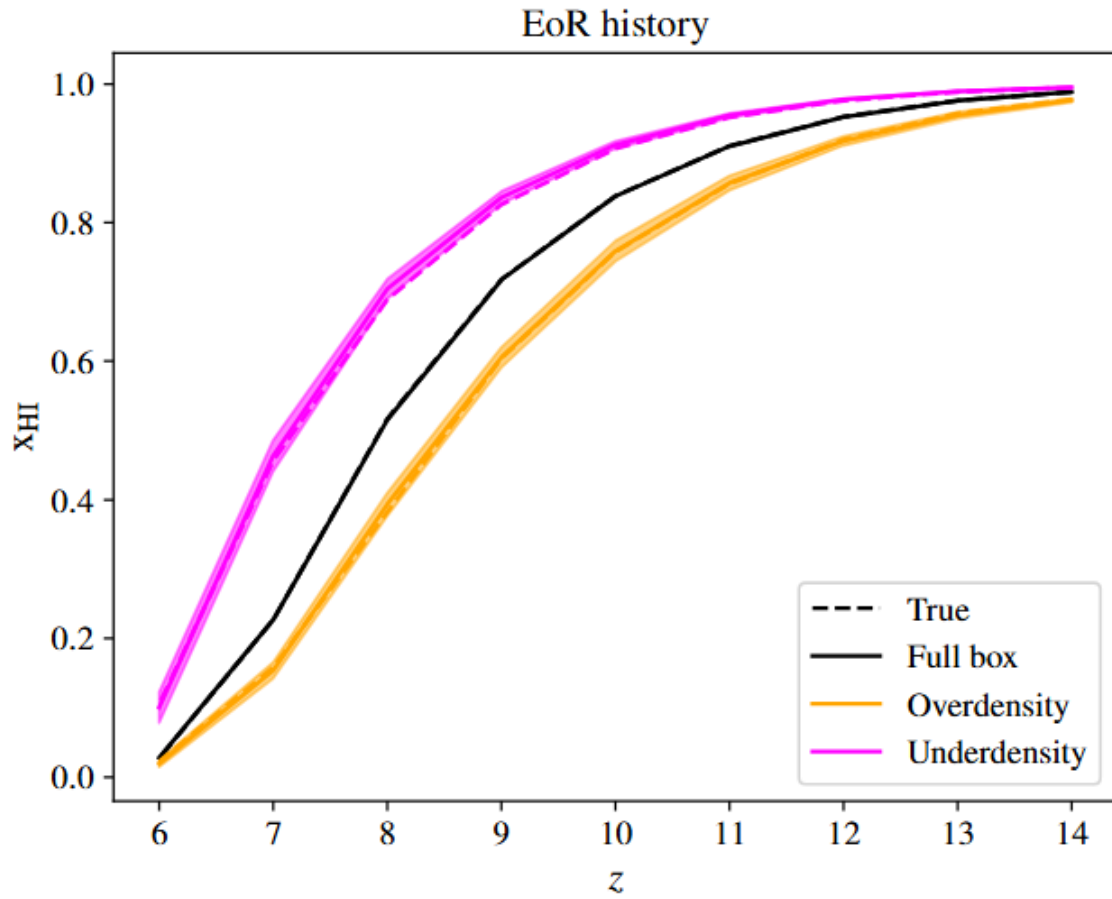
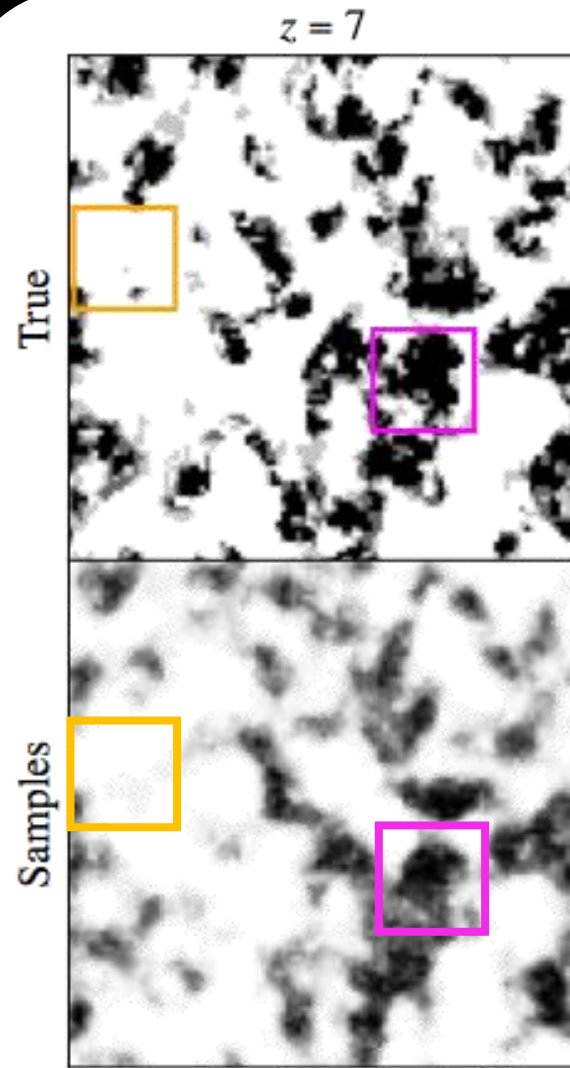
Ionization history



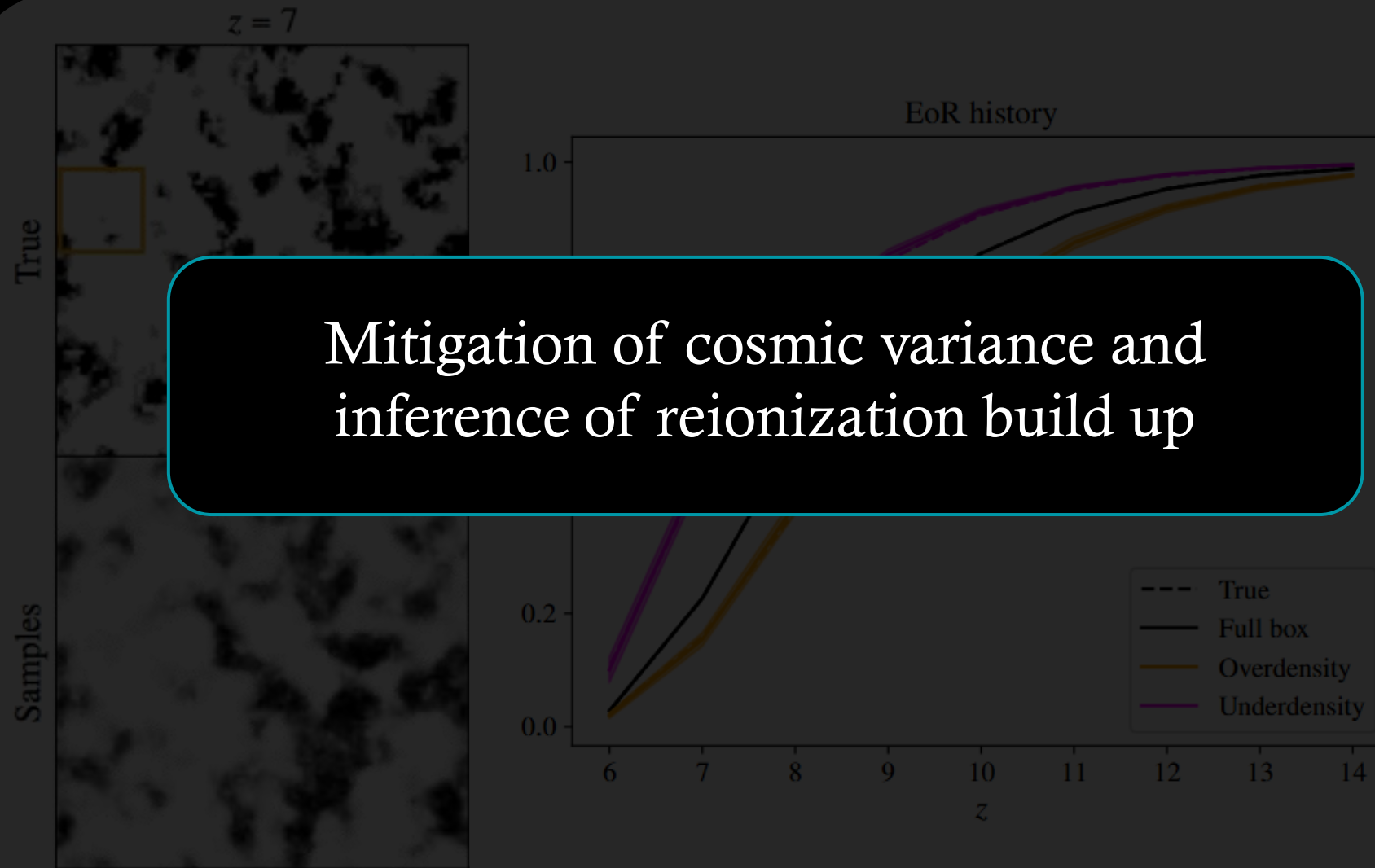
Ionization history



Ionization history

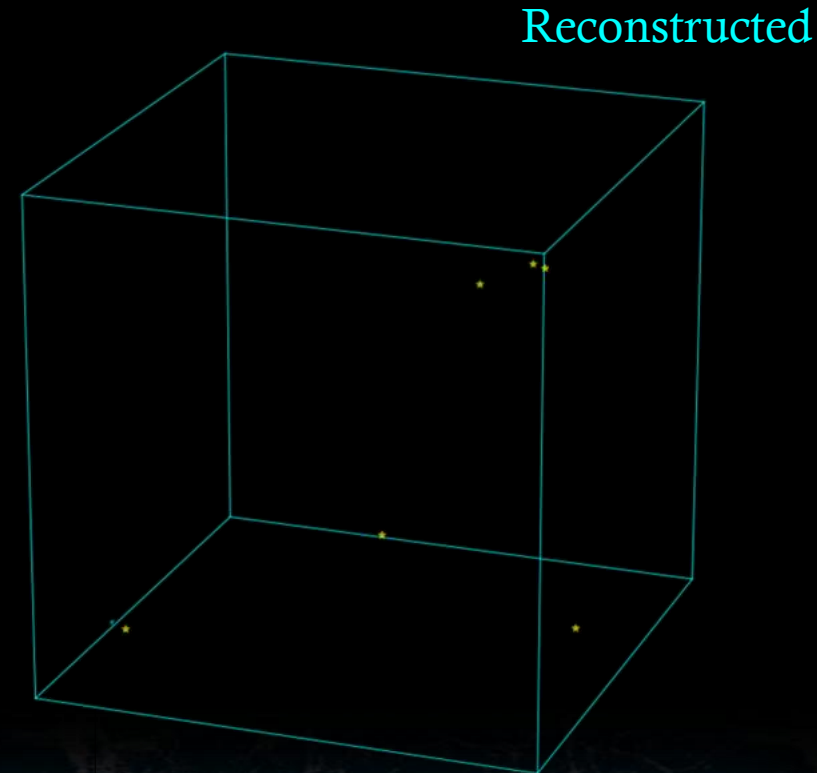
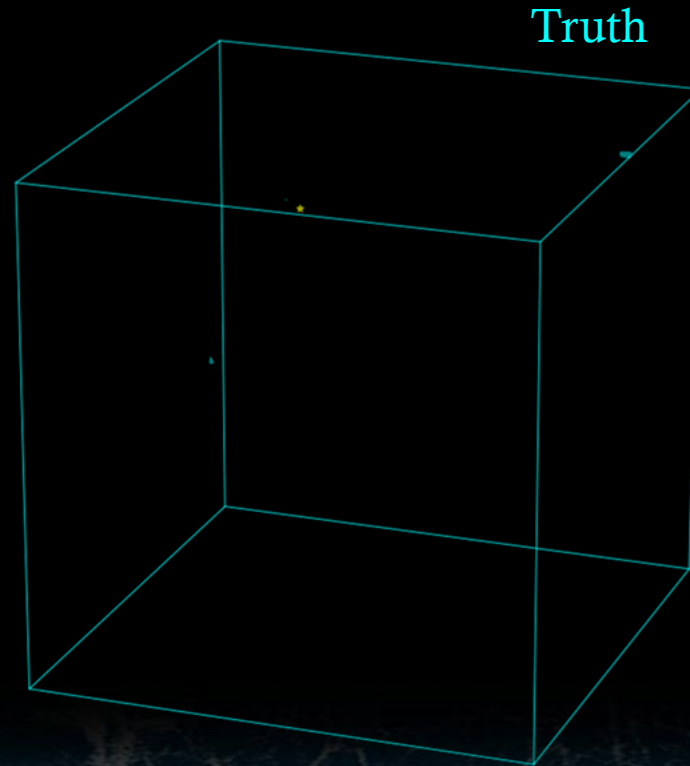


Ionization history



Summary

- ICs scales $k < 0.2 \text{ cMpc}^{-1}$ can be reconstructed with $\text{CCC} > 0.8$
- Build up of EoR morphology can be reconstructed with $< 2\%$ error on EoR history ($< 1\sigma$) for specific 50^3 cMpc^{-3} subvolumes. Mitigating cosmic variance



Extra

Results

Fiducial

- 1000h SKA w/
foregrounds
- $M_{UV} < -18$

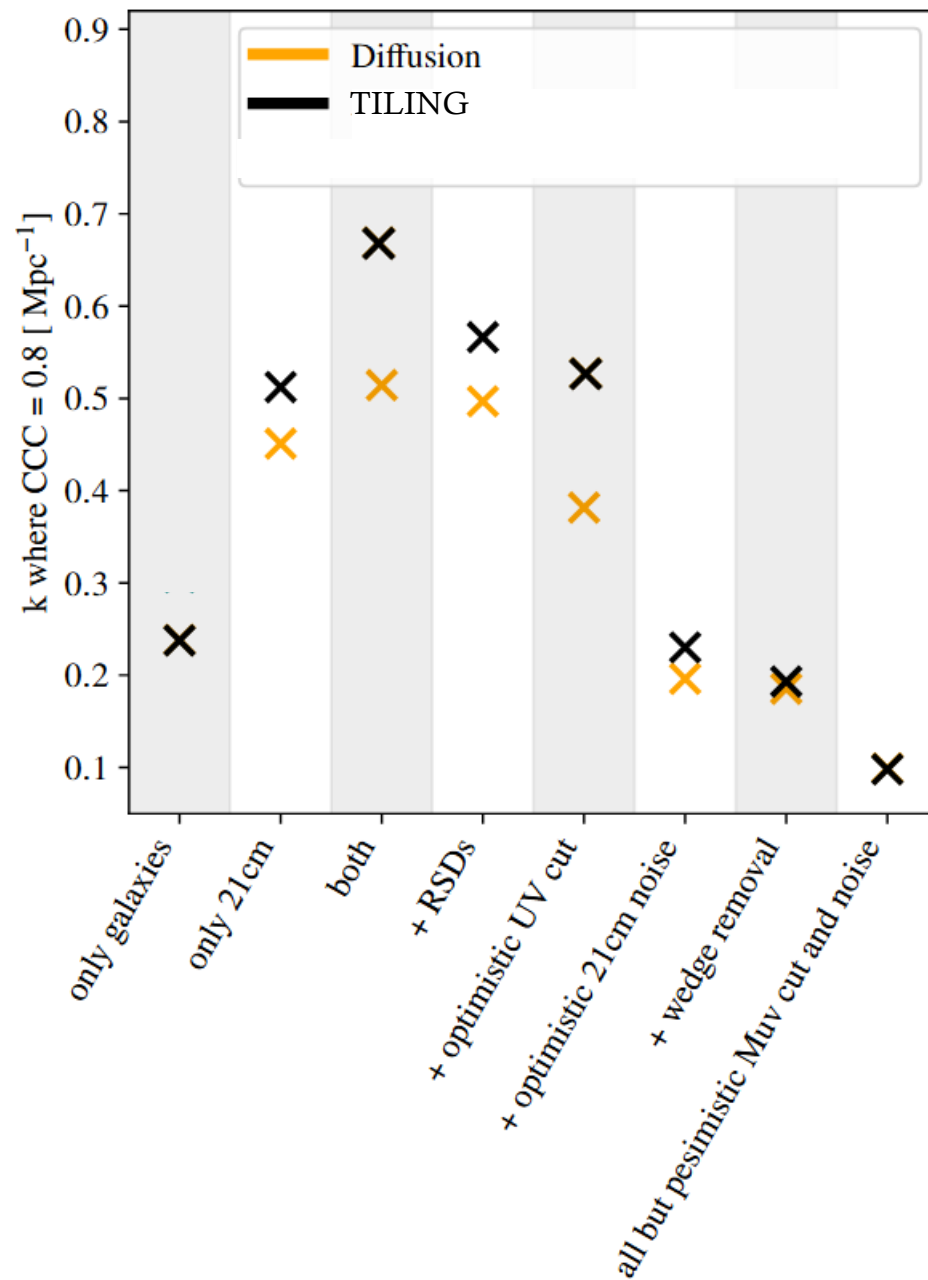
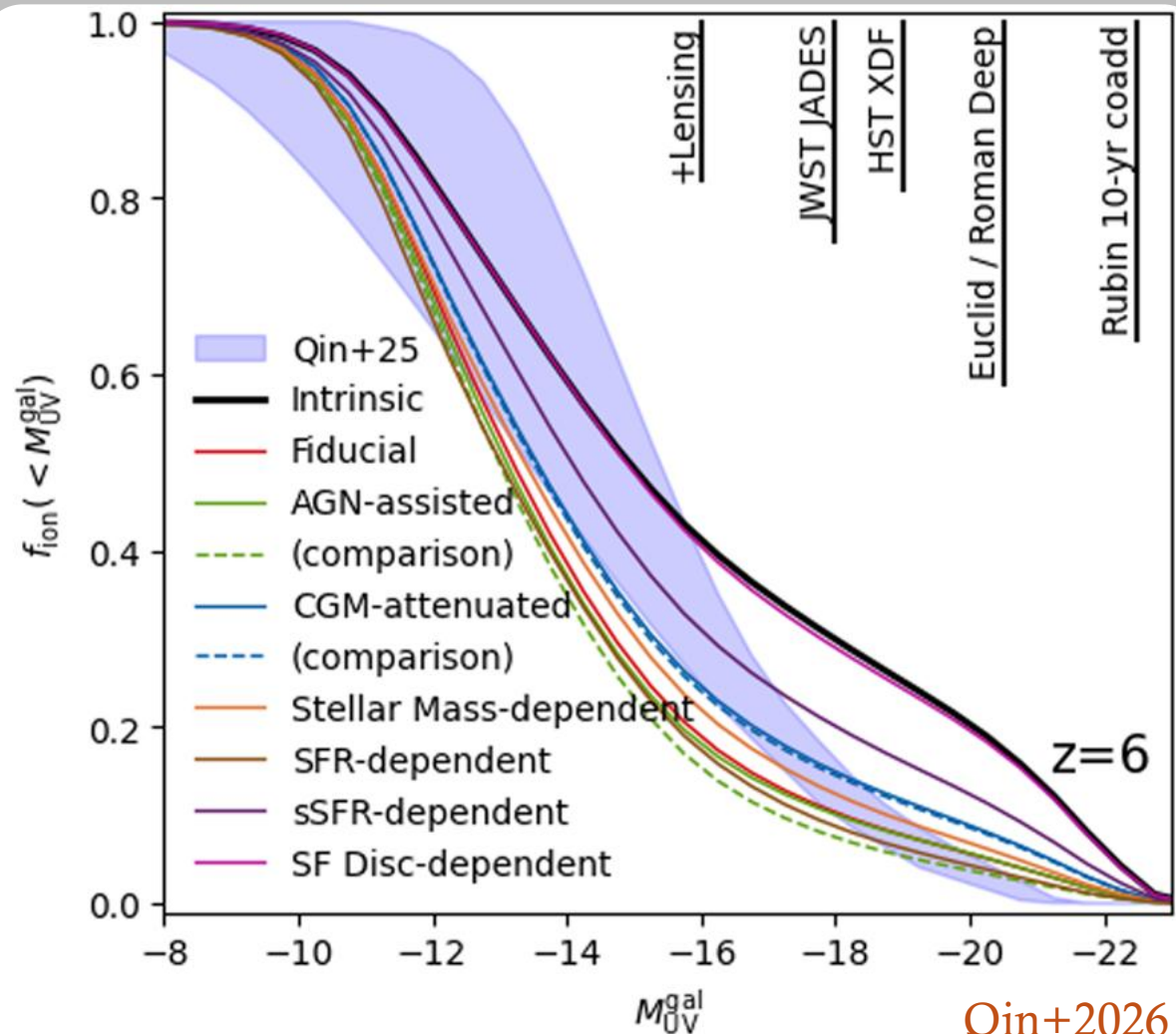


image credit: Cynthia Chiang

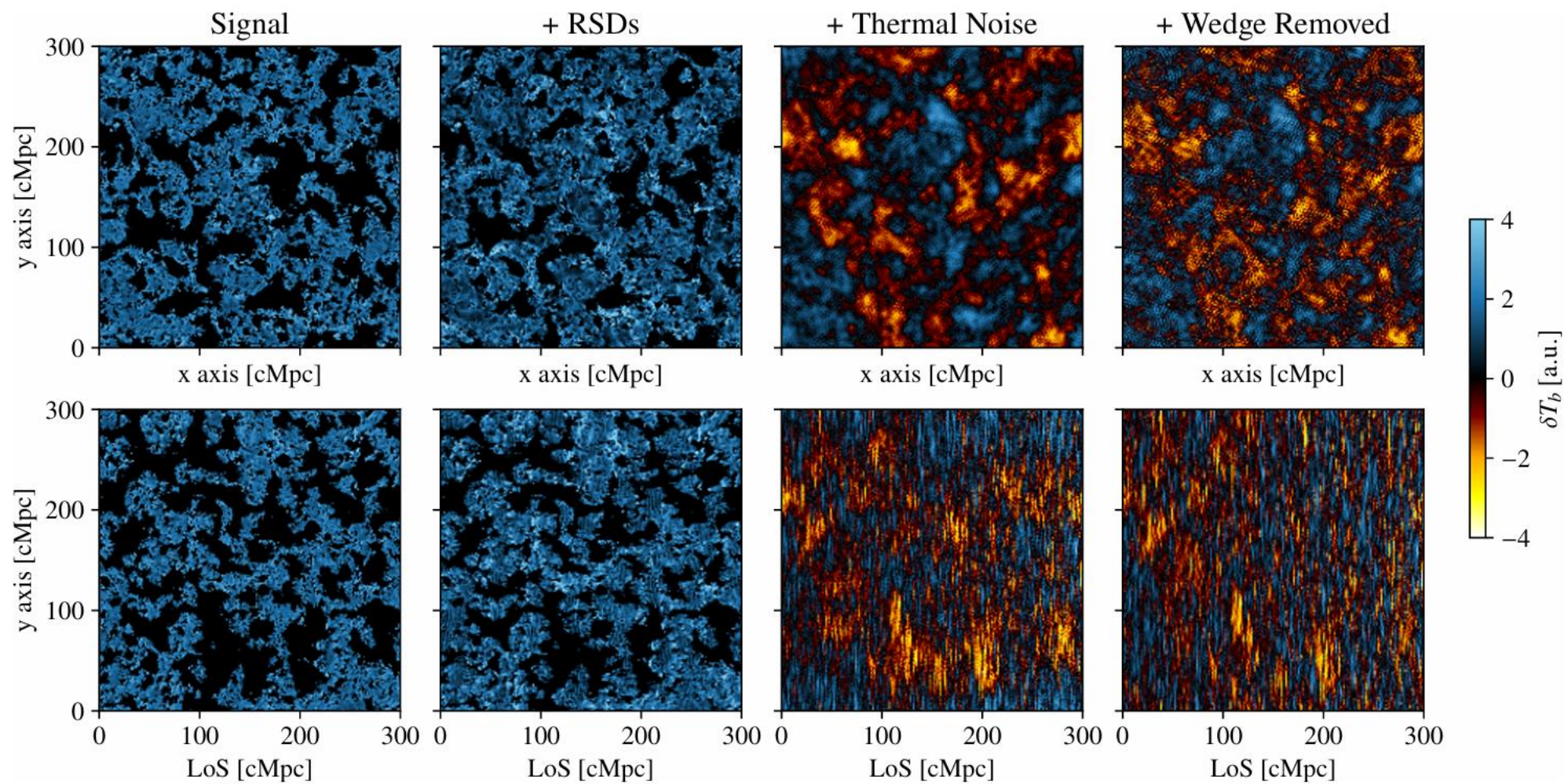
The ones we don't see are likely
the **drivers of reionization**

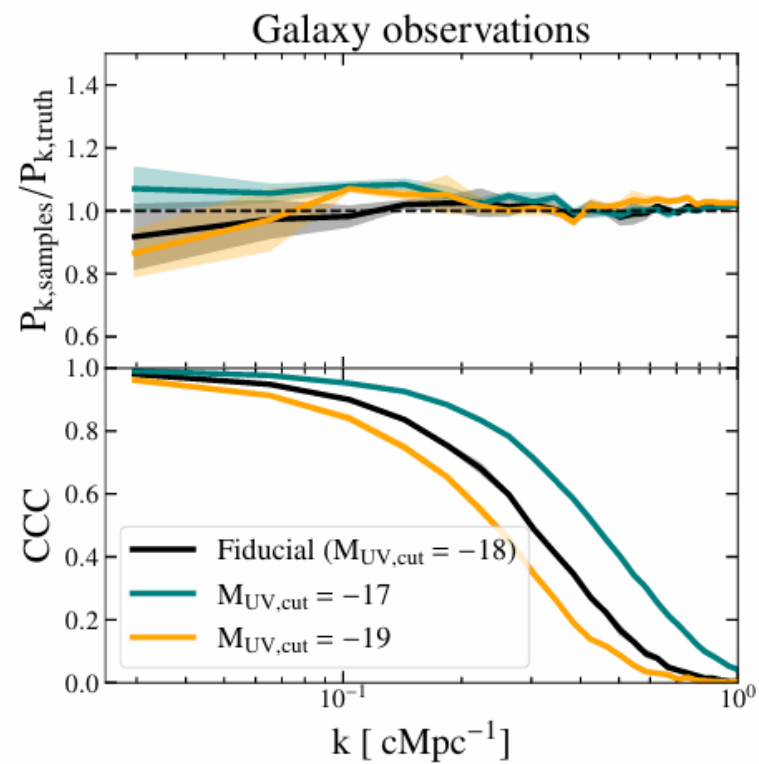
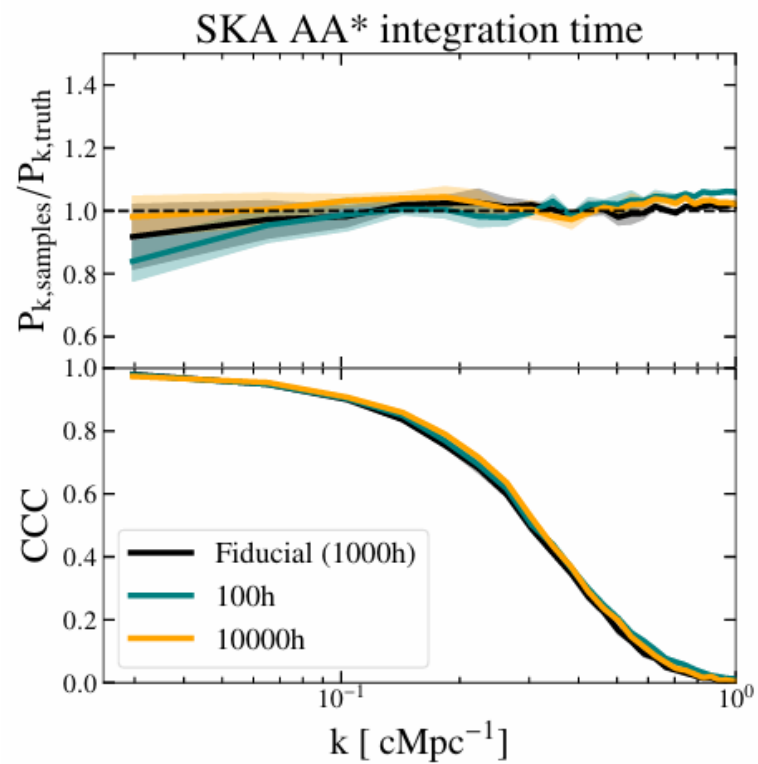
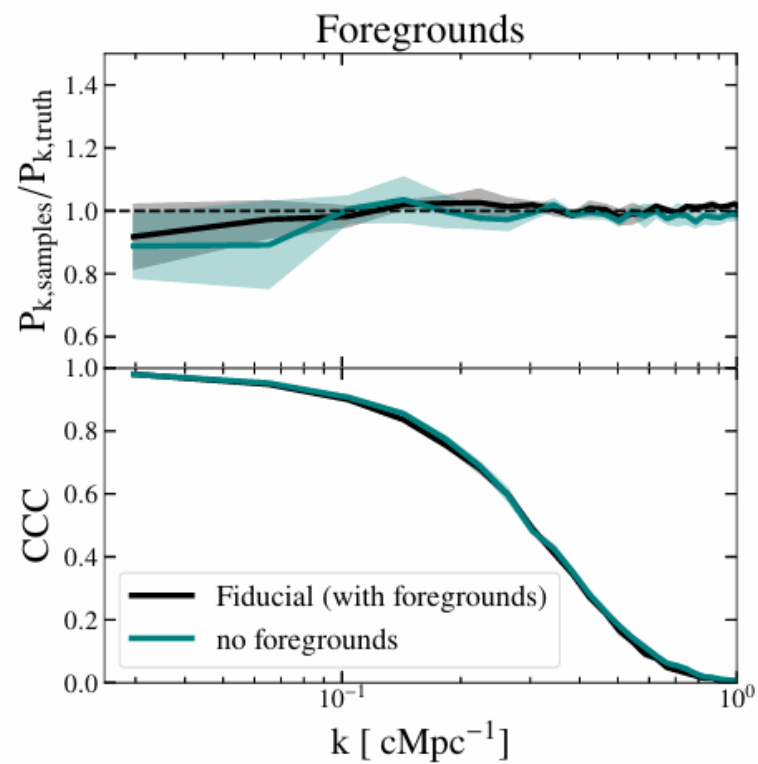


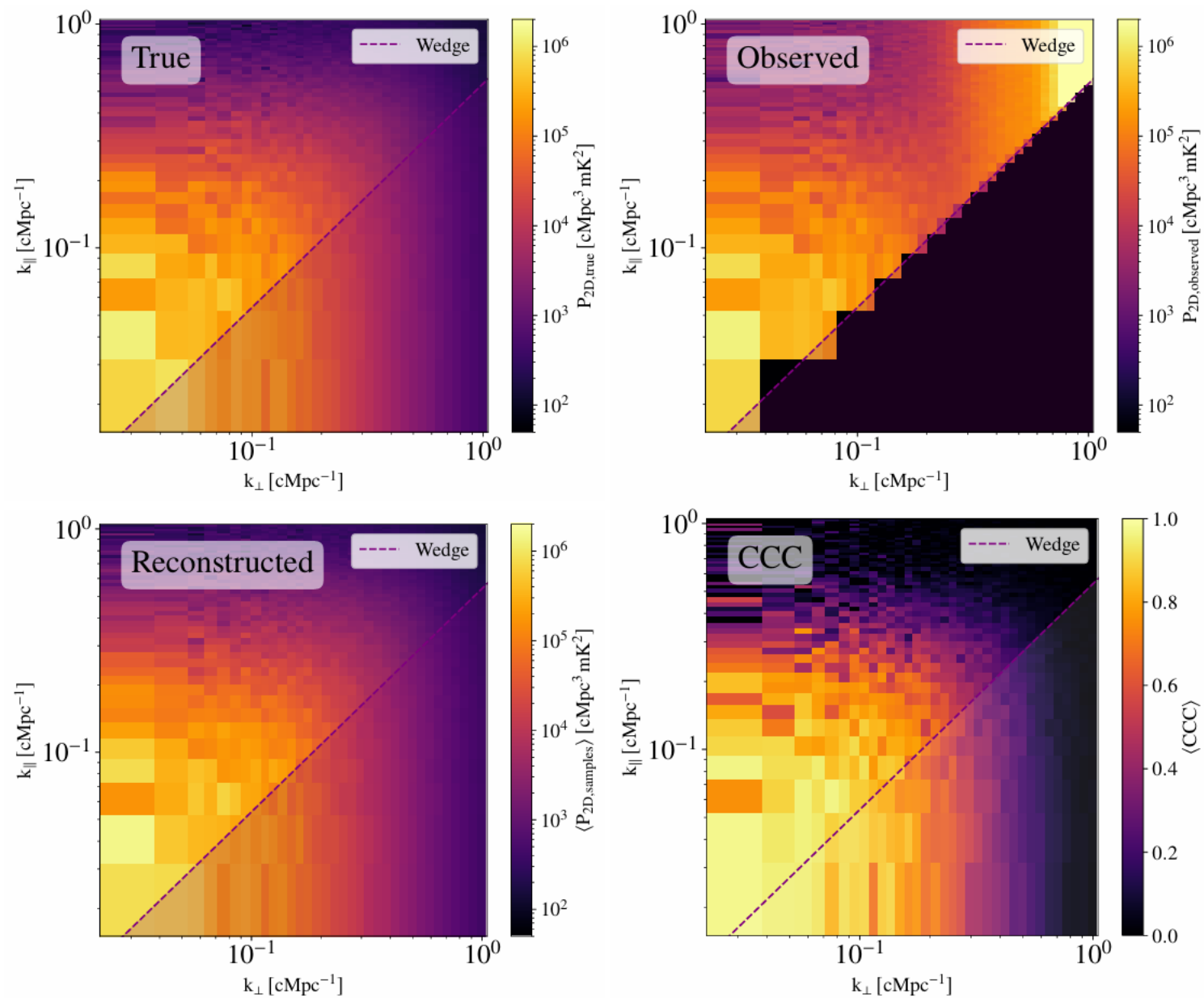
Qin+2026

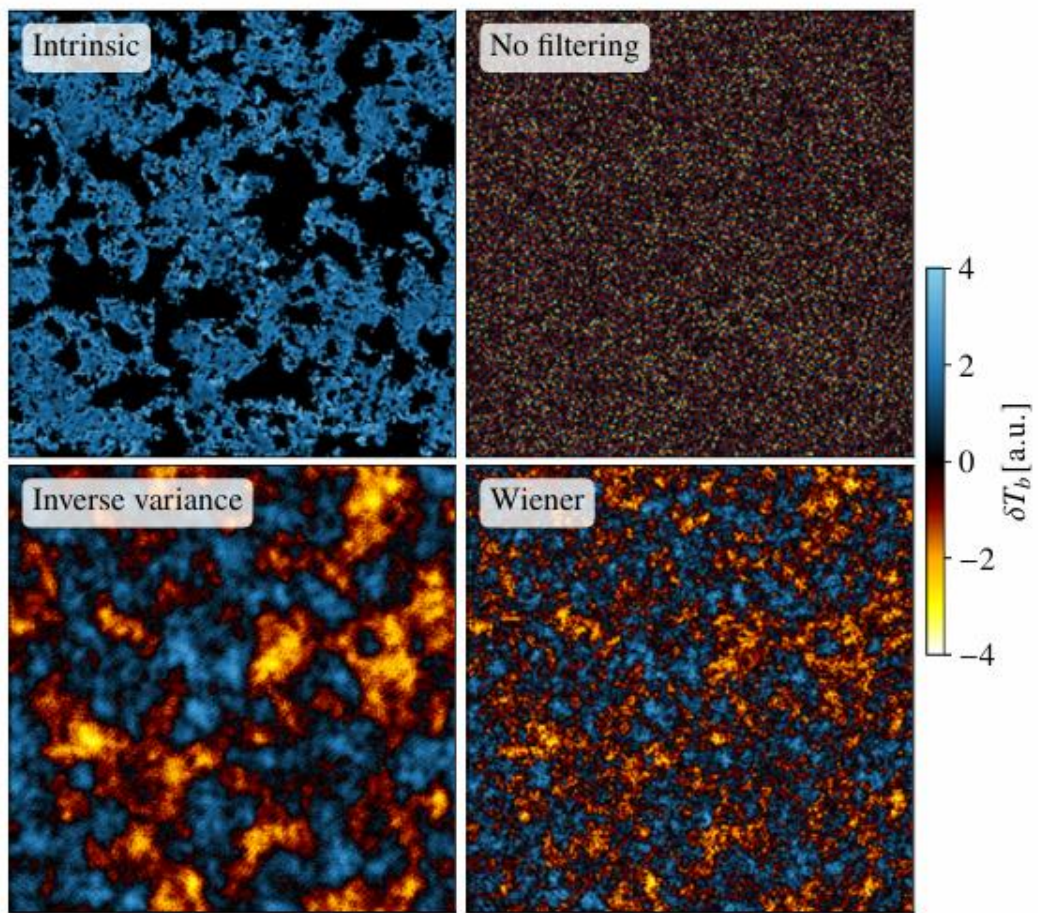
WST

Winkelstein+25,
Casey+22,
Oesch+23

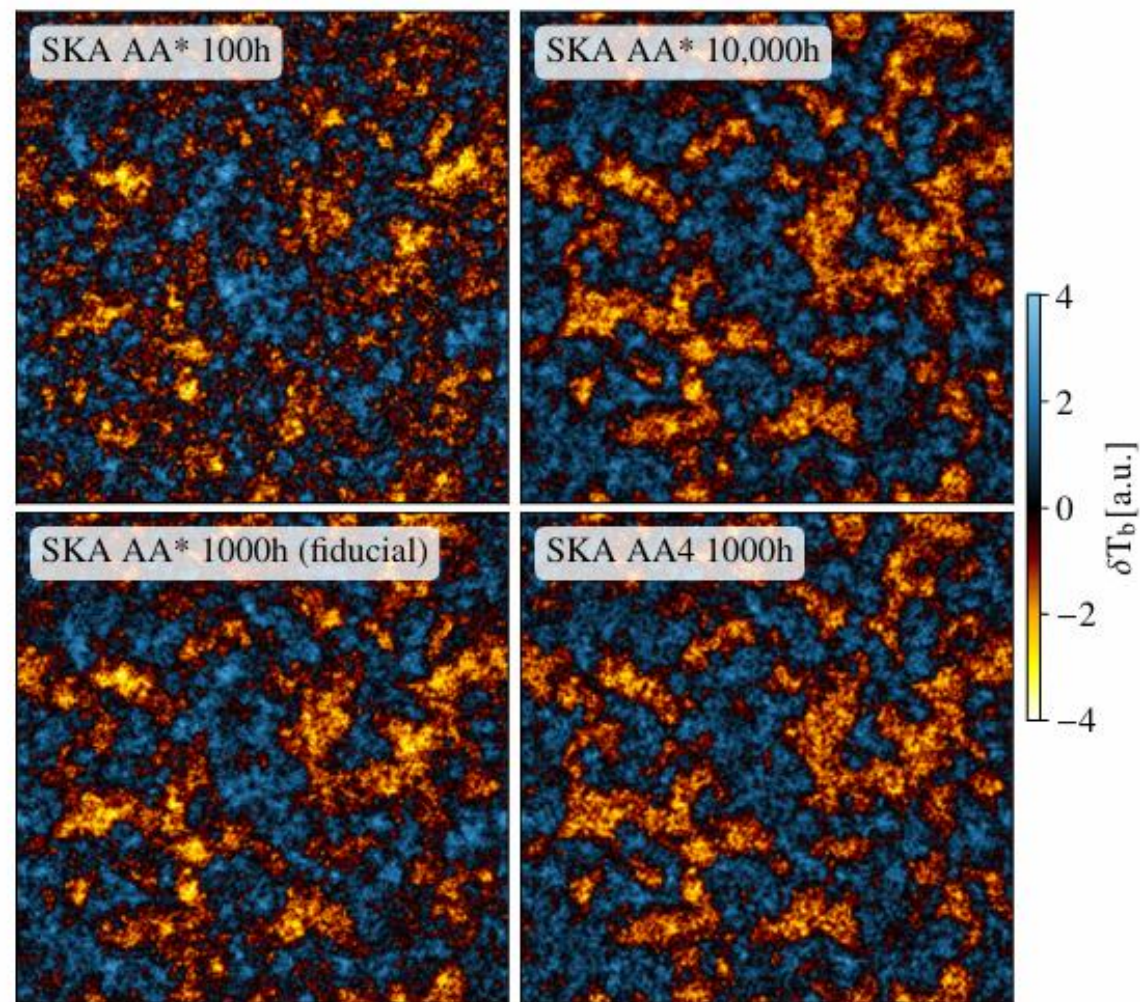








150 cMpc (0.94°)



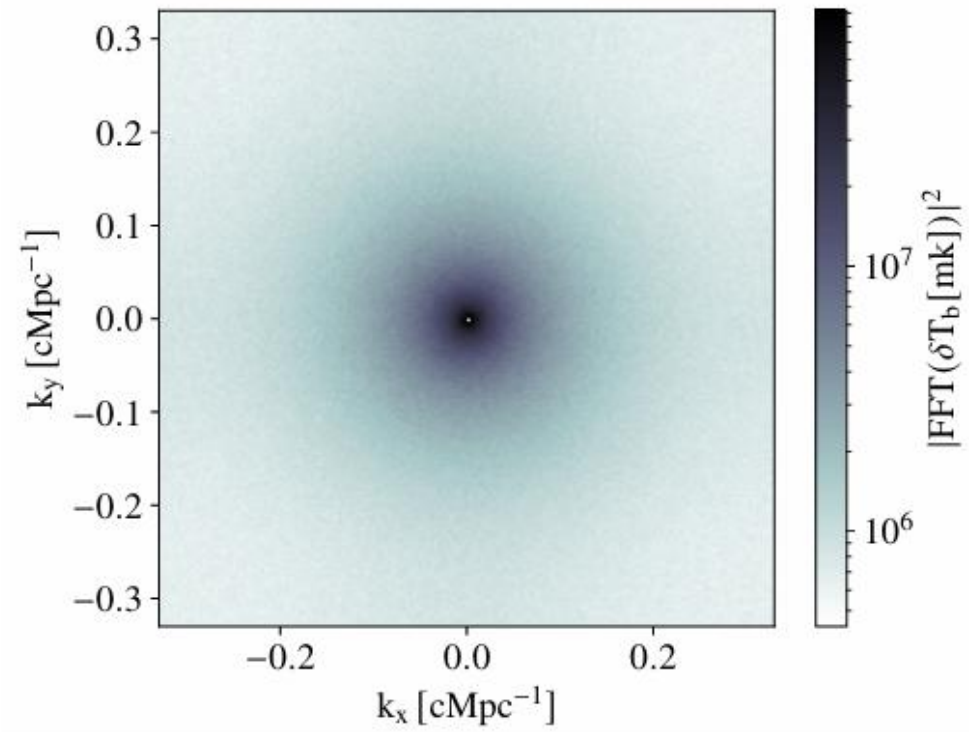


Table B.1. U-Net’s architecture based on

Layer	Input	I/O dim	Param #
Input data	-	2×200^3	-
C1	Input data	32×200^3	1,760
A1	previous	"	-
C2	"	"	27,680
B1	"	"	64
A2	"	"	-
Downsample			
D1	"	32×100^3	8,224
B2	"	"	64
A3	"	"	"
C3	"	"	27,680
B3	"	"	64
A4	"	"	"
C4	"	"	27,680
B4	"	"	64
A5	"	"	"
Downsample			
D2	"	32×50^3	8,224
B5	"	"	64
A6	"	"	"
C5	"	"	27,680
B6	"	"	64
A7	"	"	"
C6	"	"	27,680
B7	"	"	64
A8	"	"	"
Upsample			
U1	"	32×100^3	8,224
B8	"	"	64
A9	"	"	"
C7	concat A2	$64/32 \times 100^3$	55,328
B9	"	"	64
A10	"	"	"
C8	"	"	27,680
B10	"	"	64
A11	"	"	"
Upsample			
U2	"	32×200^3	8,224
B11	"	"	64
A12	"	"	"
C9	concat A5	$64/32 \times 100^3$	110,656
A13	"	"	"
C10	"	"	1,729
Layer	Meaning		
C	Convolution with kernel size 3, stride 1		
A	LeakyReLU activation function		
B	Batch normalization (Group for diffusion)		
D	Convolution with kernel size 2, stride 2		
U	Transpose with kernel size 2, stride 2		

Notes. Layers are numbered sequentially. For diffusion, a Fourier time embedding is added before every activation.

